

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250282838

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

BARBOUR; Robin et al.

Multi-Epitope Vaccine for the Treatment of Alzheimer's Disease

Abstract

The disclosure provides peptide compositions and immunotherapy compositions comprising an amyloid-beta (A β) peptide and a tau peptide. The disclosure also provides methods of treating or effecting prophylaxis of Alzheimer's disease or other diseases with beta-amyloid deposition in a subject, including methods of clearing deposits, inhibiting or reducing aggregation of A β and/or tau, blocking the uptake by neurons, clearing amyloid, and inhibiting propagation of tau seeds in a subject having or at risk of developing Alzheimer's disease or other diseases containing tau and/or amyloid-beta accumulations. The methods include administering to such patients the compositions comprising an amyloid-beta (A β) peptide and a tau peptide.

Inventors: BARBOUR; Robin (Walnut Creek, CA), KINNEY; Gene (Boca Raton, FL), ZAGO; Wagner (San Carlos, CA), NIJJAR; Tarlochan S. (Orinda, CA)

Applicant: OTHAIR PROTHENA LIMITED (Dublin, IE)

Family ID: 1000008629259

Appl. No.: 19/064783

Filed: February 27, 2025

Related U.S. Application Data

parent US continuation 18594435 20240304 parent-grant-document US 12247056 child US 19064783

parent US division 18328528 20230602 parent-grant-document US 11945849 child US 18594435

parent US division 17925813 20221116 PENDING US division PCT/US2021/033222 20210519 child US 18328528

us-provisional-application US 63027150 20200519

us-provisional-application US 63062903 20200807

us-provisional-application US 63140917 20210124

Publication Classification

Int. Cl.: **C07K14/47** (20060101); **A61K39/00** (20060101); **A61K39/39** (20060101); **A61P25/28** (20060101); **A61P37/04** (20060101)

U.S. Cl.:

CPC **C07K14/4711** (20130101); **A61K39/0007** (20130101); **A61K39/39** (20130101); **A61P25/28** (20180101); **A61P37/04** (20180101); A61K2039/53 (20130101); A61K2039/545 (20130101); A61K2039/55516 (20130101); A61K2039/55544 (20130101); A61K2039/55555 (20130101); A61K2039/55572 (20130101); A61K2039/55577 (20130101); A61K2039/575 (20130101); A61K2039/6037 (20130101); C07K2319/00 (20130101)

Background/Summary

RELATED APPLICATIONS [0001] This application is a divisional application of U.S. application Ser. No. 18/594,435, filed Mar. 4, 2024, now U.S. Pat. No. 12,247,056, which is a divisional application of U.S. application Ser. No. 18/328,528, filed Jun. 2, 2023, now U.S. Pat. No. 11,945,849, which is a divisional application of U.S. application Ser. No. 17/925,813, filed Nov. 16, 2022, which is a U.S. national phase application under 35 U.S.C. § 371 of International Application No. PCT/US2021/033222, filed May 19, 2021, and which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/140,917, filed Jan. 24, 2021, U.S. Provisional Patent Application Ser. No. 63/062,903, filed Aug. 7, 2020, and U.S. Provisional Patent Application Ser. No. 63/027,150, filed May 19, 2020, each of which is incorporated by reference herein in its entirety.

SEQUENCE LISTING STATEMENT

[0002] A computer readable form of the Sequence Listing is filed with this application by electronic submission and is incorporated into this application by reference in its entirety. The Sequence Listing is contained in the file created on Mar. 14, 2024, having the file name “20-050-WO-US-DIV_SequenceListing.xml” and is 1,171,456 bytes in size.

FIELD

[0003] The disclosure relates to the technical fields of immunology and medicine, and in particular to the treatment of Alzheimer's disease and other diseases of protein misfolding.

BACKGROUND

[0004] Alzheimer's disease (AD) is a progressive disease resulting in senile dementia. Broadly speaking, the disease falls into two categories: late onset, which occurs in old age (65+ years) and early onset, which develops well before the senile period, i.e., between 35 and 60 years. In both types of disease, the pathology is the same, but the abnormalities tend to be more severe and widespread in cases beginning at an earlier age. The disease is characterized by at least two types of lesions in the brain, neurofibrillary tangles and senile plaques. Neurofibrillary tangles are intracellular deposits of microtubule associated tau protein consisting of two filaments twisted about each other in pairs. Senile plaques (i.e., amyloid plaques) are areas of disorganized neuropil up to 150 μ m across with extracellular amyloid deposits at the center which are visible by microscopic analysis of sections of brain tissue. The accumulation of amyloid plaques within the central nervous system is also associated with Down's syndrome and other cognitive disorders, Cerebral amyloid angiopathy (CAA), and the ocular disease Age-Related Macular Degeneration.

[0005] A principal constituent of the plaques is a peptide termed A β or β -amyloid peptide. A β

peptide is a 4-kDa internal fragment of 38-43 amino acids of a larger transmembrane glycoprotein named amyloid precursor protein (APP). As a result of proteolytic processing of APP by different secretase enzymes, A β is primarily found in both a short form, 40 amino acids in length, and a long form, ranging from 42-43 amino acids in length. Part of the hydrophobic transmembrane domain of APP is found at the carboxy end of A β , and may account for the ability of A β to aggregate into plaques, particularly in the case of the long form. Accumulation of amyloid plaques in the brain eventually leads to neuronal cell death. The cognitive and physical symptoms associated with this type of neural deterioration characterize Alzheimer's disease.

[0006] Another protein reported to occur at increased levels in Alzheimer's patients relative to the general population is tau, the principal constituent of neurofibrillary tangles, which together with amyloid plaques are a hallmark characteristic of Alzheimer's disease. Tau tangles constitute abnormal fibrils measuring 10 nm in diameter occurring in pairs wound in a helical fashion with a regular periodicity of 80 nm. The tau within neurofibrillary tangles is abnormally phosphorylated (hyperphosphorylated) with phosphate groups attached to specific sites on the molecule. Severe involvement of neurofibrillary tangles is seen in the layer II neurons of the entorhinal cortex, the CA1 and subicular regions of the hippocampus, the amygdala, and the deeper layers (layers III, V, and superficial VI) of the neocortex in Alzheimer's disease. Tau pathologies are known to correlate to cognitive decline.

[0007] Accordingly, there exists the need for new therapies and reagents for the prevention or treatment of Alzheimer's disease, in particular, therapies and reagents capable of causing an immune response to the A β and Tau present in patients.

SUMMARY

[0008] In some embodiments, disclosure is directed to a polypeptide including a first peptide comprising 3-10 amino acids from residues 1-10 of SEQ ID NO:01 linked to a second peptide including 3-13 amino acids from residues 244-400 of SEQ ID NO:02. For example, the second peptide may be from the microtubule binding region (MTBR) of tau (residues 244-372 of SEQ ID NO:02). The first peptide may be N-terminal to the second peptide or the first peptide may be C-terminal to the second peptide. In addition, the first peptide may include an amino acid sequence of one of SEQ ID NOS: 3 to 38 or SEQ ID NOS: 1002 to 1057 and the second peptide may include an amino acid sequence of one of SEQ ID NOS: 39-56, 83-86, or 146-996. For example, the first polypeptide may be DAEFRHD (SEQ ID NO:06), DAEFR (SEQ ID NO: 08), or EFRHD (SEQ ID NO:21), and the second polypeptide may be 5-13 amino acids, for example QIVYKPV (SEQ ID NO:39), EIVYKSV (SEQ ID NO:42), EIVYKSP (SEQ ID NO: 43), EIVYKPV (SEQ ID NO:44), NIKHVP (SEQ ID NO:48), VKSKIGST (SEQ ID NO: 801), SKIGSTEN (SEQ ID NO:817), TENLKHQP (SEQ ID NO:695), ENLKHQPG (SEQ ID NO: 689), SKIGSTDNIKH (SEQ ID NO:985), SKIGSKDNIKH (SEQ ID NO:986), or SKIGSLDNIKH (SEQ ID NO:988).

[0009] In other embodiments, the first peptide and second peptide may be linked by a cleavable linker, which may be an amino acid sequence. A cleavable peptide linker, if present, can be 1-10 amino acids in length. In some embodiments, the linker comprises between about 1-10 amino acids, about 1-9 amino acids, about 1-8 amino acids, about 1-7 amino acids, about 1-6 amino acids, about 1-5 amino acids, about 1-4 amino acids, about 1-3 amino acids, about 2 amino acids, or one (1) amino acid. In some embodiments, the cleavable peptide linker is 1 amino acid, 2 amino acids, 3 amino acids, 4 amino acids, 5 amino acids, 6 amino acids, 7 amino acids, 8 amino acids, 9 amino acids, or 10 amino acids. For example, the linker may be arginine-arginine (Arg-Arg), arginine-valine-arginine-arginine (Arg-Val-Arg-Arg (SEQ ID NO:69)), valine-citrulline (Val-Cit), valine-arginine (Val-Arg), valine-lysine (Val-Lys), valine-alanine (Val-Ala), phenylalanine-lysine (Phe-Lys), glycine-alanine-glycine-alanine (Gly-Ala-Gly-Ala; SEQ ID NO:80), Ala-Gly-Ala-Gly (SEQ ID NO:81), or Lys-Gly-Lys-Gly (SEQ ID NO:82). In particular embodiments, the polypeptide may be DAEFRHRRQIVYKPV (SEQ ID NO:57), DAEFRHRRREIVYKSV (SEQ ID NO:58), DAEFRHRRRVKSKIGSTGGC (SEQ ID NO:997), DAEFRHRRRSKIGSTENGGC (SEQ ID

NO:998), DAEFRHRRTEHLKHQPGGC (SEQ ID NO: 999), DAEFRHRRTEHLKHQPGGGC (SEQ ID NO:1000), or DAEFRHRRSKIGSKDNIKHGGC (SEQ ID NO:1001).

[0010] In further embodiments, the polypeptide may include a linker to a carrier at a C-terminal portion of the polypeptide, or at a N-terminal portion of the polypeptide. A linker, if present, can be 1-10 amino acids in length. In some embodiments, the linker comprises between about 1-10 amino acids, about 1-9 amino acids, about 1-8 amino acids, about 1-7 amino acids, about 1-6 amino acids, about 1-5 amino acids, about 1-4 amino acids, about 1-3 amino acids, about 2 amino acids, or one (1) amino acid. In some embodiments, the linker is 1 amino acid, 2 amino acids, 3 amino acids, 4 amino acids, 5 amino acids, 6 amino acids, 7 amino acids, 8 amino acids, 9 amino acids, or 10 amino acids. For example, the linker may include an amino acid sequence of GG, GGG, AA, AAA, KK, KKK, SS and SSS. In addition, the linker to the carrier, if present at the C-terminus, may include a C-terminal cysteine (C). Alternatively, the linker to the carrier, if present at the N-terminus, may include a N-terminal cysteine (C). For example, the polypeptide may include the amino acid sequence of DAEFRHRRQIVYKPVXXC (SEQ ID NO: 70), wherein XX and C are independently optional and, if present, XX can be GG, AA, KK, SS, GAGA (SEQ ID NO:80), AGAG (SEQ ID NO:81), or KGKG (SEQ ID NO:82). For example, the polypeptide may include the amino acid sequence of DAEFRHRRREIVYKSVXXC (SEQ ID NO:79), wherein XX and C are independently optional and, if present, XX can be GG, AA, KK, SS, GAGA (SEQ ID NO:80), AGAG (SEQ ID NO:81), and KGKG (SEQ ID NO:82).

[0011] In other embodiments, the disclosure is directed to an immunotherapy composition including the polypeptides of the disclosure, wherein the polypeptide may be linked to a carrier. The carrier may include serum albumins, immunoglobulin molecules, thyroglobulin, ovalbumin, tetanus toxoid (TT), diphtheria toxoid (DT), a genetically modified cross-reacting material (CRM) of diphtheria toxin, CRM197, meningococcal outer membrane protein complex (OMPC) and *H. influenzae* protein D (HiD), rEPA (*Pseudomonas aeruginosa* exotoxin A), KLH (keyhole limpet hemocyanin), and flagellin.

[0012] Still further, embodiments of the disclosure are directed to a pharmaceutical formulation includes the polypeptides or the immunotherapy compositions of the disclosure, and including at least one adjuvant. The adjuvant may be aluminum hydroxide, aluminum phosphate, aluminum sulfate, 3 De-O-acylated monophosphoryl lipid A (MPL), QS-21, QS-18, QS-17, QS-7, TQL1055, Complete Freund's Adjuvant (CFA), Incomplete Freund's Adjuvant (IFA), oil in water emulsions (such as squalene or peanut oil), CpG, polyglutamic acid, polylysine, Adda Vax™, MF59®, and combinations thereof. In addition, the formulation may include a liposomal formulation, a diluent, or a multiple antigen presenting system (MAP). The MAP may include one or more of a Lys-based dendritic scaffold, helper T-cell epitopes, immune stimulating lipophilic moieties, cell penetrating peptides, radical induced polymerization, self-assembling nanoparticles as antigen-presenting platforms and gold nanoparticles.

[0013] Still further, embodiments of the disclosure are directed to an immunotherapy composition including a first peptide sequence comprising 3-10 amino acid residues from the first ten N-terminal residues of SEQ ID NO:01 and a second peptide sequence comprising 3-13 amino acids from residues 244-400 of SEQ ID NO:02. The first peptide may include an amino acid sequence of one of SEQ ID NOS: 3 to 38 or SEQ ID NOS: 1002 to 1057, and the second peptide may include an amino acid sequence of one of SEQ ID NOS: 39 to 56, SEQ ID NOS: 83-86 or SEQ ID NOS: 146-996. Each of the first peptide and the second peptide may include a linker to a carrier at a C-terminal portion of the polypeptide, or at a N-terminal portion of the polypeptide. When present, the linker may include an amino acid sequence selected from GG, GGG, AA, AAA, KK, KKK, SS, SSS, GAGA (SEQ ID NO:80), AGAG (SEQ ID NO:81), and KGKG (SEQ ID NO:82), and may include a C-terminal cysteine (C). In some embodiments, where the C-terminal residues in the immunogen are either IVYKPV (SEQ ID NO:194), VYKPV (SEQ ID NO:195), YKPV (SEQ ID NO:196), KPV, or PV the linker is an amino acid linker that does not have a N-terminal glycine

(e.g., GG, GAGA (SEQ ID NO:80)). The carrier may include serum albumins, immunoglobulin molecules, thyroglobulin, ovalbumin, tetanus toxoid (TT), diphtheria toxoid (DT), a genetically modified cross-reacting material (CRM) of diphtheria toxin, CRM197, meningococcal outer membrane protein complex (OMPC) and *H. influenzae* protein D (HiD), rEPA (*Pseudomonas aeruginosa* exotoxin A), KLH (keyhole limpet hemocyanin), and flagellin.

[0014] In addition, the immunotherapy composition may include at least one pharmaceutically acceptable diluent and/or a multiple antigen presenting system (MAP). The MAP may include one or more of a Lys-based dendritic scaffold, helper T-cell epitopes, immune stimulating lipophilic moieties, cell penetrating peptides, radical induced polymerization, self-assembling nanoparticles as antigen-presenting platforms and gold nanoparticles.

[0015] The immunotherapy composition may be included in a pharmaceutical composition including the immunotherapy composition and at least one adjuvant such as aluminum hydroxide, aluminum phosphate, aluminum sulfate, 3 De-O-acylated monophosphoryl lipid A (MPL), QS-21, QS-18, QS-17, QS-7, TQL1055, Complete Freund's Adjuvant (CFA), Incomplete Freund's Adjuvant (IFA), oil in water emulsions (such as squalene or peanut oil), CpG, polyglutamic acid, polylysine, Adda Vax™, MF59®, and combinations thereof.

[0016] Embodiments of the disclosure are also directed to nucleic acid sequences encoding the polypeptides and the immunotherapy compositions of the disclosure. The nucleic acids may be included in a nucleic acid immunotherapy composition including the nucleic acid and at least one adjuvant.

[0017] Still further, embodiments of the disclosure are directed to a methods for treating or effecting prophylaxis of Alzheimer's disease in a subject, and methods for inhibiting or reducing aggregation of at least one of A β and tau in a subject having or at risk of developing Alzheimer's disease. The methods include administering to the subject an immunotherapy composition, a nucleic acids immunotherapy composition, or a pharmaceutical formulation of the disclosure.

[0018] The methods of the disclosure may include repeating the administering at least a second time, at least a third time, at least a fourth time, at least a fifth time, or at least a sixth time, and may include repeating the administering at an interval of about 21 to about 28 days.

[0019] Still further, methods of the disclosure are directed to inducing an immune response in an animal. The methods include administering to the animal a polypeptide, an immunotherapy composition, a pharmaceutical formulation or a nucleic acid immunotherapy composition of the disclosure in a regimen effective to generate an immune response including antibodies that specifically bind to A β , tau, or both A β and tau. The immune response may include antibodies that specifically bind to the N-terminal region of A β and/or the microtubule region of tau.

[0020] In other embodiments, the disclosure is directed to an immunization kit including an immunotherapy composition of the disclosure and may include an adjuvant, wherein the immunotherapy composition may be in a first container and the adjuvant may be a second container.

[0021] Still further, the disclosure is directed to a kit including a nucleic acid immunotherapy composition of the disclosure and may include an adjuvant. The nucleic acid may be in a first container and the adjuvant may be in a second container.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0022] FIG. 1 shows the results of an experiment comparing the geometric mean titers of Guinea Pig serum for immunogen DAEFRHRRQIVYKPV (SEQ ID NO:57) on monomeric A β amino acids 1-28 (DAEFRHDSGYEVHHQKLFFAEDVGSNKG; SEQ ID NO:67), soluble aggregate species of A β (A β 42), full length tau, and tau MTBR epitope containing GGGSVQIVYKPV DLS

(SEQ ID NO:68) containing an N-terminal biotin.

[0023] FIG. 2A shows the results of an experiment comparing the titers of Guinea pig serum for a single peptide immunogen versus dual peptide immunogens of the disclosure on all forms of A β and full-length tau protein (DAEFRHD is SEQ ID NO:6; QIVYKPV is SEQ ID NO: 39; DAEFRHDDRQIVYKPV is SEQ ID NO:57).

[0024] FIG. 2B shows the results of an experiment comparing the titers of Guinea pig serum for a single peptide immunogen versus dual peptide immunogens of the disclosure on all forms of A β and full-length tau protein (DAEFRHD is SEQ ID NO:6; QIVYKPV is SEQ ID NO: 39; EIVYKSP is SEQ ID NO:43; DAEFRHDDRQIVYKPV is SEQ ID NO:57).

[0025] FIG. 3A shows staining of A β and Tau pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from a Guinea pig vaccinated with Immunogen 9 TABLE-US-00001 (DAEFRHDDRQIVYKPVGGC; SEQ ID NO: 59).

[0026] FIG. 3B shows staining of A β and Tau pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from a Guinea pig vaccinated with Immunogen 9 TABLE-US-00002 (DAEFRHDDRQIVYKPVGGC; SEQ ID NO: 59).

[0027] FIG. 3C shows staining of A β and Tau pathology in fresh frozen human AD brain tissue using serum (1:1500 dilution) from a Guinea pig vaccinated with Immunogen 9 TABLE-US-00003 (DAEFRHDDRQIVYKPVGGC; SEQ ID NO: 59).

[0028] FIG. 3D shows lack of staining of A β and Tau pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from an unvaccinated Guinea pig.

[0029] FIG. 3E shows lack of staining of A β and Tau pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from an unvaccinated Guinea pig.

[0030] FIG. 3F shows lack of staining of A β and Tau pathology in fresh frozen human AD brain tissue using serum (1:1500 dilution) from an unvaccinated Guinea pig.

[0031] FIG. 4A shows that Guinea pig serum from animals vaccinated with a dual immunogen peptide, DAEFRHDDRQIVYKPVGGC (SEQ ID NO:59; Immunogen 9), inhibited Abeta soluble aggregates binding to primary neurons in a dose-dependent manner.

[0032] FIG. 4B shows examples of Ab staining on primary neurons in the presence or absence of Guinea pig serum from animals vaccinated with a dual immunogen peptide, TABLE-US-00004 (SEQ ID NO: 59; Immunogen 9) DAEFRHDDRQIVYKPVGGC.

[0033] FIG. 5A shows that mice vaccinated with dual peptide antigens with various linkers produce similar titers to A β and Tau.

[0034] FIG. 5B shows that mice vaccinated with dual peptide antigens with the Tau sequence EIVYKSP (SEQ ID NO:43) produce titers to A β and Tau.

[0035] FIG. 6A shows staining of A β and Tau pathology in fresh frozen human AD brain tissue using serum (1:1000 dilution) from a mouse vaccinated with Immunogen 19.

[0036] FIG. 6B shows staining of A β and Tau pathology in fresh frozen human AD brain tissue using serum (1:1000 dilution) from a mouse vaccinated with Immunogen 19.

[0037] FIG. 6C shows lack of staining of A β and Tau pathology in fresh frozen human AD brain tissue using serum (1:1000 dilution) from an unvaccinated mouse.

[0038] FIG. 6D shows staining of Tau pathology in fresh frozen human AD brain tissue using anti-Tau antibody m6H3 (0.1 μ g/ml).

[0039] FIG. 6E shows comparable staining of Tau pathology in fresh frozen human AD brain tissue using serum (1:1500 dilution) from a Guinea pig vaccinated with Immunogen 9 TABLE-US-00005 (DAEFRHDDRQIVYKPVGGC; SEQ ID NO: 59).

[0040] FIGS. 7A and 7B show titers of dual A β -tau immunogen in two injection paradigms. Titers to A β 1-28, full-length tau and the carrier CRM protein of the two injection schedules ((FIG. 7A) showing 0, 4, 12 and 24 weeks; (FIG. 7B) showing 0, 8 and 24 weeks) over the experiment are expressed as mean titer $\pm$ SEM.

[0041] FIG. 8A shows individual animal titers to A β and tau two weeks post each injection for the

injection schedule of FIG. 7A, i.e., at weeks 2, 6, 14 and 26.

[0042] FIG. 8B shows individual animal titers to A β and tau two weeks post each injection for the injection schedule of FIG. 7B, i.e., at weeks 2, 14 and 26.

[0043] FIG. 9A shows staining of A β and Tau pathology in fresh frozen human AD brain tissue using week 26 serum (1:300 dilution) from cynomolgus monkey #1001 vaccinated with Immunogen 15 (DAEFRHDDRQIVYKPVGGC; SEQ ID NO:59) on a four injection vaccination schedule.

[0044] FIG. 9B shows a zoomed view of the rectangle indicating staining of A β pathology in FIG. 9A.

[0045] FIG. 9C shows a zoomed view of the rectangle indicating staining of Tau pathology in FIG. 9A.

[0046] FIG. 10A shows staining of A β pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from cynomolgus monkey #1003 vaccinated with Immunogen 15 TABLE-US-00006 (DAEFRHDDRQIVYKPVGGC; SEQ ID NO: 59).

[0047] FIG. 10B shows modest staining of Tau pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from cynomolgus monkey #1003.

[0048] FIG. 11A shows staining of A β pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from cynomolgus monkey #1501 vaccinated with Immunogen 15 TABLE-US-00007 (DAEFRHDDRQIVYKPVGGC; SEQ ID NO: 59).

[0049] FIG. 11B shows minimal staining of Tau pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from cynomolgus monkey #1501.

[0050] FIG. 12A shows staining of A β pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from cynomolgus monkey #2501 vaccinated with Immunogen 15 TABLE-US-00008 (DAEFRHDDRQIVYKPVGGC; SEQ ID NO: 59).

[0051] FIG. 12B shows a lack of staining of Tau pathology in fresh frozen human AD brain tissue using serum (1:300 dilution) from cynomolgus monkey #2501.

[0052] FIG. 13 shows animal titers to A β and tau 9 weeks post injection at day 0, at 4 weeks and at 8 weeks for the dual Abeta-tau constructs DAEFRHDDRVRKSKIGSTGGC (SEQ ID NO: 997); DAEFRHDDRVRKSKIGSTENGGC (SEQ ID NO:998); DAEFRHDDRRTENLKHQPGGC (SEQ ID NO:999); DAEFRHDDRRENLKHQPGGGC (SEQ ID NO: 1000); and DAEFRHDDRVRKSKIGSKDNIKHGGC (SEQ ID NO:1001).

[0053] FIG. 14 shows blocking of tau binding to Heparin by sera from A β -tau constructs TABLE-US-00009 (SEQ ID NO: 997) DAEFRHDDRVRKSKIGSTGGC; (SEQ ID NO: 998) DAEFRHDDRVRKSKIGSTENGGC; (SEQ ID NO: 999) DAEFRHDDRRTENLKHQPGGC; (SEQ ID NO: 1000) DAEFRHDDRRENLKHQPGGGC; and (SEQ ID NO: 1001) DAEFRHDDRVRKSKIGSKDNIKHGGC

[0054] FIG. 15A shows competition of tau by sera from DAEFRHDDRVRKSKIGSTGGC construct (SEQ ID NO:997).

[0055] FIG. 15B shows competition of tau by sera from DAEFRHDDRVRKSKIGSTENGGC construct (SEQ ID NO:998).

[0056] FIG. 15C shows competition of tau by sera from TABLE-US-00010 (SEQ ID NO: 999) DAEFRHDDRRTENLKHQPGGC. construct

[0057] FIG. 15D shows competition of tau by sera from TABLE-US-00011 (SEQ ID NO: 1000) DAEFRHDDRRENLKHQPGGGC. construct

[0058] FIG. 15E shows competition of tau by sera from TABLE-US-00012 (SEQ ID NO: 1001) DAEFRHDDRVRKSKIGSKDNIKHGGC. construct

[0059] FIG. 15F shows competition of tau by sera from controls.

[0060] FIG. 15G shows competition of tau by sera from

TABLE-US-00013 (SEQ ID NO: 1001) DAEFRHDDRVRKSKIGSKDNIKHGGC. construct

[0061] FIG. 16A shows Abeta/Tau Dual Vaccine Mouse Serum 2.3_031521 (1:300) Example of

staining with DAEFRHDDRVRKSKIGSTGGC (SEQ ID NO:997).

[0062] FIG. **16B** shows Abeta/Tau Dual Vaccine Mouse Serum 3.1_031521 (1:300) Example of staining with DAEFRHDDRSKIGSTENGGC (SEQ ID NO:998).

[0063] FIG. **16C** shows Abeta/Tau Dual Vaccine Mouse Serum 8.4_031521 (1:300) Example of staining with DAEFRHDDRRTENLKHQPGGC (SEQ ID NO:999).

[0064] FIG. **16D** shows Abeta/Tau Dual Vaccine Mouse Serum 9.2_031521 (1:300) Example of staining with DAEFRHDDRRENLKHQPGGGC (SEQ ID NO:1000).

[0065] FIG. **16E** shows Abeta/Tau Dual Vaccine Mouse Serum 11.1_031521 (1:300) Example of staining with DAEFRHDDRSKIGSKDNIKHGGC (SEQ ID NO: 1001).

DESCRIPTION

[0066] The disclosure provides peptide compositions and immunotherapy compositions comprising an amyloid-beta ($A\beta$) peptide and a tau peptide. The disclosure also provides methods of treating or effecting prophylaxis of Alzheimer's disease or other diseases with beta-amyloid deposition in a subject, including methods of clearing and preventing formation of deposits, inhibiting or reducing aggregation of $A\beta$ and/or tau, blocking the binding and/or uptake of $A\beta$ and/or tau by neurons, inhibiting transmission of tau species between cells, and inhibiting propagation of pathology between brain regions in a subject having or at risk of developing Alzheimer's disease or other diseases containing tau and/or amyloid-beta accumulations. The methods include administering to such patients the compositions comprising an amyloid-beta ($A\beta$) peptide and a tau peptide.

[0067] A number of terms are defined below. As used herein, the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. For example, the term “a compound” or “at least one compound” can include a plurality of compounds, including mixtures thereof.

[0068] Unless otherwise apparent from the context, the term “about” encompasses insubstantial variations, such as values within a standard margin of error of measurement (e.g., SEM) of a stated value. For example the term “about” as used herein when referring to a measurable value such as a parameter, an amount, a temporal duration, can encompass variations of $\pm 10\%$ or less, $\pm 5\%$ or less, or $\pm 1\%$ or less or less of and from the specified value. Designation of a range of values includes all integers within or defining the range, and all subranges defined by integers within the range. As used herein, statistical significance means $p \leq 0.05$.

[0069] Compositions or methods “comprising” or “including” one or more recited elements may include other elements not specifically recited. For example, a composition that “comprises” or “includes” a polypeptide sequence may contain the sequence alone or in combination with other sequences or ingredients.

[0070] An individual is at increased risk of a disease if the subject has at least one known risk-factor (e.g., age, genetic, biochemical, family history, and situational exposure) placing individuals with that risk factor at a statistically significant greater risk of developing the disease than individuals without the risk factor.

[0071] The term “patient” includes human and other mammalian subjects that receive either prophylactic or therapeutic treatment, including treatment naïve subjects. As used herein, the terms “subject” or “patient” refer to any single subject for which treatment is desired, including other mammalian subjects such as, humans, cattle, dogs, guinea pigs, rabbits, and so on. Also intended to be included as a subject are any subjects involved in clinical research trials not showing any clinical sign of disease, or subjects involved in epidemiological studies, or subjects used as controls.

[0072] The term “disease” refers to any abnormal condition that impairs physiological function. The term is used broadly to encompass any disorder, illness, abnormality, pathology, sickness, condition, or syndrome in which physiological function is impaired, irrespective of the nature of the etiology.

[0073] The term “symptom” refers to a subjective evidence of a disease, such as altered gait, as

perceived by the subject. A “sign” refers to objective evidence of a disease as observed by a physician.

[0074] As used herein, the terms “treat” and “treatment” refer to the alleviation or amelioration of one or more symptoms or effects associated with the disease, prevention, inhibition or delay of the onset of one or more symptoms or effects of the disease, lessening of the severity or frequency of one or more symptoms or effects of the disease, and/or increasing or trending toward desired outcomes as described herein.

[0075] The terms “prevention”, “prevent”, or “preventing” as used herein refer to contacting (for example, administering) the peptide(s) or immunotherapy compositions of the present disclosure with a subject before the onset of a disease, with or without A β and/or tau pathology already present (primary and secondary prevention), thereby delaying the onset of clinical symptoms and/or alleviating symptoms of the disease after the onset of the disease, compared to when the subject is not contacted with the peptide or immunotherapy compositions, and does not refer to completely suppressing the onset of the disease. In some cases, prevention may occur for limited time after administration of the peptide or immunotherapy compositions of the present disclosure. In other cases, prevention may occur for the duration of a treatment regimen comprising administering the peptide or immunotherapy compositions of the present disclosure.

[0076] The terms “reduction”, “reduce”, or “reducing” as used herein refer to decreasing the amount of A β and/or tau present in a subject or in tissue of the subject, or suppressing an increase in the amount of A β and/or tau present in a subject or in tissue of the subject, which encompasses decreasing or suppressing an increase in (e.g., decreasing the rate of increase) the amount of A β and/or tau present, accumulated, aggregated, or deposited in the subject or tissue in the subject. In certain embodiments, the decrease in or suppression of an increase in (e.g., decreasing the rate of increase) the amount of A β and/or tau present, accumulated, aggregated, or deposited in the subject refers to an amount of A β and/or tau present, accumulated, aggregated, or deposited in the central nervous system (CNS) of the subject. In certain embodiments, the decrease in or suppression of an increase in (e.g., decreasing the rate of increase) the amount of A β and/or tau present, accumulated, aggregated, or deposited in the subject refers to an amount of A β and/or tau present, accumulated, aggregated, or deposited in the periphery (e.g., peripheral circulatory system) of the subject. In certain embodiments, the decrease in or suppression of an increase in (e.g., decreasing the rate of increase) the amount of A β and/or tau present, accumulated, aggregated, or deposited in the subject refers to an amount of A β and/or tau present, accumulated, aggregated, or deposited in the brain of the subject. In some embodiments, the A β and/or tau reduced is the pathological form(s) of the A β (e.g., extracellular plaque deposits of the β -amyloid peptide (A β), neuritic amyloid plaques), and/or tau (e.g., neurofibrillary tangles of tau, dystrophic neurites). In yet other embodiments, pathological indicators of neurodegenerative disease are decreased.

[0077] The terms “epitope” or “antigenic determinant” refers to a site on an antigen to which B and/or T cells respond, or to a site on an antigen to which an antibody binds. Epitopes can be formed both from contiguous amino acids or from noncontiguous amino acids juxtaposed by tertiary folding of a protein. Epitopes formed from contiguous amino acids are typically retained on exposure to denaturing solvents whereas epitopes formed by tertiary folding are typically lost on treatment with denaturing solvents. An epitope typically includes at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, or at least 13 amino acids in a unique spatial conformation. Methods of determining spatial conformation of epitopes include, for example, x-ray crystallography and 2-dimensional nuclear magnetic resonance. See, e.g., Epitope Mapping Protocols in Methods in Molecular Biology, Vol. 66, Glenn E. Morris, Ed. (1996).

[0078] An “immunogenic agent” or “immunogen” or “antigen” is capable of inducing an immunological response against itself or modified/processed versions of itself upon administration to an animal, optionally in conjunction with an adjuvant. The terms “immunogenic agent” or

“immunogen” or “antigen” refer to a compound or composition comprising a peptide, polypeptide or protein which is “antigenic” or “immunogenic” when administered in an appropriate amount (an “immunogenically effective amount”), i.e., capable of inducing, eliciting, augmenting or boosting a cellular and/or humoral immune response and of being recognized by the products of that response (T cells, antibodies). An immunogen can be a peptide, or a combination of two or more same or different peptides, that includes at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10, at least 11, at least 12, or at least 13 amino acids in a linear or spatial conformation.

[0079] An immunogen may be effective when given alone or in combination, or linked to, or fused to, another substance (which can be administered at one time or over several intervals). An immunogenic agent or immunogen may include an antigenic peptide or polypeptide that is linked to a carrier as described herein.

[0080] A nucleic acid such as DNA or RNA that encodes an antigenic peptide, or polypeptide is referred to as a “DNA [or RNA] immunogen,” as the encoded peptide or polypeptide is expressed in vivo after administration of the DNA or RNA. The peptide or polypeptide can be recombinantly expressed from a vaccine vector, which can be naked DNA or RNA that comprises the peptide or polypeptide coding sequence operably linked to a promoter, e.g., an expression vector or cassette as described herein.

[0081] The term “adjuvant” refers to a compound that, when administered in conjunction with an antigen, augments the immune response to the antigen, but when administered alone does not generate an immune response to the antigen. Adjuvants can augment an immune response by several mechanisms including lymphocyte recruitment, stimulation of B and/or T cells, and stimulation of macrophages. An adjuvant may be a natural compound, a modified version of or derivative of a natural compound, or a synthetic compound.

[0082] The terms “peptide” and “polypeptide” are used interchangeably herein and refer to a chain of two or more consecutive amino acids. If and when a distinction is made, context makes the meaning clear. For example, if two or more peptides described herein are joined to make a dimeric or multimeric peptide, polypeptide may be used to indicate “poly” or “more than one” peptide.

[0083] The term “pharmaceutically acceptable” means that the carrier, diluent, excipient, adjuvant, or auxiliary is compatible with the other ingredients of a pharmaceutical formulation and not substantially deleterious to the recipient thereof.

[0084] The terms “immunotherapy” or “immune response” refer to the development of a beneficial humoral (antibody mediated) and/or a cellular (mediated by antigen-specific T cells or their secretion products) response directed against an A β and/or tau peptide in a recipient. Such a response can be an active response induced by administration of immunogen (e.g. an A β and/or tau peptide). A cellular immune response is elicited by the presentation of polypeptide epitopes in association with Class I or Class II MHC molecules to activate antigen-specific CD4^{sup.}+ T helper cells and/or CD8^{sup.}+ cytotoxic T cells. The response may also involve activation of monocytes, macrophages, NK cells, basophils, dendritic cells, astrocytes, microglia cells, eosinophils or other components of innate immunity. The presence of a cell-mediated immunological response can be determined by proliferation assays (CD4^{sup.}+ T cells) or CTL (cytotoxic T lymphocyte) assays. The relative contributions of humoral and cellular responses to the protective or therapeutic effect of an immunogen can be distinguished by separately isolating antibodies and T-cells from an immunized syngeneic animal and measuring protective or therapeutic effect in a second subject.

Amyloid Beta (A β)

[0085] A β (also referred to herein as beta amyloid peptide or Abeta) peptide is about a 4-kDa internal fragment of 38-43 amino acids of APP (A β 39, A β 40, A β 41, A β 42, and A β 43). A β 40, for example, consists of residues 672-711 of APP and A β 42 consists of residues 673-713 of APP. As a result of proteolytic processing of APP by different secretase enzymes in vivo or in situ, A β is found in both a “short form”, 40 amino acids in length, and a “long form”, ranging from 42-43

amino acids in length. Epitopes or antigenic determinants, as described herein, are located within the N-terminus of the A β peptide and include residues within amino acids 1-10 and 12-25 of A β , for example from residues 1-3, 1-4, 1-5, 1-6, 1-7, or 3-7, 2-4, 2-5, 2-6, 2-7, or 2-8 of A β , residues 3-5, 3-6, 3-7, 3-8, or 3-9 of A β , or residues 4-7, 4-8, 4-9, or 4-10 residues 12-24, 12-23, 12-22, 13-25, 13-24, 13-23, 13-22, 14-25, 14-24, 14-23, 14-22, 15-25, 15-24, 15-23, or 15-22 of A β . For example, from residues 12-17, 12-18, 12-19, 12-20, 12-21, 13-17, 13-18, 13-19, 13-20, 13-21, 13-22, 14-17, 14-18, 14-19, 14-20, 14-21, 14-22, 14-23, 15-17, 15-18, 15-19, 15-20, 15-21, 15-22, 15-23, or 15-24 of A β ₄₂. Additional examples of epitopes or antigenic determinants include residues 16-18, 16-19, 16-20, 16-21, 16-22, 16-23, 16-24, 16-25, 17-19, 17-20, 17-21, 17-22, 17-23, 17-24 or 17-25 of A β ₄₂. Other examples of epitopes or antigenic determinants include residues 18-20, 18-21, 18-22, 18-23, 18-24, 18-25, 19-21, 19-22, 19-23, 19-24, 19-25, 20-22, 20-23, 20-24, 20-25, 21-23, 21-24 or 21-25 of A β ₄₂.

[0086] A β (Abeta) is the principal component of characteristic plaques of Alzheimer's disease. A β is generated by processing of a larger protein APP by two enzymes, termed beta and gamma secretases. Known mutations in APP associated with Alzheimer's disease occur proximate to the site of beta or gamma secretase, or within A β . Part of the hydrophobic transmembrane domain of APP is found at the carboxy end of A β , and may account for the ability of A β to aggregate into plaques, particularly in the case of the long form. Accumulation of amyloid plaques in the brain eventually leads to neuronal cell death. The physical symptoms associated with this type of neural deterioration characterize Alzheimer's disease.

Tau

[0087] Tau is a protein with a molecular weight of about 50,000 that is normally present in nerve axons or the like, and contributes to microtubular stability. The tau proteins (or t proteins) are a group of six highly-soluble protein isoforms produced by alternative splicing from the gene MAPT (microtubule-associated protein tau). They have roles primarily in maintaining the stability of microtubules in axons and are abundant in the neurons of the central nervous system (CNS). They are less common elsewhere but are also expressed at very low levels in CNS astrocytes and oligodendrocytes. Pathologies and dementias of the nervous system such as Alzheimer's disease and Parkinson's disease are associated with tau proteins that have become hyperphosphorylated insoluble aggregates called neurofibrillary tangles. Pathogenic tau species causes toxic effects through direct binding to cells and/or accumulation inside cells and/or initiation of misfolding processes (seeding) and is can be propagated from one cell to another via cell-to-cell transmission. Toxicity could also happen by neurofibrillary tangles (NFTs), which leads to cell death and cognitive decline. Other tauopathies include, for example, progressive supranuclear palsy, corticobasal syndrome, some frontotemporal dementias, and chronic traumatic encephalopathy.

A β /Tau Polypeptides of an Immunogen

[0088] An agent used for active immunization can induce in a patient an immune response and can serve as an immunotherapy. Agents used for active immunization can be, for example, the same types of immunogens used for generating monoclonal antibodies in laboratory animals, and may include 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, or 13 or more contiguous amino acids from a region of A β and/or tau peptide. In each of the embodiments of the peptides described herein, the peptides may comprise, consist, or consist essentially of the recited sequences.

[0089] In some embodiments of the disclosure, an A β /tau immunogen can include an A β peptide comprising 3-10 amino acids from residues 1-10 or 12-25 of the N-terminal sequence of A β (SEQ ID NO:01) linked to a tau peptide comprising 3-10 amino acids from residues 244-400 of the long form of tau (SEQ ID NO:02). For example, the tau peptide may comprise 3-13 amino acids from the microtubule binding region of tau (residues 344-372 of SEQ ID NO:02).

[0090] In some embodiments of the disclosure, the A β peptide can include 3-10 amino acids from residues 1-10 or 12-25 of DAEFRHDSGYEVHHQKLVFFAEDVGSNKGAIIGLMVGGVVIA (SEQ ID NO:01). For example, the A β peptide is selected from the following:

TABLE-US-00014 (SEQ ID NO: 03) DAEFRHDSGY, (SEQ ID NO: 04) DAEFRHDSG, (SEQ ID NO: 05) DAEFRHDS, (SEQ ID NO: 06) DAEFRHD, (SEQ ID NO: 07) DAEFRH, (SEQ ID NO: 08) DAEFR, (SEQ ID NO: 09) DAEF, DAE, (SEQ ID NO: 11) AEFRHDSGY, (SEQ ID NO: 12) AEFRHDSG, (SEQ ID NO: 13) AEFRHDS, (SEQ ID NO: 14) AEFRHD, (SEQ ID NO: 15) AEFRH, (SEQ ID NO: 16) AEFR, AEF, (SEQ ID NO: 18) EFRHDSGY, (SEQ ID NO: 19) EFRHDSG, (SEQ ID NO: 20) EFRHDS, (SEQ ID NO: 21) EFRHD, (SEQ ID NO: 22) EFRH, EFR, (SEQ ID NO: 24) FRHDSGY, (SEQ ID NO: 25) FRHDSG, (SEQ ID NO: 26) FRHDS, (SEQ ID NO: 27) FRHD, FRH, (SEQ ID NO: 29) RHDSGY, (SEQ ID NO: 30) RHDSG, (SEQ ID NO: 31) RHDS, RHD, (SEQ ID NO: 33) HDSGY, (SEQ ID NO: 34) HDSG, HDS, (SEQ ID NO: 36) DSGY, DSG, SGY, (SEQ ID NO: 1002) VHHQKLVFFA, (SEQ ID NO: 1003) VHHQKLVFF, (SEQ ID NO: 1004) VHHQKLVF, (SEQ ID NO: 1005) VHHQKLV, (SEQ ID NO: 1006) VHHQKL, (SEQ ID NO: 1007) HHQKLVFFAE, (SEQ ID NO: 1008) HHQKLVFFA, (SEQ ID NO: 1009) HHQKLVFF, (SEQ ID NO: 1010) HHQKLVF, (SEQ ID NO: 1011) HHQKLV, (SEQ ID NO: 1012) HHQKL, (SEQ ID NO: 1013) HQKLVFFAED, (SEQ ID NO: 1014) HQKLVFFAE, (SEQ ID NO: 1015) HQKLVFFA, (SEQ ID NO: 1016) HQKLVFF, (SEQ ID NO: 1017) HQKLVF, (SEQ ID NO: 1018) HQKLV, (SEQ ID NO: 1019) HQKL, (SEQ ID NO: 1020) QKLVFFAEDV, (SEQ ID NO: 1021) QKLVFFAED, (SEQ ID NO: 1022) QKLVFFAE, (SEQ ID NO: 1023) QKLVFFA, (SEQ ID NO: 1024) QKLVFF, (SEQ ID NO: 1025) QKLVF, (SEQ ID NO: 1026) QKLV, QKL, (SEQ ID NO: 1028) KLVFFAEDVG, (SEQ ID NO: 1029) KLVFFAEDV, (SEQ ID NO: 1030) KLVFFAED, (SEQ ID NO: 1031) KLVFFAE, (SEQ ID NO: 1032) KLVFFA, (SEQ ID NO: 1033) KLVFF, (SEQ ID NO: 1034) KLVF, KLV, (SEQ ID NO: 1036) LVFFAEDVG, (SEQ ID NO: 1037) LVFFAEDV, (SEQ ID NO: 1038) LVFFAED, (SEQ ID NO: 1039) LVFFAE, (SEQ ID NO: 1040) LVFFA, (SEQ ID NO: 1041) LVFF, LVF, (SEQ ID NO: 1043) VFFAEDVG, (SEQ ID NO: 1044) VFFAEDV, (SEQ ID NO: 1045) VFFAED, (SEQ ID NO: 1046) VFFAE, (SEQ ID NO: 1047) VFFA, VFF, (SEQ ID NO: 1049) FFAEDVG, (SEQ ID NO: 1050) FFAEDV, (SEQ ID NO: 1051) FFAED, (SEQ ID NO: 1052) FFAE, FFA, (SEQ ID NO: 1054) FAEDVG, (SEQ ID NO: 1055) FAEDV, (SEQ ID NO: 1056) FAED, and FAE.

[0091] In certain embodiments, the A β peptide is DAEFRHD (SEQ ID NO:06), DAEFR (SEQ ID NO:08) or EFRHD (SEQ ID NO:21).

[0092] The tau peptide can correspond to a peptide comprising 3-13 amino acids from residues 244-400 of SEQ ID NO:02. In some embodiments, the fragment is unphosphorylated. In some embodiments, the fragment is phosphorylated. In some embodiments, the tau peptide comprises an amino acid sequence represented by the consensus motif (Q/E)IVYK(S/P) (SEQ ID NO: 996). In some embodiments, the tau peptide comprises an amino acid sequence represented by the consensus motif KXXSXXNX(K/H)H (SEQ ID NO: 995) where X is any amino acid. In some embodiments, the tau peptide is selected from SEQ ID NOS: 146-996. In some embodiments, the tau peptide is selected from the following:

TABLE-US-00015 (SEQ ID NO: 39) QIVYKPV, (SEQ ID NO: 40) QIVYKP, (SEQ ID NO: 41) QIVYKSV, (SEQ ID NO: 42) EIVYKSV, (SEQ ID NO: 83) QIVYKS, (SEQ ID NO: 43) EIVYKSP, (SEQ ID NO: 84) EIVYKS, (SEQ ID NO: 44) EIVYKPV, (SEQ ID NO: 85) EIVYKP, (SEQ ID NO: 45) IVYKSPV, (SEQ ID NO: 46) IVYK, (SEQ ID NO: 86) CNIKHVPG, (SEQ ID NO: 47) CNIKHVP, (SEQ ID NO: 48) NIKHVP, (SEQ ID NO: 49) HVPGGG, (SEQ ID NO: 50) HVPGG, (SEQ ID NO: 51) HKPGGG, (SEQ ID NO: 52) HKPGG, (SEQ ID NO: 53) KHVPGGG, (SEQ ID NO: 54) KHVPGG, (SEQ ID NO: 55)

HQPGGG, (SEQ ID NO: 56) HQPGG (SEQ ID NO: 801) VKSKIGST, (SEQ ID NO: 817) SKIGSTEN, (SEQ ID NO: 695) TENLKHQP, (SEQ ID NO: 689) ENLKHQPG, (SEQ ID NO: 985) SKIGSTDNIKH, (SEQ ID NO: 986) SKIGSKDNIKH, and (SEQ ID NO: 988) SKIGSLDNIKH.

In each of these embodiments, the peptide may comprise, consist, or consist essentially of the recited sequences.

[0093] In some embodiments, the A β and tau peptides are linked to form a dual A β /Tau polypeptide. The A β and tau peptides can be linked by an intra-peptide linker. For example, a polypeptide linker located between the C-terminal of the first peptide and the N terminal of the second peptide. With or without the intra-peptide linker, the A β peptide and the tau peptide may be positioned in a dual A β /tau polypeptide in any order. For example, the A β peptide may be positioned at the N-terminal portion of the dual polypeptide and the tau peptide may be positioned at the C-terminal portion of the dual polypeptide. Or, the tau peptide may be positioned at the N-terminal portion of the dual polypeptide and the A β peptide may be positioned at the C-terminal portion of the dual polypeptide side of the tau peptide. Reference to a first peptide or a second peptide herein is not intended to suggest an order of the A β or tau peptides in the polypeptide of the immunogens.

[0094] In addition, the C-terminal portion of the A β peptide, the tau peptide, or the dual A β -tau polypeptide can include a linker for conjugating the peptides or the polypeptide to a carrier. Linkers that couple the peptides or dual polypeptide to the carrier may include, for example, GG, GGG, KK, KKK, AA, AAA, SS, SSS, GAGA (SEQ ID NO:80), AGAG (SEQ ID NO: 81), KGKG (SEQ ID NO:82), and the like between the peptides or dual polypeptide and the carrier and may further include a C-terminal or a N-terminal cysteine to provide a short peptide linker (e.g., G-G-C-, K-K-C-, A-A-C-, or S-S-C-). In some embodiments, where the C-terminal residues in the immunogen are either IVYKPV (SEQ ID NO:194), VYKPV (SEQ ID NO:195), YKPV (SEQ ID NO:196), KPV, or PV the linker is an amino acid linker that does not have a N-terminal glycine (e.g., GG, GAGA (SEQ ID NO:80)). In some embodiments, the linker comprises an amino acid sequence any one of AA, AAA, KK, KKK, SS, SSS, AGAG (SEQ ID NO: 81), GG, GGG, GAGA (SEQ ID NO:80), and KGKG (SEQ ID NO:82). In some embodiments, any of the A β peptide, the tau peptide, and the dual A β /tau polypeptide may include a C-terminal cysteine without the spacer. In some embodiments, any of the A β peptide, the tau peptide, and the dual A β /tau polypeptide may include a N-terminal cysteine without the spacer.

[0095] When the A β and tau polypeptides are linked to form a dual A β /tau polypeptide, the linker may be a cleavable linker. As used herein, the term “cleavable linker” refers to any linker between the antigenic peptides that promotes or otherwise renders the A β peptide and the tau peptide more susceptible to separation from each other by cleavage (for example, by endopeptidases, proteases, low pH or any other means that may occur within or around the antigen-presenting cell) and, thereby, processing by the antigen-presenting cell, than equivalent peptides lacking such a cleavable linker. In some compositions, the cleavable linker is a protease-sensitive dipeptide or oligopeptide cleavable linker. In certain embodiments, the cleavable linker is sensitive to cleavage by a protease of the trypsin family of proteases. In some compositions, the cleavable linker comprises an amino acid sequence selected from the group consisting of arginine-arginine (Arg-Arg), arginine-valine-arginine-arginine (Arg-Val-Arg-Arg; SEQ ID NO:69), valine-citrulline (Val-Cit), valine-arginine (Val-Arg), valine-lysine (Val-Lys), valine-alanine (Val-Ala), phenylalanine-lysine (Phe-Lys), GAGA (SEQ ID NO:80), AGAG (SEQ ID NO:81), and KGKG (SEQ ID NO:82). In some compositions, the cleavable linker is arginine-arginine (Arg-Arg).

[0096] In some embodiments of the disclosure, the dual A β /tau polypeptide comprises, consists or consists essentially of an amino acid sequence selected from DAEFRHDDRQIVYKPV (SEQ ID NO:57) or DAEFRHDDRREIVYKSV (SEQ ID NO:58), or DAEFRHDDRQIVYKPVXXC (SEQ ID NO:70), wherein XX and the C-terminal cysteine are each independently optional, or

DAEFRHDDRREIVYKSVXXC (SEQ ID NO:79), wherein XX and C are independently optional and, if present, XX can be GG, AA, KK, SS, GAGA (SEQ ID NO: 80), AGAG (SEQ ID NO:81), and KGKG (SEQ ID NO:82).

[0097] In some embodiments, the dual A β /tau polypeptide is as follows:

[first peptide]-[linker 1]-[second peptide]-[linker 2]-[Cys],

wherein, if the [first peptide] is an A β peptide then the [second peptide] is a tau peptide, and if the [first peptide] is a tau peptide, then the [second peptide] is an A β peptide, each of [linker 1], [linker 2] and [Cys] are optional, and [linker 1] and [linker 2] are the same or different linkers.

[0098] In certain embodiments, the dual A β /tau polypeptide is as follows:

[Cys]-[linker 2]-[first peptide]-[linker 1]-[second peptide]

wherein, if the [first peptide] is an A β peptide then the [second peptide] is a tau peptide, and if the [first peptide] is a tau peptide, then the [second peptide] is an A β peptide, and each of [linker 1], [linker 2] and [Cys] are optional, and [linker 1] and [linker 2] are the same or different linkers.

[0099] Examples of the A β peptide include any one SEQ ID NOS 3-38 or SEQ ID NOS: 1002-1057.

[0100] Examples of the tau peptide include any one of SEQ ID NOS: 39-56, 83-86, or 146-996.

[0101] [Linker 1] is optional, and when present, may be a cleavable linker. A cleavable linker, if present, can be 1-10 amino acids in length. In some embodiments, the linker comprises between about 1-10 amino acids, about 1-9 amino acids, about 1-8 amino acids, about 1-7 amino acids, about 1-6 amino acids, about 1-5 amino acids, about 1-4 amino acids, about 1-3 amino acids, about 2 amino acids, or one (1) amino acid. In some embodiments, the cleavable linker is 1 amino acid, 2 amino acids, 3 amino acids, 4 amino acids, 5 amino acids, 6 amino acids, 7 amino acids, 8 amino acids, 9 amino acids, or 10 amino acids. In some embodiments, the linker may be a cleavable linker having an amino acid sequence selected from the group consisting of arginine-arginine (Arg-Arg), arginine-valine-arginine-arginine (Arg-Val-Arg-Arg; SEQ ID NO: 69), valine-citrulline (Val-Cit), valine-arginine (Val-Arg), valine-lysine (Val-Lys), valine-alanine (Val-Ala), phenylalanine-lysine (Phe-Lys), glycine-alanine-glycine-alanine (Gly-Ala-Gly-Ala; SEQ ID NO:80), alanine-glycine-alanine-glycine (Gly-Ala-Gly-Ala; SEQ ID NO:81), and lysine-glycine-lysine-glycine (Lys-Gly-Lys-Gly; SEQ ID NO:82).

[0102] [Linker 2] is optional, and when present is a linker that couples the polypeptide to a carrier. A linker, if present, can be 1-10 amino acids in length. In some embodiments, the linker comprises between about 1-10 amino acids, about 1-9 amino acids, about 1-8 amino acids, about 1-7 amino acids, about 1-6 amino acids, about 1-5 amino acids, about 1-4 amino acids, about 1-3 amino acids, about 2 amino acids, or one (1) amino acid. In some embodiments, the linker is 1 amino acid, 2 amino acids, 3 amino acids, 4 amino acids, 5 amino acids, 6 amino acids, 7 amino acids, 8 amino acids, 9 amino acids, or 10 amino acids. In some embodiments, the amino acid composition of a linker can mimic the composition of linkers found in natural multidomain proteins, where certain amino acids are overrepresented, underrepresented or equi-represented in natural linkers as compared to their abundance in whole protein. For example, threonine (Thr), serine (Ser), proline (Pro), glycine (Gly), aspartic acid (Asp), lysine (Lys), glutamine (Gln), asparagine (Asn), arginine (Arg), phenylalanine (Phe), glutamic acid (Glu) and alanine (Ala) are overrepresented in natural linkers. In contrast, isoleucine (Ile), tyrosine (Tyr), tryptophan (Trp), and cysteine (Cys) are underrepresented. In general, overrepresented amino acids were polar uncharged or charged residues, which constitute approximately 50% of naturally encoded amino acids, and Pro, Thr, and Gln were the most preferable amino acids for natural linkers. In some embodiments, the amino acid composition of a linker can mimic the composition of linkers commonly found in recombinant proteins, which can generally be classified as flexible or rigid linkers. For example, flexible linkers found in recombinant proteins are generally composed of small, non-polar (e.g. Gly) or polar (e.g.

Ser or Thr) amino acids whose small size provides flexibility and allows for mobility of the connecting functional domains. The incorporation of, e.g., Ser or Thr can maintain the stability of the linker in aqueous solutions by forming hydrogen bonds with the water molecules, and therefore can reduce interactions between the linker and the immunogens. In some embodiments, a linker comprises stretches of Gly and Ser residues ("GS" linker). An example of a widely used flexible linker is (Gly-Gly-Ser)_n, (Gly-Gly-Gly-Ser)_n (SEQ ID NO:1062) or (Gly-Gly-Gly-Gly-Ser)_n (SEQ ID NO: 1063), where n=1-3. Adjusting the copy number "n" can optimize a linker to achieve sufficient separation of the functional immunogen domains to, e.g., maximize an immunogenic response. Many other flexible linkers have been designed for recombinant fusion proteins that can be used herein. In some embodiments, linkers can be rich in small or polar amino acids such as Gly and Ser but also contain additional amino acids such as Thr and Ala to maintain flexibility, as well as polar amino acids such as Lys and Glu to improve solubility. See, e.g., Chen, X. et al., "Fusion Protein Linkers: Property, Design and Functionality" *Adv Drug Deliv Rev.*, 15; 65(10): 1357-1369 (2003). In certain embodiments, when present, the linker can be an amino acid sequence selected from the group consisting of as GG, GGG, KK, KKK, AA, AAA, SS, SSS, G-A-G-A (SEQ ID NO:80), A-G-A-G (SEQ ID NO:81), and K-G-K-G (SEQ ID NO: 82).

[0103] [Cys] is optional and can be helpful to conjugate the polypeptide to a carrier. When present, the Cys can be at the C-terminal portion of the polypeptide, or at the N-terminal portion of the polypeptide.

[0104] Examples of the [first peptide]-[linker 1]-[second peptide]-[linker 2]-[Cys] dual Aβ/tau polypeptide of the disclosure include the following:

TABLE-US-00016 TABLE 1 Abeta Endo tau Immunogen Immuno- SEQ pepti- SEQ SEQ gen																	
Abeta ID	dase Tau ID	C-Terminal ID	Lab ID	Sequence	NO. linker	Sequence	NO. linker	Cys	NO								
9	DAEFRHD	06	RR	QIVYKPV	39	GG	C	59	10	DAEFRHD	06	RR	QIVYKPV	39	None	C	60
18	DAEFRHD	06	RR	QIVYKPV	39	AA	C	61	19	DAEFRHD	06	RR	QIVYKPV	39	KK	C	62
	DAEFRHD	06	RR	EIVYKSP	43	GG	C	63	21	DAEFRHD	06	RR	EIVYKSP	43	AA	C	64
	DAEFRHD	06	RR	EIVYKSP	43	None	C	65	23	DAEFRHD	06	RR	EIVYKSP	43	KK	C	66
	DAEFRHD	06	RR	NIKHVP	48	GG	C	78									

Polypeptide Immunogens

[0105] The Aβ peptide, the tau peptide and, the dual Aβ/tau polypeptide are immunogens in accordance with the disclosure. In some embodiments, the peptides and the dual Aβ-tau polypeptide can be linked to a suitable carrier to help elicit an immune response. Accordingly, one or more the peptides and dual Aβ-tau polypeptides of the disclosure can be linked to a carrier. For example, each of the Aβ peptide, tau peptide and the Aβ-tau polypeptide may be linked to the carrier with or without spacer amino acids (e.g., Gly-Gly, Gly-Gly-Gly, Ala-Ala, Ala-Ala-Ala, Lys-Lys, Lys-Lys-Lys, Ser-Ser, Ser-Ser-Ser, Gly-Ala-Gly-Ala (SEQ ID NO:80), Ala-Gly-Ala-Gly (SEQ ID NO:81), or Lys-Gly-Lys-Gly (SEQ ID NO:82)). In certain embodiments, the dual Aβ-tau polypeptide can be linked to a suitable carrier using a C-terminal cysteine to provide a linker between the peptide(s) and the carrier or the dual Aβ/tau polypeptide and the carrier. In certain embodiments, the dual Aβ-tau polypeptide can be linked to a suitable carrier using an N-terminal cysteine to provide a linker between the peptide(s) and the carrier. In some embodiments, where the C-terminal residues in the immunogen are either IVYKPV (SEQ ID NO: 194), VYKPV (SEQ ID NO:195), YKPV (SEQ ID NO:196), KPV, or PV the linker is an amino acid linker that does not have a N-terminal glycine (e.g., GG, GAGA (SEQ ID NO:80)).

[0106] Suitable carriers include, but are not limited to serum albumins, keyhole limpet hemocyanin, immunoglobulin molecules, thyroglobulin, ovalbumin, tetanus toxoid, or a toxoid from other pathogenic bacteria, such as diphtheria (e.g., CRM197), *E. coli*, cholera, or *H. pylori*, or an attenuated toxin derivative. T cell epitopes are also suitable carrier molecules. Some conjugates can be formed by linking peptide immunogens of the invention to an immunostimulatory polymer molecule (e.g., tripalmitoyl-S-glycerine cysteine (Pam3Cys), mannan (a mannose polymer), or

glucan (a β 1-2 polymer)), cytokines (e.g., IL-1, IL-1 alpha and β peptides, IL-2, γ -INF, IL-10, GM-CSF), and chemokines (e.g., MIP1- α and β , and RANTES). Additional carriers include virus-like particles. In some compositions, immunogenic peptides can also be linked to carriers by chemical crosslinking. Techniques for linking an immunogen to a carrier include the formation of disulfide linkages using N-succinimidyl 3-(2-pyridylthio) propionate (SPDP), and succinimidyl 4-(N-maleimidomethyl)cyclohexane-1-carboxylate (SMCC) (if the peptide lacks a sulfhydryl group, this can be provided by addition of a cysteine residue). These reagents create a disulfide linkage between themselves and peptide cysteine residues on one protein and an amide linkage through the epsilon-amino on a lysine, or other free amino group in other amino acids. In some embodiments, chemical crosslinking can comprise use of SBAP (succinimidyl 3-(bromoacetamido) propionate), which is a short (6.2 angstrom) cross-linker for amine-to-sulfhydryl conjugation via N-hydroxysuccinimide (NHS) ester and bromoacetyl reactive groups. A variety of such disulfide/amide-forming agents are described by Jansen et al., "Immunotoxins: Hybrid Molecules Combining High Specificity and Potent Cytotoxicity" *Immunological Reviews* 62:185-216 (February 1982). Other bifunctional coupling agents form a thioether rather than a disulfide linkage. Many of these thio-ether-forming agents are commercially available and include reactive esters of 6-maleimidocaproic acid, 2-bromoacetic acid, and 2-iodoacetic acid, 4-(N-maleimidomethyl) cyclohexane-1-carboxylic acid. The carboxyl groups can be activated by combining them with succinimide or 1-hydroxyl-2-nitro-4-sulfonic acid, sodium salt. Virus-like particles (VLPs), also called pseudovirions or virus-derived particles, represent subunit structures composed of multiple copies of a viral capsid and/or envelope protein capable of self-assembly into VLPs of defined spherical symmetry in vivo. (Powilleit, et al., (2007) *PLOS ONE* 2(5):e415.) Alternatively, peptide immunogens can be linked to at least one artificial T-cell epitope capable of binding a large proportion of MHC Class II molecules, such as the pan DR epitope ("PADRE"). Pan DR-binding peptides (PADRE) are described in U.S. Pat. No. 5,736,142, WO 95/07707, and Alexander, et al, *Immunity*, 1:751-761 (1994).

[0107] Active immunogens can be presented in multimeric form in which multiple copies of an immunogen (peptide or polypeptide) are presented on a carrier as a single covalent molecule. In some embodiments, the carrier includes various forms of the dual A β /tau polypeptide. For instance, the dual A β /tau polypeptide of the immunogen can include polypeptides that have the A β antigen and the tau antigen in different orders, or may be present with or without an intrapeptide linker and/or a linker to a carrier.

[0108] In some compositions, the immunogenic peptides can also be expressed as fusion proteins with carriers. In certain compositions, the immunogenic peptides can be linked at the amino terminus, the carboxyl terminus, or internally to the carrier. In some compositions, the carrier is CRM197. In some compositions, the carrier is diphtheria toxoid.

Nucleic Acids

[0109] The disclosure further provides nucleic acids encoding any of the amyloid-beta (A β) peptides and the tau peptides as disclosed herein. The nucleic acid immunotherapy compositions, as disclosed herein, comprise, consist of, or consist essentially of, a first nucleic acid sequence encoding an amyloid-beta (A β) peptide, and a second nucleic acid sequence encoding a tau peptide as disclosed herein. For example, the A β peptide is a sequence that is 3-10 amino acid residues in length and from the first ten N-terminal residues of SEQ ID NO:01, and the tau peptide is a sequence that is 3-13 amino acids in length and from residues 244-400 of SEQ ID NO:02.

Accordingly, a nucleic acid encoding any of SEQ ID NOS: 3-38 or SEQ ID NOS: 1002-1057 may be combined with a nucleic acid encoding any of SEQ ID NOS: 39-56, 83-86, or 146-996 to provide an immunogen and a component of pharmaceutical composition of the disclosure.

Likewise, one or more nucleic acids encoding any of A β and tau sequences may include the codons for an RR- N-terminal or -RR C-terminal dipeptide. In certain embodiments, the A β and tau peptide sequences may be encoded by the same nucleic acid sequence or by separate nucleic acid

sequences. In some embodiments, the nucleic acid sequences may also encode a linker to a carrier and/or a C-terminal cysteine as described herein. In addition, when a single nucleic acid sequence encodes both peptides, the sequence may also encode an intra-peptide linker as described herein. The nucleic acid compositions described herein (pharmaceutical compositions) can be used in methods for treating or effecting prophylaxis and/or prevention of Alzheimer's disease. In another embodiment, the nucleic acid immunotherapy compositions as disclosed herein provide compositions for reducing pathogenic forms of A β and/or tau in the subject and/or in the tissue of the subject. In some embodiments, the A β and/or tau reduced by the immunotherapy compositions is the pathological form(s) of the A β (e.g. extracellular plaque deposits of the β -amyloid peptide (A β); neuritic amyloid plaques), and/or tau (e.g. flame-shaped neurofibrillary tangles of tau; neurofibrillary tangles of tau). In yet other embodiment, pathological indicators of neurodegenerative disease are decreased by the nucleic acid immunotherapy compositions. In another embodiment, the nucleic acid immunotherapy compositions as disclosed herein provide compositions for reducing brain A β and brain tau.

[0110] A nucleic acid such as DNA that encodes an immunogen and is used as a vaccine can be referred to as a “DNA immunogen” or “DNA vaccine” as the encoded polypeptides are expressed in vivo after administration of the DNA. DNA vaccines are intended to induce antibodies against the proteins of interest they encode in a subject by: integrating DNA encoding the proteins of interest into a vector (a plasmid or virus); administering the vector to the subject; and expressing the proteins of interest in the subject in which the vector has been administered to stimulate the immune system of the subject. A DNA vaccine remains in the body of the subject for a long time after the administration, and continues to slowly produce the encoded proteins. Thus, excessive immune responses can be avoided. DNA vaccines can also be modified using a genetic engineering techniques. Optionally, such nucleic acids further encode a signal peptide and can be expressed with the signal peptide linked to peptide. Coding sequences of nucleic acids can be operably linked with regulatory sequences to ensure expression of the coding sequences, such as a promoter, enhancer, ribosome binding site, transcription termination signal, and the like. The nucleic acids encoding A β and tau can occur in isolated form or can be cloned into one or more vectors. The nucleic acids can be synthesized by, for example, solid state synthesis or PCR of overlapping oligonucleotides. Nucleic acids encoding A β and tau peptides and polypeptides with and without linkers or cleavable linkers and with or without protein based carriers can be joined as one contiguous nucleic acid, e.g., within an expression vector.

[0111] DNA is more stable than RNA, but involves some potential safety risks such as induction of anti-DNA antibodies, thus in some embodiments, the nucleic acid can be RNA. RNA nucleic acid that encodes an immunogen and is used as a vaccine can be referred to as a “RNA immunogen” or “RNA vaccine” or “mRNA vaccine” as the encoded polypeptides are expressed in vivo after administration of the RNA. Ribonucleic acid (RNA) vaccines can safely direct a subject's cellular machinery to produce one or more polypeptide(s) of interest. In some embodiments, a RNA vaccine can be a non-replicating mRNA (messenger-RNA) or a virally derived, self-amplifying RNA. mRNA-based vaccines encode the antigens of interest and contain 5' and 3' untranslated regions (UTRs), whereas self-amplifying RNAs encode not only the antigens, but also the viral replication machinery that enables intracellular RNA amplification and abundant protein expression. In vitro transcribed mRNA can be produced from a linear DNA template using a T7, a T3 or an Sp6 phage RNA polymerase. The resulting product can contain an open reading frame that encodes the peptides of interest as disclosed herein, flanking 5'- and 3'-UTR sequences, a 5' cap and a poly(A) tail. In some embodiments, a RNA vaccine can comprise trans-amplifying RNA (for example, see Beissert et al., *Molecular Therapy* January 2020 28(1):119-128). In certain embodiments, RNA vaccines encode an A β peptide and a tau peptide as disclosed herein, and are capable of expressing the A β and a tau peptides, in particular if transferred into a cell such as an immature antigen presenting cell. RNA may also contain sequences which encode other

polypeptide sequences such as immune stimulating elements. In some embodiments, the RNA of a RNA vaccine can be modified RNA. The term “modified” in the context of the RNA can include any modification of RNA which is not naturally present in RNA. For example, modified RNA can refer to RNA with a 5'-cap; however, RNA may comprise further modifications. A 5'-cap can be modified to possess the ability to stabilize RNA when attached thereto. In certain embodiments, a further modification may be an extension or truncation of the naturally occurring poly(A) tail or an alteration of the 5'- or 3'-untranslated regions (UTR). In some embodiments, the RNA e.g. or mRNA vaccine is formulated in an effective amount to produce an antigen specific immune response in a subject. For example, the RNA vaccine formulation is administered to a subject in order to stimulate the humoral and/or cellular immune system of the subject against the A β and tau antigens, and thus may further comprise one or more adjuvant(s), diluents, carriers, and/or excipients, and is applied to the subject in any suitable route in order to elicit a protective and/or therapeutic immune reaction against the A β and tau antigens.

[0112] Basic texts disclosing general methods of molecular biology, all of which are incorporated by reference, include: Sambrook, J et al., *Molecular Cloning: A Laboratory Manual*, 2nd Edition, Cold Spring Harbor Press, Cold Spring Harbor, N.Y., 1989; Ausubel, F M et al. *Current Protocols in Molecular Biology*, Vol. 2, Wiley-Interscience, New York, (current edition); Kriegler, *Gene Transfer and Expression: A Laboratory Manual* (1990); Glover, D M, ed, *DNA Cloning: A Practical Approach*, vol. I & II, IRL Press, 1985; Albers, B. et al., *Molecular Biology of the Cell*, 2nd Ed., Garland Publishing, Inc., New York, N.Y. (1989); Watson, J D et al., *Recombinant DNA*, 2nd Ed., Scientific American Books, New York, 1992; and Old, R W et al., *Principles of Gene Manipulation: An Introduction to Genetic Engineering*, 2nd Ed., University of California Press, Berkeley, Calif. (1981).

[0113] Techniques for the manipulation of nucleic acids, such as, e.g., generating mutations in sequences, sub-cloning, labeling probes, sequencing, hybridization and the like are well described in the scientific and patent literature. See, e.g., Sambrook, ed., *MOLECULAR CLONING: A LABORATORY MANUAL* (2ND ED.), Vols. 1-3, Cold Spring Harbor Laboratory, (1989); *CURRENT PROTOCOLS IN MOLECULAR BIOLOGY*, Ausubel, ed. John Wiley & Sons, Inc., New York (1997); *LABORATORY TECHNIQUES IN BIOCHEMISTRY AND MOLECULAR BIOLOGY: HYBRIDIZATION WITH NUCLEIC ACID PROBES*, Part I. Tijssen, ed. Elsevier, N.Y. (1993).

[0114] Nucleic acids, vectors, capsids, polypeptides, and the like can be analyzed and quantified by any of a number of general means well known to those of skill in the art. These include, e.g., analytical biochemical methods such as NMR, spectrophotometry, radiography, electrophoresis, capillary electrophoresis, high performance liquid chromatography (HPLC), thin layer chromatography (TLC), and hyperdiffusion chromatography, various immunological methods, e.g. fluid or gel precipitin reactions, immunodiffusion, immuno-electrophoresis, radioimmunoassays (RIAs), enzyme-linked immunosorbent assays (ELISAs), immunofluorescence assays, Southern analysis, Northern analysis, dot-blot analysis, gel electrophoresis (e.g., SDS-PAGE), RT-PCR, quantitative PCR, other nucleic acid or target or signal amplification methods, radiolabeling, scintillation counting, and affinity chromatography.

Pharmaceutical Compositions

[0115] Each of the peptides and immunogens described herein can be presented in a pharmaceutical composition that is administered with pharmaceutically acceptable adjuvants and pharmaceutically acceptable excipients. The adjuvant increases the titer of induced antibodies and/or the binding affinity of induced antibodies relative to the situation if the peptide were used alone. A variety of adjuvants can be used in combination with an immunogen of the disclosure to elicit an immune response. Some adjuvants augment the intrinsic response to an immunogen without causing conformational changes in the immunogen that affect the qualitative form of the response. An adjuvant may be a natural compound, a modified version of or derivative of a natural

compound, or a synthetic compound.

[0116] Some adjuvants include aluminum salts, such as aluminum hydroxide and aluminum phosphate, 3 De-O-acylated monophosphoryl lipid A (MPL™) (see GB 2220211 (RIBI ImmunoChem Research Inc., Hamilton, Montana, now part of Corixa). As used herein, MPL refers to natural and synthetic versions of MPL. Examples of synthetic versions include PHAD®, 3D-PHAD® and 3D(6A)-PHAD® (Avanti Polar Lipids, Alabaster, Alabama).

[0117] QS-21 is a triterpene glycoside or saponin isolated from the bark of the *Quillaja saponaria* Molina tree found in South America (see Kensil et al., in *Vaccine Design: The Subunit and Adjuvant Approach* (eds. Powell & Newman, Plenum Press, NY, 1995)) QS-21 products include Stimulon® (Antigenics, Inc., New York, NY; now Agenus, Inc. Lexington, MA) and QS-21 Vaccine Adjuvant (Desert King, San Diego, CA). QS-21 has been disclosed, characterized, and evaluated in U.S. Pat. Nos. 5,057,540, and 8,034,348, the disclosures of which are herein incorporated by reference. Additionally, QS-21 has been evaluated in numerous clinical trials in various dosages. See, NCT00960531 (clinicaltrials.gov/ct2/show/study/NCT00960531), Hüll et al., *Curr Alzheimer Res.* 2017 July; 14(7): 696-708 (evaluated 50 mcg of QS-21 in with various doses of vaccine ACC-001); Gilman et al., “Clinical effects of Abeta immunization (AN1792) in patients with AD in an interrupted trial” *Neurology*. 2005 May 10; 64(9):1553-62; Wald et al., “Safety and immunogenicity of long HSV-2 peptides complexed with rhHsc70 in HSV-2 seropositive persons” *Vaccine* 2011; 29(47):8520-8529; and Cunningham et al., “Efficacy of the Herpes Zoster Subunit Vaccine in Adults 70 Years of Age or Older.” *NEJM*. 2016 Sep. 15; 375(11):1019-32. QS-21 is used in FDA approved vaccines including SHINGRIX. SHINGRIX contains 50 mcg of QS-21. In certain embodiments, the amount of QS-21 is from about 10 µg to about 500 µg.

[0118] TQL1055 is an analogue of QS-21 (Adjuvance Technologies, Lincoln, NE). The semi-synthetic TQL1055 has been characterized in comparison to QS-21 as having high purity, increased stability, decreased local tolerability, decreased systemic tolerability. TQL1055 has been disclosed, characterized, and evaluated in US20180327436 A1, WO2018191598 A1, WO2018200656 A1, and WO2019079160 A1, the disclosures of which are herein incorporated by reference.

US20180327436 A1 teaches that 2.5 fold more TQ1055 was superior to 20 µg QS-21 but there was not an improvement over 50 µg TQ1055. However, unlike QS-21 there was no increase in either weight loss or hemolysis of RBC as the TQL1055 dose increased. WO2018200656 A1 teaches that with an optimal amount of TQ1055, one can lower the amount of antigen and achieve superior titers. In certain embodiments, the amount of TQL1055 is from about 10 µg to about 500 µg.

[0119] Other adjuvants are oil in water emulsions (such as squalene or peanut oil), optionally in combination with immune stimulants, such as monophosphoryl lipid A (see Stoute et al., *N. Engl. J. Med.* 336, 86-91 (1997)), pluronic polymers, and killed mycobacteria. Ribi adjuvants are oil-in-water emulsions. Ribi contains a metabolizable oil (squalene) emulsified with saline containing Tween 80. Ribi also contains refined mycobacterial products which act as immunostimulants and bacterial monophosphoryl lipid A. Other adjuvants can be CpG oligonucleotides (see WO 98/40100), cytokines (e.g., IL-1, IL-1 alpha and β peptides, IL-2, γ-INF, IL-10, GM-CSF), chemokines (e.g., MIP1-α and β, and RANTES), saponins, RNA, and/or TLR agonists (for example, TLR4 agonists such as MPL and synthetic MPL molecules), aminoalkyl glucosaminide phosphate and other TLR agonists. Adjuvants can be administered as a component of a therapeutic composition with an active agent or can be administered separately, before, concurrently with, or after administration of the therapeutic agent.

[0120] In various embodiments of the disclosure, the adjuvant is QS-21 (Stimulon™). In some compositions, the adjuvant is MPL. In certain embodiments, the amount of MPL is from about 10 µg to about 500 µg. In some compositions, the adjuvant is TQL1055. In certain embodiments, the amount of TQL1055 is from about 10 µg to about 500 µg. In some compositions, the adjuvant is QS21. In certain embodiments, the amount of QS21 is from about 10 µg to about 500 µg. In some compositions, the adjuvant is a combination of MPL and QS-21. In some compositions, the

adjuvant is a combination of MPL and TQL1055. In some compositions, the adjuvant can be in a liposomal formulation.

[0121] In addition, some embodiments of the disclosure can comprise a multiple antigen presenting system (MAP). Multiple antigen-presenting peptide vaccine systems have been developed to avoid the adverse effects associated with conventional vaccines (i.e., live-attenuated, killed or inactivated pathogens), carrier proteins and cytotoxic adjuvants. Two main approaches have been used to develop multiple antigen presenting peptide vaccine systems: (1) the addition of functional components, e.g., T-cell epitopes, cell-penetrating peptides, and lipophilic moieties; and (2) synthetic approaches using size-defined nanomaterials, e.g., self-assembling peptides, non-peptidic dendrimers, and gold nanoparticles, as antigen-displaying platforms. Use of a multiple antigenic peptide (MAP) system can improve the sometimes poor immunogenicity of subunit peptide vaccines. In a MAP system, multiple copies of antigenic peptides are simultaneously bound to the α - and ϵ -amino groups of a non-immunogenic Lys-based dendritic scaffold, helping to confer stability from degradation, thus enhancing molecular recognition by immune cells, and induction of stronger immune responses compared with small antigenic peptides alone. In some compositions, the MAP comprises one or more of a Lys-based dendritic scaffold, helper T-cell epitopes, immune stimulating lipophilic moieties, cell penetrating peptides, radical induced polymerization, self-assembling nanoparticles as antigen-presenting platforms and gold nanoparticles.

[0122] Pharmaceutical compositions for parenteral administration are preferably sterile and substantially isotonic and manufactured under GMP conditions. Pharmaceutical compositions can be provided in unit dosage form (i.e., the dosage for a single administration). Pharmaceutical compositions can be formulated using one or more physiologically acceptable carriers, diluents, excipients or auxiliaries. The formulation depends on the route of administration chosen. For injection, the peptides of the disclosure can be formulated in aqueous solutions, preferably in physiologically compatible buffers such as Hank's solution, Ringer's solution, or physiological saline or acetate buffer (to reduce discomfort at the site of injection). The solution can contain formulatory agents such as suspending, stabilizing and/or dispersing agents. Alternatively, peptide compositions can be in lyophilized form for constitution with a suitable vehicle, e.g., sterile pyrogen-free water, before use.

[0123] Peptides (and optionally a carrier fused to the peptide(s)) can also be administered in the form of a nucleic acid encoding the peptide(s) and expressed in situ in a subject. A nucleic acid segment encoding an immunogen is typically linked to regulatory elements, such as a promoter and enhancer that allow expression of the DNA segment in the intended target cells of a subject. For expression in blood cells, as is desirable for induction of an immune response, promoter and enhancer elements from, for example, light or heavy chain immunoglobulin genes or the CMV major intermediate early promoter and enhancer are suitable to direct expression. The linked regulatory elements and coding sequences are often cloned into a vector.

[0124] DNA and RNA can be delivered in naked form (i.e., without colloidal or encapsulating materials). Alternatively a number of viral vector systems can be used including retroviral systems (see, e.g., Boris-Lawrie and Teumin, *Cur. Opin. Genet. Develop.* 3(1):102-109 (1993)); adenoviral vectors (see, e.g., Bett et al, *J. Virol.* 67(10): 5911-21 (1993)); adeno-associated virus vectors (see, e.g., Zhou et al., *J. Exp. Med.* 179(6):1867-75 (1994)), viral vectors from the pox family including vaccinia virus and the avian pox viruses, viral vectors from the alpha virus genus such as those derived from Sindbis and Semliki Forest Viruses (see, e.g., Dubensky et al., *J. Virol.* 70(1):508-519 (1996)), Venezuelan equine encephalitis virus (see U.S. Pat. No. 5,643,576) and rhabdoviruses, such as vesicular stomatitis virus (see WO 96/34625) and papillomaviruses (WO 94/12629; Ohe et al., *Human Gene Therapy* 6(3):325-333 (1995); and Xiao & Brandsma, *Nucleic Acids. Res.* 24(13):2620-2622 (1996)).

[0125] DNA and RNA encoding an immunogen, or a vector containing the same, can be packaged into liposomes, nanoparticles or lipoproteins complexes. Suitable other polymers, include, for

example, protamine liposomes, polysaccharide particles, cationic nanoemulsion, cationic polymer, cationic polymer liposome, cationic lipid nanoparticles, cationic lipid, cholesterol nanoparticles, cationic lipid-cholesterol, PEG nanoparticle, or dendrimer nanoparticles. Additional suitable lipids and related analogs are described by U.S. Pat. Nos. 5,208,036, 5,264,618, 5,279,833, and 5,283,185, each of which are herein incorporated by reference in their entirety. Vectors and DNA encoding an immunogen can also be adsorbed to or associated with particulate carriers, examples of which include polymethyl methacrylate polymers and polylactides and poly(lactide-co-glycolides), (see, e.g., McGee et al., *J. Micro Encap.* March-April 1997; 14(2):197-210).

[0126] Pharmaceutically acceptable carrier compositions can also include additives, including, but not limited to, water, pharmaceutically acceptable organic solvents, collagen, polyvinyl alcohol, polyvinylpyrrolidone, carboxyvinyl polymers, carboxymethylcellulose sodium, sodium polyacrylate, sodium alginate, water-soluble dextran, carboxymethyl starch sodium, pectin, methylcellulose, ethylcellulose, xanthan gum, gum arabic, casein, agar, polyethylene glycol, diglycerine, glycerine, propylene glycol, petrolatum, paraffin, stearyl alcohol, stearic acid, human serum albumin, mannitol, sorbitol, lactose, and surfactants acceptable as pharmaceutical additives.

Subjects Amenable to Treatment

[0127] The presence of A β plaques and/or neurofibrillary tangles has been found in several diseases including Alzheimer's disease, Down's syndrome, mild cognitive impairment, cerebral amyloid angiopathy, primary age-related tauopathy, postencephalitic parkinsonism, posttraumatic dementia or dementia pugilistica, Pick's disease, type C Niemann-Pick disease, supranuclear palsy, frontotemporal dementia, frontotemporal lobar degeneration, argyrophilic grain disease, globular glial tauopathy, amyotrophic lateral sclerosis/parkinsonism dementia complex of Guam, corticobasal degeneration (CBD), dementia with Lewy bodies, Lewy body variant of Alzheimer's disease (LBVAD), chronic traumatic encephalopathy (CTE), globular glial tauopathy (GGT), Parkinson's disease, progressive supranuclear palsy (PSP), dry age-related macular degeneration (AMD), and inclusion-body myositis.

[0128] The compositions and methods of the disclosure can be used in treatment or prophylaxis of any of these diseases. Because of the widespread association between neurological diseases and A β and/or tau, the compositions and methods of the disclosure can be used in treatment or prophylaxis of any subject showing elevated levels of A β and/or tau (e.g., in the CSF) compared with a mean value in individuals without neurological disease. The compositions and methods of the disclosure can also be used in treatment or prophylaxis of neurological disease in individuals having a mutation in A β and/or tau associated with neurological disease. The methods are particularly suitable for treatment or prophylaxis of Alzheimer's disease.

[0129] Subjects amenable to treatment include individuals at risk of disease but not showing symptoms, as well as patients presently showing symptoms, including treatment naïve subjects that have not been previously treated for disease. Subjects at risk of disease include those in an aging population, asymptomatic subjects with A β and/or tau pathologies and having a known genetic risk of disease. Such individuals include those having relatives who have experienced this disease, and those whose risk is determined by analysis of genetic or biochemical markers. Genetic markers of risk include mutations in A β and/or tau, as well as mutations in other genes associated with neurological disease. For example, the ApoE4 allele in heterozygous and even more so in homozygous form is associated with risk of Alzheimer's disease (AD). Other markers of risk of Alzheimer's disease include mutations in the APP gene, particularly mutations at position 717 and positions 670 and 671 referred to as the Hardy and Swedish mutations respectively, mutations in the presenilin genes, PS1 and PS2, a family history of AD, hypercholesterolemia or atherosclerosis. Individuals presently suffering from Alzheimer's disease can be recognized by PET imaging, from characteristic dementia, as well as the presence of risk factors described above. In addition, a number of diagnostic tests are available for identifying individuals who have AD. These include measurement of CSF or blood tau or phospho-tau and A β 42 levels. Elevated tau or phospho-tau

and decreased A β 42 levels signify the presence of AD. Some mutations associated with Parkinson's disease, for example, Ala30Pro or Ala53Thr, or mutations in other genes associated with Parkinson's disease such as leucine-rich repeat kinase (LRRK2 or PARK8). Subjects can also be diagnosed with any of the neurological diseases mentioned above by the criteria of the DSM IV TR.

[0130] In asymptomatic subjects, treatment can begin at any age (e.g., 10, 20, 30, or more). Usually, however, it is not necessary to begin treatment until a subject reaches 20, 30, 40, 50, 60, 70, 80, or 90 years of age. Treatment typically entails multiple dosages over a period of time. Treatment can be monitored by assaying antibody levels over time. If the response falls, a booster dosage is indicated. In the case of potential Down's syndrome patients, treatment can begin antenatally by administering therapeutic agent to the mother or shortly after birth.

Methods of Treatments and Uses

[0131] The disclosure provides methods of inhibiting or reducing aggregation of Abeta and/or tau in a subject having or at risk of developing a neurodegenerative disease (e.g., Alzheimer's disease). The methods include administering to the subject the compositions as disclosed herein. A therapeutically effective amount is a dosage that, when given for an effective period of time, achieves the desired immunological or clinical effect. Dosage regimens may be adjusted to provide the optimum therapeutic response. For example, several divided doses may be administered at set intervals (e.g., weekly, monthly) or the dose may be proportionally reduced as indicated by the exigencies of the therapeutic situation.

[0132] In prophylactic applications, the compositions described herein can be administered to a subject susceptible to, or otherwise at risk of a disease (e.g., Alzheimer's disease) in a regimen (dose, frequency and route of administration) effective to reduce the risk, lessen the severity, or delay the onset of at least one sign or symptom of the disease. In particular, the regimen is effective to inhibit or delay A β plaque formation and/or inhibit or delay tau or phospho-tau and paired filaments formed from it in the brain, and/or inhibit or delay its toxic effects and/or inhibit/or delay development of behavioral deficits. In therapeutic applications, the compositions described herein are administered to a subject suspected of, or a patient already suffering from a disease (e.g., Alzheimer's disease) in a regimen (dose, frequency and route of administration) effective to ameliorate or at least inhibit further deterioration of at least one sign or symptom of the disease. In particular, the regimen is preferably effective to reduce or at least inhibit further increase of levels of A β plaques and/or tau, phosphor-tau, or paired filaments formed from it, associated toxicities and/or behavioral deficits.

[0133] A regimen is considered therapeutically or prophylactically effective if an individual treated achieves an outcome more favorable than the mean outcome in a control population of comparable subjects not treated by methods of the invention, or if a more favorable outcome is demonstrated in treated subjects versus control subjects in a controlled clinical trial (e.g., a phase II, phase II/III or phase III trial) at the $p < 0.05$ or 0.01 or even 0.001 level.

[0134] Effective doses of vary depending on many different factors, such as means of administration, target site, physiological state of the patient, whether the patient is an ApoE carrier, whether the patient is human or an animal, other medications administered, and whether treatment is prophylactic or therapeutic.

[0135] In some embodiments, the effective amount is a total dose of 25 μ g to 1000 μ g, or 50 μ g to 1000 μ g. In some embodiments, the effective amount is a total dose of 100 μ g. In some embodiments, the effective amount is a dose of 25 μ g administered to the subject a total of two times. In some embodiments, the effective amount is a dose of 100 μ g administered to the subject a total of two times. In some embodiments, the effective amount is a dose of 400 μ g administered to the subject a total of two times. In some embodiments, the effective amount is a dose of 500 μ g administered to the subject a total of two times. In some embodiments, a RNA (e.g., mRNA) vaccine is administered to a subject by intradermal, intramuscular injection, or by intranasal

administration.

[0136] In some embodiments, the amount of an agent for active immunotherapy varies from 1 to 1,000 micrograms (μg), or from 0.1-500 μg , or from 10 to 500 μg , or from 50 to 250 μg per patient and can be from 1-100 or 1-10 μg per injection for human administration. The timing of injections can vary significantly from once a day, to once a week, to once a month, to once a year, to once a decade. A typical regimen consists of an immunization followed by booster injections at time intervals, such as 6 week intervals or two months. Another regimen consists of an immunization followed by one or more booster injections 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 months later. Another regimen entails an injection every two months for life. Alternatively, booster injections can be on an irregular basis as indicated by monitoring of immune response. The frequency of administration may be once or more as long as the side effects are within a clinically acceptable range.

[0137] In some embodiments, the compositions or methods as disclosed herein comprise administering to a subject a nucleic acid vaccine comprising one or more DNA or RNA polynucleotides having an open reading frame encoding a first peptide and a second peptide wherein a dosage of between 10 $\mu\text{g}/\text{kg}$ and 400 $\mu\text{g}/\text{kg}$ of the nucleic acid vaccine is administered to the subject. In some embodiments the dosage of the RNA polynucleotide is 1-5 μg , 5-10 μg , 10-15 μg , 15-20 μg , 10-25 μg , 20-25 μg , 20-50 μg , 30-50 μg , 40-50 μg , 40-60 μg , 60-80 μg , 60-100 μg , 50-100 μg , 80-120 μg , 40-120 μg , 40-150 μg , 50-150 μg , 50-200 μg , 80-200 μg , 100-200 μg , 120-250 μg , 150-250 μg , 180-280 μg , 200-300 μg , 50-300 μg , 80-300 μg , 100-300 μg , 40-300 μg , 50-350 μg , 100-350 μg , 200-350 μg , 300-350 μg , 320-400 μg , 40-380 μg , 40-100 μg , 100-400 μg , 200-400 μg , or 300-400 μg per dose. In some embodiments, the nucleic acid is administered to the subject by intradermal or intramuscular injection. In some embodiments, the nucleic acid is administered to the subject on day zero. In some embodiments, a second dose of the nucleic acid is administered to the subject on day seven, or fourteen, or twenty one.

[0138] The compositions described herein are preferably administered via a peripheral route (i.e., one in which the administered composition results in a robust immune response and/or the induced antibody population crosses the blood brain barrier to reach an intended site in the brain, spinal cord, or eye). For peripheral diseases, the induced antibodies leave the vasculature to reach the intended peripheral organs. Routes of administration include oral, subcutaneous, intranasal, intradermal, or intramuscular. Some routes for active immunization are subcutaneous and intramuscular. Intramuscular administration and subcutaneous administration can be made at a single site or multiple sites. Intramuscular injection is most typically performed in the arm or leg muscles. In some methods, agents are injected directly into a particular tissue where deposits have accumulated.

[0139] The number of dosages administered can be adjusted to result in a more robust immune response (for example, higher titers). For acute disorders or acute exacerbations of a chronic disorder, between 1 and 10 doses are often sufficient. Sometimes a single bolus dose, optionally in divided form, is sufficient for an acute disorder or acute exacerbation of a chronic disorder. For chronic disorders, a vaccine/immunotherapy as disclosed herein can be administered at regular intervals, e.g., weekly, fortnightly, monthly, quarterly, every six months for at least 1, 5 or 10 years, or the life of the patient.

[0140] An effective amount of a DNA or RNA encoded immunogen can be between about 1 nanogram and about 1 gram per kilogram of body weight of the recipient, or about between about 0.1 $\mu\text{g}/\text{kg}$ and about 10 mg/kg , or about between about 1 $\mu\text{g}/\text{kg}$ and about 1 mg/kg . Dosage forms suitable for internal administration preferably contain (for the latter dose range) from about 0.1 μg to 100 μg of active ingredient per unit. The active ingredient may vary from 0.5 to 95% by weight based on the total weight of the composition. Alternatively, an effective dose of dendritic cells loaded with the antigen is between about 10×10^4 and 10×10^8 cells. Those skilled in the art of immunotherapy will be able to adjust these doses without undue experimentation.

[0141] The nucleic acid compositions may be administered in a convenient manner, e.g., injection by a convenient and effective route. Routes can include, but are not limited to, intradermal “gene gun” delivery or intramuscular injection. The modified dendritic cells are administered by subcutaneous, intravenous or intramuscular routes. Other possible routes include oral administration, intrathecal, inhalation, transdermal application, or rectal administration.

[0142] Depending on the route of administration, the composition may be coated in a material to protect the compound from the action of enzymes, acids and other natural conditions which may inactivate the compound. Thus, it may be necessary to coat the composition with, or co-administer the composition with, a material to prevent its inactivation. For example, an enzyme inhibitors of nucleases or proteases (e.g., pancreatic trypsin inhibitor, diisopropylfluorophosphate and trasylol) or in an appropriate carrier such as liposomes (including water-in-oil-in-water emulsions) as well as conventional liposomes (Strejan et al., *J. Neuroimmunol* 7(1):27-41, 1984).

[0143] The immunotherapeutic compositions disclosed herein may also be used in combination with other treatments for diseases associated with the accumulation of A β or tau, for example, anti-A β antibodies such as antibodies that specifically bind to any of the A β epitopes disclosed herein. For example, aducanumab or any of the antibodies disclosed in, for example, U.S. Patent Publication No. 20100202968 and U.S. Pat. No. 8,906,367, and/or anti-tau antibodies such as antibodies that specifically bind to any of the tau epitopes disclosed herein, ABBV-8E12, gosuranemab, zagotenemab, RG-6100, BIIB076 or any of the antibodies disclosed in WO2014/165271, U.S. Pat. No. 10,501,531, WO2017/191559, WO2017/191560, WO2017/191561, US20190330314, US20190330316, and WO2018/204546. In some combination therapy methods, the patient receives passive immunotherapy prior to the active immunotherapy methods disclosed herein. In other methods, the patient receives passive and active immunotherapy during the same period of treatment. Alternatively, patients may receive active immunotherapy prior to passive immunotherapy. Combinations may also include small molecule therapies and non-immunogenic therapies such as RAZADYNE® (galantamine), EXELON® (rivastigmine), and ARICEPT® (donepezil) and other compositions that improve the function of nerve cells in the brain.

[0144] The compositions of the disclosure may be used in the manufacture of medicaments for the treatment regimens described herein.

Treatment Regimens

[0145] Desired outcomes of the methods of treatment as disclosed herein vary according to the disease and patient profile and are determinable to those skilled in the art. Desired outcomes include an improvement in the patient's health status. Generally, desired outcomes include measurable indices such as reduction or clearance of pathologic amyloid fibrils, decreased or inhibited amyloid aggregation and/or deposition of amyloid fibrils, and increased immune response to pathologic and/or aggregated amyloid fibrils. Desired outcomes also include amelioration of amyloid disease-specific symptoms. As used herein, relative terms such as “improve,” “increase,” or “reduce” indicate values relative to a control, such as a measurement in the same individual prior to initiation of treatment described herein, or a measurement in a control individual or group. A control individual is an individual afflicted with the same amyloid disease as the individual being treated, who is about the same age as the individual being treated (to ensure that the stages of the disease in the treated individual and the control individual are comparable), but who has not received treatment using the disclosed immunotherapy/vaccine formulations. Alternatively, a control individual is a healthy individual, who is about the same age as the individual being treated. Changes or improvements in response to therapy are generally statistically significant and described by a p-value less than or equal to 0.1, less than 0.05, less than 0.01, less than 0.005, or less than 0.001 may be regarded as significant.

[0146] Effective doses of the compositions as disclosed herein, for the treatment of a subject vary depending upon many different factors, including means of administration, target site,

physiological state of the patient, whether the patient is human or an animal, other medications administered, if any, and whether treatment is prophylactic or therapeutic. Treatment dosages can be titrated to optimize safety and efficacy. The amount of immunogen can also depend on whether adjuvant is also administered, with higher dosages being required in the absence of adjuvant. The amount of an immunogen for administration sometimes varies from 1-500 µg per patient and more usually from 5-500 µg per injection for human administration. Occasionally, a higher dose of 1-2 mg per dosage is used. Typically, about 10, 20, 50 or 100 µg is used for each human dosage. The timing of dosages can vary significantly from once a day, to once a year, to once a decade. On any given day that a dosage of immunogen is given, the dosage is greater than 1 µg/patient and usually greater than 10 µg/patient if adjuvant is also administered, and greater than 10 µg/patient and usually greater than 100 µg/patient in the absence of adjuvant. A typical regimen consists of an immunization followed by booster dosage(s) at 6-week intervals. Another regimen consists of an immunization followed by booster dosage(s) 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 months later. Another regimen entails dosage(s) every two months for life. Alternatively, booster dosage(s) can be on an irregular basis as indicated by monitoring of immune response.

[0147] When administered in combination with a second treatment for Alzheimer's disease, such as, Razadyne® (galantamine), Exelon® (rivastigmine), and Aricept® (donepezil), the second treatment can be administered according the product label or as necessary in view of the treatment with the compositions of the disclosure.

Kits

[0148] The disclosure further provides kits (e.g., containers) comprising the compositions disclosed herein and related materials, such as instructions for use (e.g., package insert). The instructions for use may contain, for example, instructions for administration of the compositions and optionally one or more additional agents. The containers of peptide and/or nucleic acid compositions may be unit doses, bulk packages (e.g., multi-dose packages), or sub-unit doses.

[0149] Package insert refers to instructions customarily included in commercial packages of therapeutic products that contain information about the indications, usage, dosage, administration, contraindications and/or warnings concerning the use of such therapeutic products. Kits can also include a second container comprising a pharmaceutically acceptable buffer, such as bacteriostatic water for injection (BWFI), phosphate-buffered saline, Ringer's solution and dextrose solution. It can also include other materials desirable from a commercial and user standpoint, including other buffers, diluents, filters, needles, and syringes.

Uses

[0150] Each of the peptides, polypeptides, immunogens, and pharmaceutical compositions described herein may be for use in treating one or more of the diseases as described herein. In addition, each of the peptides, polypeptides, immunogens, and pharmaceutical compositions described herein may be for use in methods for treating one or more of the diseases as described herein. Each of the peptides, polypeptides, immunogens, and pharmaceutical compositions described herein may be used in a method for manufacturing a medicament for treating or use in treating one or more of the diseases as described herein.

[0151] The following are provided for exemplification purposes only and are not intended to limit the scope of the invention described in broad terms above.

[0152] All U.S. and international patent applications identified herein are incorporated by reference in their entirety.

EXAMPLES

Example 1: Immunization of Guinea Pigs

[0153] Guinea pigs were injected intramuscularly with 50 µg of a test immunogen, 25 µg QS21 in 200 µl of Addavax on day 0, 21, 49 and 77. Bleeds were done 7 days post immunization. The peptides tested included DAEFRHD (SEQ ID NO:06), QIVYKPV (SEQ ID NO: 39), and DAEFRHDDRQIVYKPV (SEQ ID NO:57). The specific immunogens were DAEFRHDC (SEQ

ID NO:71), QIVYKPVGGC (SEQ ID NO:72), and DAEFRHRRQIVYKPVGGC (SEQ ID NO:59). The peptides were coupled through the C-terminal cysteine to CRM-197 with a maleimide linkage.

[0154] Female Guinea Pigs were at least 5 weeks old at the start of the study having an approximate body weight of 350-500 g. Appropriate animal housing and research procedures for animal husbandry and care were conducted in an accredited facility in accordance with the guidelines of the U.S. Department of Agriculture's (USDA) and the Assessment and Accreditation of Laboratory Animal Care (AAALAC) International.

[0155] The immunogen concentration was 0.5 mg/ml. Prior to each administration of the test immunogen, approximately a 3 cm^{sup}.2 area on each hind limb was shaved and wiped with ethanol for visualization of the injection site. Each animal received a test immunogen dose of 200 microliters (0.25 micrograms/microliter) divided into two separate sites each of 100 microliter per injection (i.e., animals received 50 µg of immunogen in 100 µl PBS+25 µg of QS21 in 100 µl MF59). A 25 G-27 G needle was inserted intramuscularly into the hind limb, approximately 0.25-0.5 cm deep, and injected at 100 microliters per site. Injection sites were rotated each administration between four separate sites per hind limb and separated by at least 2 cm.

Example 2: Measurement of Antibody Titers

[0156] Whole blood samples were collected into clot activator tubes via jugular vein at 250-350 microliters per collection at weeks 1, 4, and 8. The maximum volume of whole blood was collected into clot activator tubes via cardiac puncture at termination on week 12. All blood samples were allowed to clot at room temperature for greater than 30 minutes, centrifuged at ambient temperature (approximately 20-25° C.) at 3,000 RPM for 10-15 minutes, and serum supernatant was transferred individually into clean cryovials. Serum supernatant was stored frozen at -80° C. (+12° C.).

Titer Guinea Pig Bleed on Soluble Aβ Aggregates

[0157] 2.5 µg/ml of a soluble aggregate Aβ prep (HFIP film of Aβ 42 was resuspended and incubated overnight with shaking, then spun to remove insoluble aggregate) was coated on to the plate at 100 µl per well in PBS and incubated overnight at room temperature. Plates were blocked for 1 hour with 1% BSA in PBS. Plates were aspirated and to row A, 200 µl of 0.1% BSA in PBS Tween was added to 1-4. In 1 neg. GP serum was added at 1/100 while 2-4 contained 1/100 test serum. Rows B-H contained 100 µl of 0.1% BSA in PBS Tween. Rows were serially diluted ½ down the plate giving dilution of 1/100 to 1/12800. Wells were incubated 2 hours at room temperature, then were washed and a 1/5000 dilution of anti Guinea Pig IgG HRP in 0.1% BSA in PBS Tween was prepared and then 100 µl added to the washed well. This incubated for 1 hour and was washed. OPD substrate was prepared using Thermo-Fisher OPD tablets at 1 tablet per 10 mls. Thermo fisher substrate buffer was added at 1/10 and each well had 100 µl added and incubated for 15 minutes. 50 µl of 2N H.sub.2SO.sub.4 was added to stop the reaction and plates were read on a molecular devices spectromax at 490 nM. Titer defined as the dilution giving 50% maximum OD and was extrapolated if it fell between dilutions

Titer Guinea Pig Bleeds on Tau

[0158] 2 µg/ml recombinant WT Tau 4R2N was coated on to the plate using 100 µl per well in PBS and incubated overnight at room temperature. Plates were blocked for 1 hour with 1% BSA in PBS. Plates were aspirated and to row A 200 µl of 0.1% BSA in PBS Tween was added. In column 1 neg. GP serum was added at 1/100 while the rest of the row contained 1/100 test serums. Rows were serially diluted ½ down the plate giving dilution of 1/100 to 1/12800. Wells incubated 2 hours at room temperature then were washed and a 1/5000 dilution of anti-Guinea Pig IgG HRP in 0.1% BSA in PBS Tween was prepared and then 100 µl added to the washed well. This incubated for 1 hour and was washed. OPD substrate was prepared using Thermo-Fisher OPD tablets at 1 tablet per 10 mls. Thermo fisher substrate buffer was added at 1/10 and each well had 100 µl added and was incubated for 15 minutes. 50 µl of 2N H.sub.2SO.sub.4 was added to stop the reaction and plates were read on a molecular devices spectromax at 490 nM. Titer was defined as the dilution giving

50% maximum OD and was extrapolated if it fell between dilutions.

Titer Guinea Pig Bleed on A β 1-28 or A β 1-15

[0159] A β 1-15 and A β 1-28 were both used at different parts of the study. Both of these will not form aggregates. 2 μ g/ml A β monomers were coated at coated on to the plate 100 μ l per well in PBS and incubated overnight at room temperature. Plates were blocked for 1 hour with 1% BSA in PBS. Plates were aspirated and to row A 200 μ l of 0.1% BSA in PBS Tween was added. In column 1 neg. GP serum was added at 1/100 while the rest of the row contained 1/100 test serums. Rows were serially diluted $\frac{1}{2}$ down the plate giving dilution of 1/100 to 1/12800. Wells incubated 2 hours at room temperature then were washed and a 1/5000 dilution of anti Guinea Pig IgG HRP in 0.1% BSA in PBS Tween was prepared and then 100 μ l added to the washed well. This incubated for 1 hour and was washed. OPD substrate was prepared using Thermo-Fisher OPD tablets at 1 tablet per 10 mls. Thermo fisher substrate buffer was added at 1/10 and each well had 100 μ l added and was incubated for 15 minutes. 50 μ l of 2N H.sub.2SO.sub.4 was added to stop the reaction and plates were read on a molecular devices spectromax at 490 nM. Titer was defined as the dilution giving 50% maximum OD and was extrapolated if it fell between dilutions.

Titering the Tau Against a Peptide Containing its Epitope

[0160] Thermofisher neutavidin plates were rehydrated with 0.05% Tween in TBS and aspirated. Peptide GGGSVQIVYKPVDLS (SEQ ID NO:68) containing a biotin was made up in a 1/500 dilution in 0.1% BSA in PBS tween. Adding 100 μ l per well for 1 hour and then washed. To row A of the plate 200 μ l of 0.1% BSA in PBS Tween was added. In column 1 neg. GP serum was added at 1/100 while the rest of the rows contained 1/100 test serums. Rows were serially diluted $\frac{1}{2}$ down the plate giving dilution of 1/100 to 1/12800. Wells incubated 2 hours at room temperature then were washed and a 1/5000 dilution of anti Guinea Pig IgG HRP in 0.1% BSA in PBS Tween was prepared and then 100 μ l added to the washed well. This incubated for 1 hour and was washed. OPD substrate was prepared using Thermo-Fisher OPD tablets at 1 tablet per 10 mls. Thermo fisher substrate buffer was added at 1/10 and each well had 100 μ l added and was incubated for 15 minutes. 50 μ l of 2N H.sub.2SO.sub.4 was added to stop the reaction and plates were read on a molecular devices spectromax at 490 nM. Titer defined as the dilution giving 50% maximum OD and was extrapolated if it fell between dilutions.

[0161] FIG. 1 shows the results comparing the geometric mean titers of Guinea Pig serum for immunogen DAEFRHRRQIVYKPV (SEQ ID NO:57) titered on monomeric A β 1-28, soluble aggregates of A β , full length tau, and tau MTBR epitope containing TABLE-US-00017 (SEQ ID NO: 68) GGGSVQIVYKPVDLS.

[0162] FIG. 2 shows the results for a single peptide immunogen A β and tau peptides (DAEFRHD (SEQ ID NO:06) and QIVYKPV (SEQ ID NO:39)) versus dual peptide immunogen DAEFRHRRQIVYKPV (SEQ ID NO:57) on A β 1-28 and full-length tau.

Example 3: Staining of Alzheimer's Brain Tissue with Sera from Guinea Pigs Immunized with a Vaccine as Disclosed Herein

[0163] Autopsy blocks of fresh frozen human brain tissue (~ 0.5 g) were embedded in optimal cutting temperature compound (OCT compound) and cut using a cryostat to generate 10 μ m sections. The sections were placed into a solution of glucose oxidase and beta D-glucose, in the presence of sodium azide, to block endogenous peroxidase. Once tissue sections were prepared, the staining with the specified Guinea pig sera from Guinea pigs immunized with a vaccine as disclosed herein was carried out at two dilutions (1:300 and 1:1500), using a rabbit anti-guinea pig secondary antibody and a DAKO DAB Detection Kit as per the manufacturer's instructions. The staining was processed using an automated Leica Bond Stainer. The results indicate that sera from Guinea pigs immunized with a vaccine as disclosed herein comprises antibodies specific to A β and tau in human brain tissue of Alzheimer's patients (see FIG. 3A-3F)

Example 4: Serum from Vaccinated Animals Blocks Soluble A β Aggregates from Binding to Neurons

[0164] E18 primary rat hippocampal neurons were cultured as described previously (Zago, et al. "Neutralization of Soluble, Synaptotoxic Amyloid β Species by Antibodies Is Epitope Specific," J Neurosci. 2012 Feb. 22; 32(8): 2696-2702). Soluble A β aggregate was pre-incubated with or without guinea pig vaccine serum on culture DIV14-21 to block soluble A β aggregate from neuritic binding. Guinea pig serum was isolated from animals vaccinated with dual immunogen peptide: DAEFRHDDRQIVYKPVGGC (SEQ ID NO:59; Immunogen 9). Fresh unlabeled, biotinylated or (9:1) soluble A β was prepared one day prior and incubated overnight at 4° C. Each diluted serum sample (1:1000, 1:300, and 1:100) and soluble A β solution was prepared at 2 $\times$ the final concentration in one-half of final treatment volume using NeuroBasal-no phenol red (NB-NPR) medium. This was combined with one-half final volume of 2 $\times$ soluble A β and with one-half final volume of 2 $\times$ diluted guinea pig vaccine serum to make up a 1 $\times$ final concentration in total final treatment volume, which was mixed well and then pre-incubated for 30 minutes at 37° C. E18 neurons were rinsed with NB-NPR at 150 μ L/well before adding binding treatment. Guinea pig serum from vaccinated animals/A β treatment was added to E18 neurons at 60 μ L/well, and then incubated for 30 minutes at 37° C. under normal incubator conditions (5% CO₂; 9% (2). Cells were rinsed twice using 150 μ L/well of NB-NPR, and then fixed in 4% paraformaldehyde in 1 $\times$ DPBS for 20 minutes. Cells were permeabilized using 0.1 TX-100 for 5 minutes, and blocked using 10% normal goat serum (NGS) for 1 hour at room temperature (RT). Cells were incubated with MAP2 & NeuN primary antibodies in 100 μ L/well, 1 $\times$ DPBS containing 1% BSA+1% NGS overnight at 4° C. The next day, cells were rinsed twice in 150 μ L/well 1 $\times$ DPBS for 5 minutes each wash. Secondary antibodies were added for 1 hour at RT in 100 μ L/well 1 $\times$ DPBS+1% BSA+1% NGS. High-content imaging (HCL) analysis was performed to quantify soluble aggregate A β neuritic binding using Operetta HCL CLS instrument (Perkin Elmer; modified Neurite Outgrowth algorithm: 40 $\times$ H.sub.20 objective; 40 fields per well; (n=3) per condition; data shown as mean (+/-) SD); MAP2 & NeuN (Abcam) neuronal markers used to each trace neurite tree and count cell body number per optical field; Neuritic A β soluble aggregate spots detected using streptavidin-488 or polyclonal A β antibody (Thermo; Millipore); and data reported as A β soluble aggregate spots/neuron (or as Integrated Intensity)).

[0165] Approximately 80-150 neurons were observed per well for each condition tested. The results demonstrate that Guinea pig serum from animals vaccinated with a dual immunogen peptide, DAEFRHDDRQIVYKPVGGC (SEQ ID NO:59; Immunogen 9), inhibited Abeta binding to neurons in a dose dependent manner (see FIG. 4).

Example 5: Mice Vaccinated with Dual Peptide Antigens Produce Titers to A β and Tau

[0166] Swiss Webster Female mice were injected on day 0, 14 and 28 with 25 μ g of a dual peptide immunogen (Table 2) and 25 μ g QS21 (Desert King) in PBS total 200 μ L/injection. Each mouse received 200 μ L subcutaneously. Mice were bled on day 21 and 35.

TABLE-US-00018

TABLE 2	Dual peptide immunogen sequence	Immunogen (SEQ ID NO.)
9	DAEFRHDDRQIVYKPVGGC	(SEQ ID NO: 59)
10	DAEFRHDDRQIVYKPVCC	(SEQ ID NO: 60)
18	DAEFRHDDRQIVYKPVAAAC	(SEQ ID NO: 61)
19	DAEFRHDDRQIVYKPVKKC	(SEQ ID NO: 62)
20	DAEFRHDDRREIVYKSPGGC	(SEQ ID NO: 63)
21	DAEFRHDDRREIVYKSPAAC	(SEQ ID NO: 64)
22	DAEFRHDDRREIVYKSPC	(SEQ ID NO: 65)
23	DAEFRHDDRREIVYKSPKKC	(SEQ ID NO: 66)

[0167] The immunogens containing Tau peptide EIVYKSP (SEQ ID No:43); see FIG. 5B) showed greater variability overall in tau titers (with the exception of immunogen 23) than those with the tau MTBR sequence QIVYKPV (SEQ ID NO:39); therefore the base microtubule binding region (MTBR) peptide containing DAEFRHDDRQIVYKPV (SEQ ID NO:57) was also tested (FIG. 5A). Titers of three immunogens with different linkers (no linker, AA and KK) compared to the GGC linker are shown in FIG. 5.

[0168] Titers observed from the second bleed are set forth in Tables 3 and 4:

TABLE-US-00019 TABLE 3 Titer Against Abeta 1-28 (at 50% Max OD) geometric Immunogen mouse # geometric mean SD number 1 2 3 4 mean factor 9 4000 2000 6000 1000 2632.15 2.20 10 13000 2000 4000 1000 3193.44 2.98 18 2000 2800 1200 5000 2407.6 1.80 19 7200 2500 1500 1800 2640.34 2.02 20 4000 2000 6000 1000 2470 1.96 21 13000 2000 4000 1000 1665 2.71 22 300 7000 2000 7000 2329 4.43 23 350 3400 4000 7000 2403 3.74

TABLE-US-00020 TABLE 4 Titer Against Tau (at 50% Max OD) geometric Immunogen mouse # geometric mean SD number 1 2 3 4 mean factor 9 400 800 200 800 475.7 1.94 10 400 3200 3200 1500 1574.4 2.67 18 1500 100 6000 100 547.7 7.72 19 10000 1000 13000 1500 3738.9 3.68 20 300 3200 No titer 1000 313 12.07 21 1500 100 6000 100 278.3 7.75 22 100 4000 No titer 13000 477.5 27.33 23 300 3000 1300 500 874.8 2.78

[0169] Titers observed in guinea pigs immunized with Immunogen 9

DAEFRHDDRQIVYKPVGGC (SEQ ID NO:59) are set forth in Table 5 below.

TABLE-US-00021 TABLE 5 Titer Against Titer Abeta Titer Titer Against bio- Guinea Against soluble Against GGSQIVYKPV DLS Pig Abeta 1-28 aggregate Tau (SEQ ID NO: 73) 1 3100 1500 1600 1000 2 3200 4000 6400 6400 3 3300 2000 3200 4000

Example 6: Serum from Vaccinated Animals Stains A β Plaques and Tau Pathologies in Human Brain Tissue

[0170] Fresh frozen human brain tissues from autopsied Alzheimer's disease donors or non-diseased controls was embedded in OCT, and cut in a cryostat to generate 10 μ m frozen sections. The tissue sections were incubated in a solution of glucose oxidase and beta D-glucose, in the presence of sodium azide, to block endogenous peroxidase. The staining with sera from vaccinated mice. Mice were vaccinated with the following dual antigen peptides:

DAEFRHDDRQIVYKPVGGC (SEQ ID NO:59, Immunogen 9); DAEFRHDDRQIVYKPVC (SEQ ID NO:60, Immunogen 10); DAEFRHDDRQIVYKPVAAC (SEQ ID NO:61, Immunogen 18); and DAEFRHDDRQIVYKPVKKC (SEQ ID NO:62, Immunogen 19) or control mice.

Staining was then carried out at 1:1000 dilution, in an automated Leica Bond Rx Stainer (Leica Biosystems). Antibody binding was detected using the Bond Polymer Refine Detection Kit (DS9800, Leica Biosystems), which is based on an anti-mouse polymer detection, DAB visualization and hematoxylin nuclear counter-staining. After cover-slipping, the stained tissue slides were digitally imaged with a Hamamatsu NanoZoomer 2.0HT slide scanner (Hamamatsu Corporation) with an NDP.scan, 2.5.85 software. The digitized images were viewed and analyzed using the NDP.view, 2.7.43.0 software.

[0171] Results demonstrate A β plaques and tau neurofibrillary tangles were identified based on their typical histopathological characteristics. Such pathologies were absent from tissues incubated with control mouse serum. Also, non-diseased tissue had no such pathological staining after incubation with the sera from vaccinated mice. FIGS. 6A-E and Table 6 summarize the results of A β and Tau staining using mouse sera from animals vaccinated with dual antigen peptides (1:1000 dilution of sera used to stain human brain tissue from AD patients).

TABLE-US-00022 TABLE 6 A β Tau Serum ID staining pathology Background 9-1 + Clear 9-2 + Dark 9-3 +++++ Clear 9-4 ++ Clear 10-1 +++++ OK 10-2 +++ Clear 10-3 +++++ * OK 10-4 +++++ * Dark 18-1 ++ Dark 18-3 + Clear 19-1 +++++ +++ OK 19-2 ++ * OK 19-3 + ++ Dark 19-4 ++ OK 9: DAEFRHDDRQIVYKPVGGC (SEQ ID NO: 59) 10: DAEFRHDDRQIVYKPVC (SEQ ID NO: 60) 18: DAEFRHDDRQIVYKPVAAC (SEQ ID NO: 61) 19: DAEFRHDDRQIVYKPVKKC (SEQ ID NO: 62) * possible tau pathology, too weak to clearly identify staining as tau

Example 7: Immunization and Determination of Titers in Cynomolgus Monkeys

[0172] The study described in this example was designed to assess the development of titers to A β and tau in cynomolgus monkeys using the dual antigen peptide DAEFRHDDRQIVYKPVGGC (SEQ ID NO:59). We also assessed two injection schedules and persistence of titers.

Immunizations

[0173] Several dual A β -tau linear immunogens with a dendritic cell cleavage site were screened in mice for balanced titer on A β and tau proteins (see Example 5, above). A subset of immunogens were further evaluated in guinea-pig (see Examples 1-5, above) and cynomolgus monkey (this example). Two groups of four monkeys were immunized intramuscularly with 50 μ g of immunogen (SEQ ID NO:59; DAEFRHDDRQIVYKPVGGC) coupled to CRM197 carrier protein, and 50 μ g of the adjuvant QS21. Group 1 was injected on week 0, 4, 12 and 24 with bleeds taken every 2 weeks through week 38. Group 2 was injected at week 0, 8, 24 with bleeds taken every 2 weeks through week 38. Serum titer levels were determined against A β and full-length tau. Fresh frozen human AD or control brain sections were stained with sera from immunized and control animals. The activity of the guinea-pig immune sera was also assessed on soluble A β oligomer binding in primary rat hippocampal neurons.

Titer Protocol

Measurement of Antibody Titer to A β 1-28 in Cynomolgus Monkey

[0174] Cynomolgus monkey bleeds were titered by enzyme-linked immunosorbent assay (ELISA). Plates were coated overnight at 2 μ g/mL with A β 1-28 (SEQ ID NO:67) in phosphate-buffered saline (PBS) and then blocked 1 hour with 1% bovine serum albumin (BSA) in PBS. Pre-bleed cynomolgus monkey was used as a negative control while known positive anti-serum from previous mouse studies was used as a positive control at the same dilutions of test serum. Bleeds were diluted in PBS/0.1% BSA/0.1% Tween 20 (PBS/BSA/T) starting at 1/100 and serially diluted 1:2 down the plate. Plates were washed with TBS/Tween 20 and goat anti-monkey immunoglobulin G (IgG) (heavy+light chains) horseradish peroxidase (HRP) (IgG [H+L] HRP; Invitrogen) was added and incubated 1 hour at room temperature. Plates were washed in TBS/Tween 20, and antibody binding was detected with o-phenylenediamine dihydrochloride (OPD) substrate (Thermo Fisher Scientific, Waltham, MA) following manufacturer's instructions. Plates were read at 490 nM on a Molecular Devices Spectromax. Titer was defined as the dilution giving 50% Maximum OD or 4 $\times$ background (defined in graphs and tables) extrapolation was used if it fell in between dilutions.

Measurement of Antibody Titer to Tau in Cynomolgus Monkey

[0175] Cynomolgus monkey bleeds were titered by enzyme-linked immunosorbent assay (ELISA) against full-length recombinant tau (Proteos, Kalamazoo, MI; SEQ ID NO:02). Plates were coated overnight at 2 μ g/mL tau in phosphate-buffered saline (PBS) and then blocked 1 hour with 1% bovine serum albumin (BSA) in PBS. Pre-bleed guinea pig serum was used as a negative control while known positive anti-serum from previous mouse studies was used as a positive control at the same dilutions of test serum. Bleeds were diluted in PBS/0.1% BSA/0.1% Tween 20 (PBS/BSA/T) starting at 1/100 and serially diluted 1:2 down the plate. Plates were washed with TBS/Tween 20 and goat anti-monkey immunoglobulin G (IgG) (heavy+light chains) horseradish peroxidase (HRP) (IgG [H+L] HRP; Invitrogen) was added and incubated 1 hour at room temperature. Plates were washed in TBS/Tween 20, and antibody binding was detected with o-phenylenediamine dihydrochloride (OPD) substrate (Thermo Fisher Scientific, Waltham, MA) following manufacturer's instructions. Plates were read at 490 nM on a Molecular Devices Spectromax. Titer was defined as the dilution giving 50% Maximum OD or 4 $\times$ background (defined in graphs and tables) extrapolation was used if it fell in between dilutions.

Titer Cynomolgus Monkey Bleeds on the Carrier Protein CRM197

[0176] Cynomolgus monkey bleeds were titered by enzyme-linked immunosorbent assay (ELISA) against CRM197 (FinaBio, Maryland) carrier protein. Plates were coated overnight at 2 μ g/mL tau in phosphate-buffered saline (PBS) and then blocked 1 hour with 1% bovine serum albumin (BSA) in PBS. Pre-bleed guinea pig serum was used as a negative control while known positive anti-serum from previous mouse studies was used as a positive control at the same dilutions of test serum. Bleeds were diluted in PBS/0.1% BSA/0.1% Tween 20 (PBS/BSA/T) starting at 1/100 and serially diluted 1:2 down the plate. Plates were washed with TBS/Tween 20 and goat anti-monkey

immunoglobulin G (IgG) (heavy+light chains) horseradish peroxidase (HRP) (IgG [H+L] HRP; Invitrogen) was added and incubated 1 hour at room temperature. Plates were washed in TBS/Tween 20, and antibody binding was detected with o-phenylenediamine dihydrochloride (OPD) substrate (Thermo Fisher Scientific, Waltham, MA) following manufacturer's instructions. Plates were read at 490 nM on a Molecular Devices Spectromax. Titer was defined as the dilution giving 50% Maximum OD or 4× background (defined in graphs and tables) extrapolation was used if it fell in between dilutions.

Results

[0177] Titers levels were similar for Aβ and tau. Monkeys subject to the group 1 vaccination schedule had titers with a greater area under the curve than those of group 2 but did not show a significant improvement in maximum titer under these conditions (FIG. 7A, FIG. 7B). Overall titer levels were lower than the guinea pig study described above. This change may be attributable to a change of adjuvant from QS21 in ADDVAX in the Guinea pig to QS21 in phosphate buffered saline (PBS) in cynomolgus monkeys to match approved vaccines. Persistence of titer was modest for both four injection and three injection vaccination schedules (FIG. 8A, FIG. 8B). These results demonstrate that these immunogens can generate a balanced immune response to Aβ and tau in primates. Optimization of dosing schedule and/or formulation may improve titer levels and persistence.

Example 8: Immunohistochemistry on AD and Control Brain Sections

[0178] Sera from immunized monkeys were evaluated for the ability to bind pathological Aβ plaques and tau tangles in human brain tissue from subjects with AD. Binding both Aβ plaques and tau tangles is expected reduce plaque burden and reduce tau transmission which in turn should reduce the signs and symptoms of AD.

[0179] Autopsy blocks of fresh frozen human brain tissue were embedded in optimal cutting temperature compound (OCT compound) and cut using a cryostat to generate 10 micron sections. The sections were placed into a solution of glucose oxidase and beta D-glucose, in the presence of sodium azide, to block endogenous peroxidase. Once tissue sections were prepared, the staining with the specified cynomolgus immune sera was carried out at two dilutions (1:300 in 5% goat serum with 0.25% Triton for 1 hour at RT). Binding was detected with a mouse anti-monkey IgG secondary antibody purified unlabeled (Mybiosource 3 mg/mL) for 1 hour at RT, and a goat anti-mouse IgG secondary antibody (Jackson, 1:200) for 1 hour at RT, avidin-biotin complex (ABC; Vector PK-4000) and a DAKO DAB Detection Kit according to manufacturer instructions. The staining was processed using an automated Leica Bond Stainer.

[0180] Table 7 shows a summary of the staining in the cynomolgus monkey study. All animal except 1002 had positive staining for Aβ. Only animal 1001 had strong tau staining (FIG. 9A-FIG. 9C), with 1003 (FIG. 10A-FIG. 10B) and 1501 (FIG. 11A-FIG. 11B) demonstrating modest staining. Animal 2501, for example, did not demonstrate significant tau staining (FIG. 12A-FIG. 12B). There did not appear to be a correlation with tau staining and tau titers as evidenced by animal 1001, which did not have a strong tau titer but stained tau strongly.

TABLE-US-00023

TABLE 7	Animal	Aβ	Tau	Aβ	Tau	Number	Staining	Titer	Titer
1001	++	800	150	1002	-	25	25	1003	+++
15000	6000	1501	+++	+/-	8000	3200	2001	+++	-
9000	350	2003	++	-	200	1500	2102	+++	-
8000	800	2501	+++	-	8000	1600			

[0181] Immune sera bound avidly to Aβ plaques and tau tangles in human AD brain sections at concentrations expected to be achieved in CNS under in vivo conditions. Monkeys with the strongest titers at week 26 stained Aβ, and one monkey with low to moderate tau titers stained both Aβ and tau.

[0182] These results demonstrate that the serum antibody binding of the cynomolgus monkey antibody response to be balanced and bind to pathological Aβ plaques and tau tangles.

Example 9: Serum Blocking of Soluble Aβ Aggregates from Binding to Neurons and T-Cell Activity

[0183] Primary hippocampal neurons isolated from E18 rat were cultured as described above. Fresh unlabeled, biotinylated or (9:1) soluble A β was prepared one day prior to assay and incubated overnight at 4° C. Neurons were rinsed with NB-NPR at 150 μ L/well before adding A β /serum treatment. An IgG cut of cynomolgus monkey bleeds from vaccinated animals was added to E18 neurons at 60 μ L/well, and then incubated for 30 minutes at 37° C. under normal incubator conditions (5% CO₂; 9% O₂). Cells were rinsed twice using 150 μ L/well of NB-NPR, and then fixed in 4% paraformaldehyde in 1 $\times$ DPBS for 20 minutes. Cells were permeabilized using 0.1 TX-100 for 5 minutes, and blocked using 10% normal goat serum (NGS; Thermofisher) for 1 hour at room temperature (RT). Cells were incubated with MAP2 & NeuN primary antibodies in 100 μ L/well, in 1 $\times$ DPBS containing 1% BSA+1% NGS overnight at 4° C.

[0184] The next day, cells were rinsed twice in 150 μ L/well 1 $\times$ DPBS for 5 minutes each wash. Secondary antibodies were added for 1 hour at RT in 100 μ L/well 1 $\times$ DPBS+1% BSA+1% NGS. A β soluble aggregate spots were detected using streptavidin-488 or polyclonal A β antibody (Thermofisher; Millipore); High-content imaging (HCl) analysis was performed to quantify soluble aggregate A β neuritic binding using Operetta HCl CLS instrument (Perkin Elmer; modified Neurite Outgrowth algorithm: 40 $\times$ H.sub.2O objective; 40 fields per well; (n=3) per condition; data shown as mean (+/-) SD); MAP2 & NeuN (Abcam) neuronal markers were used to each trace neurite tree and count cell body number per optical field; Neuritic data were reported as A β soluble aggregate spots/neuron (or as Integrated Intensity)). Approximately 80-150 neurons were observed per well for each condition tested (Zago, et al., 2012).

[0185] T-cell reactivity was also investigated. Immune sera inhibited the binding of soluble A β aggregates to hippocampal neurons. Immunogens did not elicit cytotoxic T-cell responses to A β or tau.

Example 10: Animals Vaccinated with Dual Peptide Antigens Produce Titers to A β and Tau

[0186] The study described in this example was designed to assess dual A β and tau antigen peptides in mice. The dual A β -tau constructs in this example effectively demonstrated: high titers for both antigens; blocking of tau binding to heparin; and staining/binding to A β and tau peptides in brain tissue from human Alzheimer's patients. Furthermore, titers against tau for an engineered tau immunogen (Dual #11) were comparable, and in some cases, better than other tau immunogens despite including a non-native tau sequence. This demonstrates that engineered immunogens are useful in vaccine constructs.

TABLE-US-00024

TABLE	8	Dual	Abeta	Endo	tau	Con-	SEQ	pepti-	SEQ	struct	Abeta	ID	dase
Tau	ID	C-Terminal	ID	#	Sequence	NO.	linker	Sequence	NO.	linker	Cys	2	DAEFRHD
VKSKIGST	801	GG C	3	DAEFRHD	06	RR	SKIGSTEN	817	GG C	8	DAEFRHD	06	RR
TENLKHQP	695	GG C	9	DAEFRHD	06	RR	ENLKHQPG	689	GG C	11	DAEFRHD	06	RR
SKIGSKDNIKH	986	GG C											

TABLE-US-00025

TABLE	9	Dual	Construct	SEQ	ID	ID	#	Sequence	NO:	2
DAEFRHDDR	VKSKIGSTGGC	997	3	DAEFRHDDR	SKIGSTENG	998	8			
DAEFRHDDR	TENLKHQPGGC	999	9	DAEFRHDDR	ENLKHQPGGC	1000	11			
DAEFRHDDR	SKIGSKDNIKHGGC	1001								

[0187] Conjugation—Peptides were made by Biopeptide (San Diego, CA) CRM-bromoacetate was received from Fina Biosolutions (Rockville, MD) peptides were coupled to CRM following the below protocol.

[0188] 1M Tris HCL pH 8.0, MilliQ DI Water and 50 mM Borate, 100 mM NaCl, 5 mM EDTA pH 8.5 were sterile filtered and degassed. 1 mg of each peptide was dissolved in 0.2 mls of degassed water, then 0.1 mls degassed Tris_HCL was added. This was followed by 0.2 mls of the stock CRM-Bromoacetate (1 mg total), finally 0.5 mls of the Borate buffer was added. This incubated 24 hours at 4 degrees C. on a nutator to provide mixing. Samples were desalted into PBS and 5 μ L was run on a 10% Tris gel to confirm conjugation.

[0189] Injection of animals: Four (4) female Swiss webster mice were used in each group.

Immunogen preparation was 25 µg Immunogen, 25 µg QS21 and 150 µl of 0.02% Tween 80/PBS per injection. Each mouse received 200 µl subcutaneously. Mice were injected at day 0, at 4 weeks from day 0, and at 8 weeks from day 0, with bleeds taken for titer at 5 weeks from day 0 and animals sacrificed and terminal bleed collected at 9 weeks from day 0 (see FIG. 13).

[0190] Titer Assay: Mouse serum was titered by enzyme-linked immunosorbent assay (ELISA). Plates were coated overnight at 2 µg/mL with either Abeta 1-28 (Anaspec, San Jose, CA), or recombinant tau (Proteos, Kalamazoo, MI in phosphate-buffered saline (PBS), and then blocked 1 hour with 1% bovine serum albumin (BSA) in PBS. Normal mouse serum was used as a negative control while known positive anti-serum from previous mouse studies was used as a positive control at the same dilutions as test serum. Bleeds were diluted in PBS/0.1% BSA/0.1% Tween 20 (PBS/BSA/T) starting at 1/100 and serially diluted 1:2 down the plate. Plates were washed with TBS/Tween 20 and goat anti-mouse immunoglobulin G (IgG) (heavy+light chains) horseradish peroxidase (HRP) (ThermoFisher) 1/5000 was added and incubated 1 hour at room temperature. Plates were washed in TBS/Tween 20, and antibody binding was detected with o-phenylenediamine dihydrochloride (OPD) substrate (Thermo Fisher Scientific, Waltham, MA) following manufacturer's instructions. Plates were read at 490 nM on a Molecular Devices Spectromax. Titer was defined as the dilution giving 4× background (defined in graphs and tables); extrapolation was used if it fell in between dilutions.

MTBR Binding

[0191] Certain antibodies that bind to MTBR can bind to more than one MTBR region due to the homology of the various MTBR regions. Antiserum is titered on four MTBR regions using peptides of MTBR 1-4 purchased from Anaspec (San Jose, CA)

TABLE-US-00026 MTBR peptide 1 (SEQ ID NO: 1058)

QTAPVPMPLDKNVKSKIGSTENLKHQPGGGK MTBR peptide 2 (SEQ ID NO: 1059) VQIINKKLDLSNVQSKCGSKDNIKHVPGGGG MTBR peptide 3 (SEQ ID NO:1060) VQIVYKPVDSLKVTSKCGSLGNIHHKPGGGGQ MTBR peptide 4 (SEQ ID NO: 1061) VEVKSEKLDFKDRVQSKIGSLDNITHVPGGGN

[0192] Mouse serum is titered by enzyme-linked immunosorbent assay (ELISA). Plates are coated overnight at 2 µg/mL with each the various MTBR peptides in phosphate-buffered saline (PBS) and then blocked 1 hour with 1% bovine serum albumin (BSA) in PBS. Normal mouse serum is used as a negative control. Bleeds are diluted in PBS/0.1% BSA/0.1% Tween 20 (PBS/BSA/T) starting at 1/100 and serially diluted 1:2 down the plate. Plates are washed with TBS/Tween 20 and goat anti-mouse immunoglobulin G (IgG) (heavy+light chains) horseradish peroxidase (HRP) (ThermoFisher) 1/5000 is added and incubated 1 hour at room temperature. Plates are washed in TBS/Tween 20, and antibody binding is detected with o-phenylenediamine dihydrochloride (OPD) substrate (Thermo Fisher Scientific, Waltham, MA) following manufacturer's instructions. Plates are read at 490 nM on a Molecular Devices Spectromax. Titer is defined as the dilution giving 4× background (defined in graphs and tables); extrapolation is used if it falls in between dilutions.

Blocking of Tau Binding to Heparin

[0193] As a potential surrogate marker for the ability of the serum to block uptake of tau into cells an ELISA measuring the blocking of tau binding to Heparin plates was developed. Recombinant tau was biotinylated in-house. Heparin coated plates (Bioworld, Dublin, OH) were blocked with 2% BSA/PBS for 1 hour. In a separate deep well polypropylene 96 well plate (ThermoFisher) Serum was diluted from 1/25-1/3200 in 2% BSA/PBS, 60 µl total, to this 60 µl 200 ng/ml biotinylated tau in 2% BSA/PBS was added for a final concentration of serum 1/50-1/6400 and tau at 100 ng/ml. The mixture of serum and tau incubated for 2 hours then 100 µl/well was transferred to the blocked heparin plates and incubated 1 hour. Plates were washed in 0.1% Tween 20/TBS and goat anti-mouse immunoglobulin G (IgG) (heavy+light chains) horseradish peroxidase (HRP) (ThermoFisher) 1/5000 was added and incubated 1 hour at room temperature. Plates were washed in TBS/Tween 20 and 100 µl ThermoFisher TMB added and incubated 8 minutes stopped

H.sub.2SO.sub.4 and read at 450.

IHC Binding of Serum

[0194] Autopsy blocks of fresh frozen human brain tissue were embedded in optimal cutting temperature compound (OCT compound) and cut using a cryostat to generate 10 μ m sections. The sections were placed into a solution of glucose oxidase and beta D-glucose, in the presence of sodium azide, to block endogenous peroxidase. Once tissue sections were prepared, the staining with the specified mouse immune sera was carried out at 1:500 in 5% goat serum with 0.25% triton for 1 hour at RT. To image the binding to plaques and tangles, Biotin-SP-Conjugated Goat anti mouse IgG from Jackson (Lot #115-065-166) at 1:200 dilution were incubated with the sections. DAKO DAB Detection Kit as per the manufacturer's instructions. The staining was processed using an automated Leica Bond Stainer. The results indicate that sera from Guinea pigs immunized with a vaccine as disclosed herein comprises antibodies specific to Abeta and tau in human brain tissue of Alzheimer's patients

Serum Blocking of Abeta Binding to Neurons

[0195] E18 primary rat hippocampal neurons are cultured as described previously (Zago, et al. "Neutralization of Soluble, Synaptotoxic Amyloid β Species by Antibodies Is Epitope Specific," *J Neurosci.* 2012 Feb. 22; 32(8): 2696-2702). Soluble A β aggregate is pre-incubated with or without vaccine serum on culture DIV14-21 to block soluble A β aggregate from neuritic binding. Fresh unlabeled, biotinylated or (9:1) soluble A β is prepared one day prior and incubated overnight at 4° C. Each diluted serum sample and soluble A β solution is prepared at 2 $\times$ the final concentration in one-half of final treatment volume using NeuroBasal-no phenol red (NB-NPR) medium. This is combined with one-half final volume of 2 $\times$ soluble A β and with one-half final volume of 2 $\times$ diluted vaccine serum to make up a 1 $\times$ final concentration in total final treatment volume, which is mixed well and then pre-incubated for 30 minutes at 37° C. E18 neurons are rinsed with NB-NPR at 150 μ L/well before adding binding treatment. Serum from vaccinated animals/A β treatment to is added to E18 neurons at 60 μ L/well, and then incubated for 30 minutes at 37° C. under normal incubator conditions (5% CO.sub.2; 9% O.sub.2). Cells are rinsed twice using 150 μ L/well of NB-NPR, and then fixed in 4% paraformaldehyde in 1 $\times$ DPBS for 20 minutes. Cells are permeabilized using 0.1 TX-100 for 5 minutes, and blocked using 10% normal goat serum (NGS) for 1 hour at room temperature (RT). Cells are incubated with MAP2 & NeuN primary antibodies in 100 μ L/well, 1 $\times$ DPBS containing 1% BSA+1% NGS overnight at 4° C. The next day, cells are rinsed twice in 150 μ L/well 1 $\times$ DPBS for 5 minutes each wash. Secondary antibodies are added for 1 hour at RT in 100 μ L/well 1 $\times$ DPBS+1% BSA+1% NGS. High-content imaging (HCI) analysis is performed to quantify soluble aggregate A β neuritic binding using Operetta HCI CLS instrument (Perkin Elmer; modified Neurite Outgrowth algorithm: 40 $\times$ H.sub.2O objective; 40 fields per well; (n=3) per condition; data shown as mean (+/-) SD); MAP2 & NeuN (Abcam) neuronal markers is used to each trace neurite tree and count cell body number per optical field; Neuritic A β soluble aggregate spots detected using streptavidin-488 or polyclonal A β antibody (Thermo; Millipore); and data reported as A β soluble aggregate spots/neuron (or as Integrated Intensity)). Approximately 80-150 neurons are observed per well for each condition tested.

Results

[0196] Data for Abeta and tau titers and staining is summarized in Table 10 and Table 11.

TABLE-US-00027 TABLE 10 50% Titer Titer blocking IHC IHC Animal Abeta Tau tau binding
Abeta Tau Dual 2.1 35,000 35,000 1:300 ++++ + Construct 2.2 21,000 75,000 1:225 ++++ + 2 2.3 12,000 21,000 1:550 +++ +++++ 2.4 20,000 7,000 1:200 ++ +++++ Dual 3.1 22,000 12,000 1:275 +++ +++++ Construct 3.2 20,000 2,000 1:200 ~ +++++ 3 3.3 7,000 40,000 1:125 ++++ ++ 3.4 1,000 25,000 1:200 + +++++ Dual 8.1 15,000 100,000 1:275 ++ (or +½) +++++½ Construct 8.2 7,000 6,800 1:675 ++ (or +½) +++ 8 8.3 60,000 10,000 1:200 +++ - 8.4 30,000 20,000 1:200 ++++ +++++ Dual 9.1 150,000 50,000 1:525 + +++++½ Construct 9.2 30,000 21,000 1:325 ++ +++++ 9 9.3 13,000 30,000 1:325 +++ +++++ 9.4 72,000 30,000 1:625 +++ +++ Dual 11.1 30,000 6,000 1:375 ++++

++½ Construct 11.2 10,000 40,000 1:875 ~ ++++ 11 11.3 11,000 20,000 1:325 ++++½ ++ 11.4 10,000 10,000 1:275 ++++ +
 TABLE-US-00028 TABLE 11 Dual Abeta/Tau Tau Mouse Cell Sera A-beta bodies Neurites Notes
 2.1 ++++ + + 2.2 ++++ ++ + Presence of nuclear staining 2.3 +++ +++++ +++ 2.4 ++ +++++
 +++++ 3.1 +++ +++++ +++++ 3.2 - +++++ +++++ 3.3 ++++ ++ +++ Strong nuclear staining 3.4 +
 +++++ ++ 8.1 ++ +++++ +++++ 8.2 ++ +++ +++ 8.3 +++ - - 8.4 +++++ +++++ +++++ 9.1 + +++++
 +++++ 9.2 ++ +++++ +++ 9.3 +++ +++++ +++++ 9.4 +++ +++ +++++ 11.1 +++++ ++ +++ 11.2 + +++++ +
 11.3 +++++ ++ - 11.4 +++++ + - Some nuclear staining

Titers

[0197] As seen in FIG. 13 and Tables 10-13, all groups most animals produced high and balanced Abeta and tau titers, though some had a less variable titer response.

TABLE-US-00029 TABLE 12 Titers for the dual Abeta/tau constructs. Abeta tau Mouse Mouse
 Mouse Mouse Mouse Mouse Mouse Mouse Mouse 1 2 3 4 1 2 3 4 Dual Construct 2 Bleed 1 30000 25000
 25000 12000 150 5000 3000 4000 Bleed 2 35000 21000 12000 20000 35000 75000 21000 7000
 Dual Construct 3 Bleed 1 6400 800 15000 100 4000 6400 5000 2000 Bleed 2 22000 22000 7000
 1000 12000 2000 40000 2500 Dual Construct 8 Bleed 1 1500 2000 20000 18000 10000 25000 200
 8000 Bleed 2 15000 7000 60000 30000 100000 6800 1000 20000 Dual Construct 9 Bleed 1 2000
 12000 13000 6400 800 6000 7000 2000 Bleed 2 150000 30000 13000 72000 5000 21000 30000
 30000 Dual Construct 11 Bleed 1 6000 2000 5000 7000 6400 7000 2500 4000 30000 10000 11000
 10000 6000 40000 20000 10000

TABLE-US-00030 TABLE 13 Remaining tau bound after serum incubation
 DAEFRHDDRVSISKIGSTGGC (SEQ ID NO: 997) dilution Mouse 1 Mouse 2 Mouse 3
 Mouse 4 50 19.57 29.12 16.66 12.70 100 28.93 38.25 26.51 25.21 200 41.16
 53.48 29.64 46.53 400 59.88 89.43 43.15 81.31 800 95.14 139.21 65.00 134.97
 DAEFRHDDRNLKHQPGGGC (SEQ ID NO: 1000) dilution Mouse 1 Mouse 2
 Mouse 3 Mouse 4 50 16.44 21.90 24.26 15.35 100 22.07 47.55 39.93 19.39 200 37.10
 44.06 48.36 43.65 400 58.32 86.61 86.41 51.98 800 93.54 98.76 115.50 86.00
 DAEFRHDDRISKIGSTENGGC (SEQ ID NO: 998) dilution Mouse 1 Mouse 2 Mouse 3
 Mouse 4 50 16.84 27.29 28.01 28.56 100 25.18 39.02 53.18 42.60 200
 55.66 73.61 93.32 75.38 400 92.87 120.00 131.15 125.27 800 110.16 132.77 130.90
 135.93 DAEFRHDDRISKIGSKDNIKHGGC (SEQ ID NO: 1001) dilution Mouse 1
 Mouse 2 Mouse 3 Mouse 4 50 15.25 13.15 14.73 17.81 100 19.45 11.20 26.64 37.79
 200 37.65 14.59 56.17 54.57 400 74.17 39.69 78.69 91.10 800 90.35 62.16 96.30 109.00
 DAEFRHDDRRTENLKHQPGGGC (SEQ ID NO: 999) dilution Mouse 1 Mouse 2 Mouse
 3 Mouse 4 50 20.01 12.36 35.04 42.63 100 37.62 28.44 44.45 47.62 200 54.86
 47.97 66.72 61.84 400 114.93 48.51 91.94 87.67 800 110.60 81.25 100.01 97.15 Negative
 Control dilution Mouse 1 Mouse 2 Mouse 3 Mouse 4 50 83.35 71.98 84.30 83.96
 100 114.78 94.03 103.43 98.98 200 104.03 100.60 97.80 96.83 400 86.73 111.16 95.31
 91.72 800 96.20 110.05 99.38 94.45

Blocking of Tau Binding to Heparin

[0198] All constructs showed the ability to block the binding of tau to heparin, in particular, construct DAEFRHDDRVSISKIGSTGGC (SEQ ID NO:997) and construct DAEFRHDDRNLKHQPGGGC (SEQ ID NO:1000) showing the most robust blocking. See FIG. 15, Table 10 and Table 11.

IHC on Alzheimer Brain

[0199] Most animals had staining of pathological Abeta and tau, see FIG. 16.

Example 11: Immunogens

[0200] Immunogens were selected for evaluation in vaccine peptide constructs. Constructs comprise Aβ immunogens and tau immunogens. Some immunogens comprise a tau peptide comprising 3-10 amino acids from tau. Other immunogens comprise an engineered tau

immunogen.

Engineered Tau Immunogens

[0201] Certain immunogenic peptides were designed and selected to (i) raise antibodies that bind within the within microtubule binding repeats (MTBRs) of human Tau protein, (ii) be less likely to generate an unwanted T cell-mediated autoimmune response, and (iii) be less likely to raise antibodies that would cross-react with other human proteins.

[0202] First, sequence analysis and 3D modeling of Tau MTBRs was conducted to identify amino acid residues that may be important for raising antibodies that bind MTBR. The results of these analyses were used to design synthetic Tau immunogenic peptides with conserved residues and shuffled interspersed residues. Resulting engineered synthetic peptides are listed in Table 14.

TABLE-US-00031 TABLE 14 Engineered Tau immunogenic peptides

Engineered tau immunogenic SEQ ID	peptides	sequence NO:
SKIGSTENLKH	978	SKIGSTENIKH 979
SKIGSKDNLKH	980	SKIGSKENIKH 981
SKIGSLENLKH	982	SKIGSLENIKH 983
SKIGSTDNLKH	984	SKIGSTDNIKH 985
SKIGSKDNIKH	986	SKIGSLDNLKH 987
SKIGSLDNIKH	988	SKIGSTGNLKH 989
SKIGSTGNIKH	990	SKIGSKGNLKH 991
SKIGSKGNIKH	992	SKIGSLGNLKH 993
SKIGSLGNIKH	994	

[0203] Next, to assess the potential of an unwanted T cell-mediated autoimmune response, the engineered peptides were subjected to in silico analysis to predict MHC II binding using the IEDB (Immune Epitope Database) from National Institute of Allergy and Infectious Diseases/La Jolla Immunology Institute. MHC class II binding is considered a good indicator of a sequence containing a T-cell epitope. A panel of alleles were used for MHC II binding prediction. Engineered peptides with a predicted half maximal inhibitory concentration (IC₅₀) above a specified cutoff were considered to have a low probability of MHC II binding and were selected for further analysis.

[0204] Finally, the engineered peptides with low predicted MHC II binding were evaluated to predict if the anti-Tau MTBR antibodies that would be raised by the peptides could have unwanted cross-reactivity with other human proteins. Sequences of the engineered peptides were subjected to bioinformatic analysis against a non-redundant human proteome database to determine homology with human proteins. Engineered peptide sequences with low homology to secreted or cell-surface proteins were selected as top candidates to be used as antigens. Top candidate engineered Tau immunogenic peptides are listed in Table 15.

TABLE-US-00032 TABLE 15 Top candidate engineered Tau immunogenic peptides.

Engineered tau immunogenic SEQ ID	peptides	sequence NO:
SKIGSTDNIKH	985	
SKIGSKDNIKH	986	SKIGSLDNIKH 988

[0205] The dual A β -tau constructs comprising an engineered tau immunogen (e.g., Dual #11) demonstrated results that were comparable, and in some cases, better than other tau immunogens despite including a non-native tau sequence. FIG. 13 shows high titers for both A β and tau antigens, including an engineered tau sequence. FIG. 14 shows blocking of tau binding to heparin. FIG. 15G shows resulting tau antibody binding to MTBR region peptides. FIG. 16 E shows staining/binding to A β and tau peptides in brain tissue from human Alzheimer's patients. This demonstrates that engineered immunogens are useful in vaccine constructs

CONCLUSION

[0206] Dual immunogen A β -tau vaccine constructs were developed and it was shown that these constructs raised balanced titers to A β and tau in mice, guinea-pigs, and cynomolgus monkeys. The antibodies were immunoreactive with both A β plaques and neurofibrillary tau tangles in human AD brain sections and blocked the binding of soluble A β aggregates (oligomers) to neurons without eliciting T-cell responses for A β or tau. These results support the development of a single-agent, dual-immunogen vaccine with the ability to target the pathogenic forms of A β and tau. These results support the development of a dual A β -tau vaccine with the ability to target pathogenic A β and tau for the prevention and/or treatment of AD.

[0207] Although various specific embodiments of the present invention have been described herein, it is to be understood that the invention is not limited to those precise embodiments and that various changes or modifications can be affected therein by one skilled in the art without departing from the scope and spirit of the invention.

[0208] In each of the embodiments of the peptide described herein, the peptide may comprise, consist, or consist essentially of the recited sequences. Thus, incorporated in this disclosure (see Table 16) are the following sequences that can be part of the compositions comprising an amyloid-beta (A β) peptide and a tau peptide as disclosed herein.

TABLE-US-00033 TABLE 16 SEQUENCES SEQ ID NO: 01 - Abeta 1-42
DAEFRHDSGYEVHHQKLVFFAEDVGSNKGAIIGLMVGGVVIA SEQ ID NO: 02 -
TAU (UNIPROTKB - P10636 (*Homo sapiens*) >P10636-8 (Isoform Tau-F)
10 20 30 40
MAEPRQEFEV MEDHAGTYGL GDRKDQGGYT MHQDQEGDTD
50 60 70 80 AGLKESPLQT
PTEDGSEEPG SETSDAKSTP TAEDVTAPLV 90
100 110 120 DEGAPGKQAA AQPHTEIPEG
TTAEEAGIGD TPSLEDEAAG 130
140 150 160 HVTQARMVSK SKDGTGSDDK
KAKGADGKTK IATPRGAAPP 170
180 190 200 GQKGQANATR IPAKTPPAPK
TPPSSGEPPK SGDRSGYSSP 210 220
230 240 GSPGTPGSRS RTPSLPTPPT REPKKVAVVR TPPKSPSSAK
250 260 270 280
SRLQTAPVPM PDLKNVFSKI GSTENLKHQP GGGKVQIINK
290 300 310 320 KLDLSNVQSK
CGSKDNIKHV PGGGSVQIVY KPVDLSKVTS 330
340 350 360 KCGSLGNIHH KPGGGQVEVK
SEKLDFKDRV QSKIGSLDNI 370 380
390 400 THVPGGGNKK IETHKLTFRE NAKAKTDHGA
EIVYKSPVVS 410 420
430 440 GDTSPRHLSN VSSTGSIDMV DSPQLATLAD
EVSASLAKQG L A-beta immunogens: DAEFRHDSGY (SEQ ID NO: 03) DAEFRHDSG
(SEQ ID NO: 04) DAEFRHDS (SEQ ID NO: 05) DAEFRHD (SEQ ID NO: 06)
DAEFRH (SEQ ID NO: 07) DAEFR (SEQ ID NO: 08) DAEF (SEQ ID NO: 09)
DAE AEFRHDSGY (SEQ ID NO: 11) AEFRHDSG (SEQ ID NO: 12) AEFRHDS
(SEQ ID NO: 13) AEFRHD (SEQ ID NO: 14) AEFRH (SEQ ID NO: 15) AEFR
(SEQ ID NO: 16) AEF EFRHDSGY (SEQ ID NO: 18) EFRHDSG (SEQ ID NO:
19) EFRHDS (SEQ ID NO: 20) EFRHD (SEQ ID NO: 21) EFRH (SEQ ID NO:
22) EFR FRHDSGY (SEQ ID NO: 24) FRHDSG (SEQ ID NO: 25) FRHDS (SEQ
ID NO: 26) FRHD (SEQ ID NO: 27) FRH RHDSGY (SEQ ID NO: 29) RHDSG
(SEQ ID NO: 30) RHDS (SEQ ID NO: 31) RHD HDSGY (SEQ ID NO: 33)
HDSG (SEQ ID NO: 34) HDS DSGY (SEQ ID NO: 36) DSG SGY VHHQKLVFFA
(SEQ ID NO: 1002) VHHQKLVFF (SEQ ID NO: 1003) VHHQKLVF (SEQ ID
NO: 1004) VHHQKLV (SEQ ID NO: 1005) VHHQKL (SEQ ID NO: 1006)
HHQKLVFFAE (SEQ ID NO: 1007) HHQKLVFFA (SEQ ID NO: 1008) HHQKLVFF
(SEQ ID NO: 1009) HHQKLVF (SEQ ID NO: 1010) HHQKLV (SEQ ID NO:
1011) HHQKL (SEQ ID NO: 1012) HQKLVFFAED (SEQ ID NO: 1013)
HQKLVFFAE (SEQ ID NO: 1014) HQKLVFFA (SEQ ID NO: 1015) HQKLVFF
(SEQ ID NO: 1016) HQKLVF (SEQ ID NO: 1017) HQKLV (SEQ ID NO: 1018)
HQKL (SEQ ID NO: 1019) QKLVFFAEDV (SEQ ID NO: 1020) QKLVFFAED

(SEQ ID NO: 1021) QKLVFFAE (SEQ ID NO: 1022) QKLVFFA (SEQ ID NO: 1023) QKLVFF (SEQ ID NO: 1024) QKLVF (SEQ ID NO: 1025) QKLV (SEQ ID NO: 1026) QKL KLVFFAEDVG (SEQ ID NO: 1028) KLVFFAEDV (SEQ ID NO: 1029) KLVFFAED (SEQ ID NO: 1030) KLVFFAE (SEQ ID NO: 1031) KLVFFA (SEQ ID NO: 1032) KLVFF (SEQ ID NO: 1033) KLVF (SEQ ID NO: 1034) KLV LVFFAEDVG (SEQ ID NO: 1036) LVFFAEDV (SEQ ID NO: 1037) LVFFAED (SEQ ID NO: 1038) LVFFAE (SEQ ID NO: 1039) LVFFA (SEQ ID NO: 1040) LVFF (SEQ ID NO: 1041) LVF VFFAEDVG (SEQ ID NO: 1043) VFFAEDV (SEQ ID NO: 1044) VFFAED (SEQ ID NO: 1045) VFFAE (SEQ ID NO: 1046) VFFA (SEQ ID NO: 1047) VFF FFAEDVG (SEQ ID NO: 1049) FFAEDV (SEQ ID NO: 1050) FFAED (SEQ ID NO: 1051) FFAE (SEQ ID NO: 1052) FFA FAEDVG (SEQ ID NO: 1054) FAEDV (SEQ ID NO: 1055) FAED (SEQ ID NO: 1056) FAE
Tau immunogens: QIVYKPV (SEQ ID NO: 39) QIVYKP (SEQ ID NO: 40) QIVYKSV (SEQ ID NO: 41) EIVYKSV (SEQ ID NO: 42) QIVYKS (SEQ ID NO: 83) EIVYKSP (SEQ ID NO: 43) EIVYKS (SEQ ID NO: 84) EIVYKPV (SEQ ID NO: 44) EIVYKP (SEQ ID NO: 85) IVYKSPV (SEQ ID NO: 45) IVYK (SEQ ID NO: 46) CNIKHVPG (SEQ ID NO: 86) CNIKHVP (SEQ ID NO: 47) NIKHVP (SEQ ID NO: 48) HVPGGG (SEQ ID NO: 49) HVPGG (SEQ ID NO: 50) HKPGGG (SEQ ID NO: 51) HKPGG (SEQ ID NO: 52) KHVPGGG (SEQ ID NO: 53) KHVPGG (SEQ ID NO: 54) HQPGGG (SEQ ID NO: 55) HQPGG (SEQ ID NO: 56) VQIINK (SEQ ID NO: 146) VQIINKK (SEQ ID NO: 147) VQIINKKL (SEQ ID NO: 148) QIINK (SEQ ID NO: 149) QIINKK (SEQ ID NO: 150) QIINKKL (SEQ ID NO: 151) EAAGHVTQC (SEQ ID NO: 152) EAAGHVTQAR (SEQ ID NO: 153) AAGHVTQAC (SEQ ID NO: 154) AGHVTQARC (SEQ ID NO: 155) AGHVTQAR (SEQ ID NO: 156) GYTMHQD (SEQ ID NO: 157) QGGYTMHC (SEQ ID NO: 158) QGGYTMHQD (SEQ ID NO: 159) GGYTMHQC (SEQ ID NO: 160) VPGGGSVQIV (SEQ ID NO: 161) PGGGSVQIV (SEQ ID NO: 162) GGSVQIV (SEQ ID NO: 163) GGSVQIV (SEQ ID NO: 164) GSVQIV (SEQ ID NO: 165) SVQIV (SEQ ID NO: 166) VQIV (SEQ ID NO: 167) QIV PGGGSVQIVY (SEQ ID NO: 169) GGSVQIVY (SEQ ID NO: 170) GGSVQIVY (SEQ ID NO: 171) GSVQIVY (SEQ ID NO: 172) SVQIVY (SEQ ID NO: 173) VQIVY (SEQ ID NO: 174) QIVY (SEQ ID NO: 175) IVY GGSVQIVYK (SEQ ID NO: 177) GGSVQIVYK (SEQ ID NO: 178) GSVQIVYK (SEQ ID NO: 179) SVQIVYK (SEQ ID NO: 180) VQIVYK (SEQ ID NO: 181) QIVYK (SEQ ID NO: 182) VYK GGSVQIVYKP (SEQ ID NO: 184) GSVQIVYKP (SEQ ID NO: 185) SVQIVYKP (SEQ ID NO: 186) VQIVYKP (SEQ ID NO: 187) IVYKP (SEQ ID NO: 188) VYKP (SEQ ID NO: 189) YKP GSVQIVYKPV (SEQ ID NO: 191) SVQIVYKPV (SEQ ID NO: 192) VQIVYKPV (SEQ ID NO: 193) IVYKPV (SEQ ID NO: 194) VYKPV (SEQ ID NO: 195) YKPV (SEQ ID NO: 196) KPV SVQIVYK PVD (SEQ ID NO: 198) VQIVYK PVD (SEQ ID NO: 199) QIVYK PVD (SEQ ID NO: 200) IVYK PVD (SEQ ID NO: 201) VYK PVD (SEQ ID NO: 203) YK PVD (SEQ ID NO: 204) K PVD (SEQ ID NO: 205) PVD VQIVYK PVDL (SEQ ID NO: 207) QIVYK PVDL (SEQ ID NO: 208) IVYK PVDL (SEQ ID NO: 209) VYK PVDL (SEQ ID NO: 210) YK PVDL (SEQ ID NO: 211) K PVDL (SEQ ID NO: 212) PVDL (SEQ ID NO: 213) VDL QIVYK PVDLS (SEQ ID NO: 215) IVYK PVDLS (SEQ ID NO: 216) VYK PVDLS (SEQ ID NO: 217) YK PVDLS (SEQ ID NO: 218) K PVDLS (SEQ ID NO: 219) PVDLS (SEQ ID NO: 220) VDL (SEQ ID NO: 221) IVYK PVDLSK (SEQ ID NO: 223) VYK PVDLSK (SEQ ID NO: 224) YK PVDLSK (SEQ ID NO: 225) K PVDLSK (SEQ ID NO: 226) PVDLSK (SEQ ID NO: 227) VDL (SEQ ID NO: 228) DLSK (SEQ ID NO: 229) LSK

YKPVLDLSKV (SEQ ID NO: 231) YKPVLDLSKV (SEQ ID NO: 232) KPVLDLSKV
(SEQ ID NO: 233) PVDLSKV (SEQ ID NO: 234) VDLSKV (SEQ ID NO: 235)
DLSKV (SEQ ID NO: 236) LSKV (SEQ ID NO: 237) SKV YKPVLDLSKVT (SEQ
ID NO: 239) KPVLDLSKVT (SEQ ID NO: 240) PVDLSKVT (SEQ ID NO: 241)
VDLSKVT (SEQ ID NO: 242) AKTDHGAEIV (SEQ ID NO: 243) KTDHGAEIV
(SEQ ID NO: 244) TDHGAEIV (SEQ ID NO: 245) DHGAEIV (SEQ ID NO:
246) HGAEIV (SEQ ID NO: 247) GAEIV (SEQ ID NO: 248) AEIV (SEQ ID
NO: 249) EIV KTDHGAEIVY (SEQ ID NO: 251) TDHGAEIVY (SEQ ID NO: 252)
DHGAEIVY (SEQ ID NO: 253) HGAEIVY (SEQ ID NO: 254) GAEIVY (SEQ ID
NO: 255) AEIVY (SEQ ID NO: 256) EIVY (SEQ ID NO: 257) IVY TDHGAEIVYK
(SEQ ID NO: 259) DHGAEIVYK (SEQ ID NO: 260) HGAEIVYK (SEQ ID
NO: 261) GAEIVYK (SEQ ID NO: 262) AEIVYK (SEQ ID NO: 263) EIVYK
(SEQ ID NO: 264) IVYK (SEQ ID NO: 265) DHGAEIVYKS (SEQ ID NO: 266)
HGAEIVYKS (SEQ ID NO: 267) GAEIVYKS (SEQ ID NO: 268) AEIVYKS (SEQ
ID NO: 269) EIVYKS (SEQ ID NO: 270) IVYKS (SEQ ID NO: 271) VYKS
(SEQ ID NO: 272) YKS HGAEIVYKSP (SEQ ID NO: 274) GAEIVYKSP (SEQ
ID NO: 275) AEIVYKSP (SEQ ID NO: 276) EIVYKSP (SEQ ID NO: 277)
IVYKSP (SEQ ID NO: 278) VYKSP (SEQ ID NO: 279) YKSP (SEQ ID NO:
280) KSP GAEIVYKSPV (SEQ ID NO: 282) AEIVYKSPV (SEQ ID NO: 283)
EIVYKSPV (SEQ ID NO: 284) IVYKSPV (SEQ ID NO: 285) VYKSPV (SEQ ID
NO: 286) YKSPV (SEQ ID NO: 287) KSPV (SEQ ID NO: 288) SPV AEIVYKSPVV
(SEQ ID NO: 290) EIVYKSPVV (SEQ ID NO: 291) IVYKSPVV (SEQ ID NO:
292) VYKSPVV (SEQ ID NO: 293) YKSPVV (SEQ ID NO: 294) KSPVV (SEQ
ID NO: 295) SPVV (SEQ ID NO: 296) PVV EIVYKSPVVS (SEQ ID NO: 298)
IVYKSPVVS (SEQ ID NO: 299) VYKSPVVS (SEQ ID NO: 300) YKSPVVS (SEQ
ID NO: 301) KSPVVS (SEQ ID NO: 302) SPVVS (SEQ ID NO: 303) PVVS
(SEQ ID NO: 304) VVS IVYKSPVVSG (SEQ ID NO: 306) VYKSPVVSG (SEQ
ID NO: 307) YKSPVVSG (SEQ ID NO: 308) KSPVVSG (SEQ ID NO: 309)
SPVVSG (SEQ ID NO: 310) PVVSG (SEQ ID NO: 311) VVSG (SEQ ID NO:
312) VSG VYKSPVVSGD (SEQ ID NO: 314) YKSPVVSGD (SEQ ID NO: 315)
KSPVVSGD (SEQ ID NO: 316) SPVVSGD (SEQ ID NO: 317) PVVSGD (SEQ
ID NO: 318) VVSGD (SEQ ID NO: 319) VSGD (SEQ ID NO: 320) SGD
YKSPVVSGDT (SEQ ID NO: 322) KSPVVSGDT (SEQ ID NO: 323) SPVVSGDT
(SEQ ID NO: 324) PVVSGDT (SEQ ID NO: 325) VVSGDT (SEQ ID NO: 326)
VSGDT (SEQ ID NO: 327) SGDT (SEQ ID NO: 328) GDT KSPVVSGDTS (SEQ
ID NO: 330) SPVVSGDTS (SEQ ID NO: 331) PVVSGDTS (SEQ ID NO: 332)
VVSGDTS (SEQ ID NO: 333) VSGDTS (SEQ ID NO: 334) SGDTS (SEQ ID
NO: 335) GDTTS (SEQ ID NO: 336) DTS SPVVSGDTSP (SEQ ID NO: 338)
PVVSGDTSP (SEQ ID NO: 339) VVSGDTSP (SEQ ID NO: 340) VSGDTSP (SEQ
ID NO: 341) SGDTSP (SEQ ID NO: 342) GDTSP (SEQ ID NO: 343) DTSP
(SEQ ID NO: 344) TSP PVVSGDTSPR (SEQ ID NO: 346) VVSGDTSPR (SEQ
ID NO: 347) VSGDTSPR (SEQ ID NO: 348) SGDTSPR (SEQ ID NO: 349)
GDTSPR (SEQ ID NO: 350) DTSPR (SEQ ID NO: 351) TSPR (SEQ ID NO:
352) SPR HQPGGGKVQI (SEQ ID NO: 354) QPGGGKVQI (SEQ ID NO: 355)
PGGGKVQI (SEQ ID NO: 356) GGGKVQI (SEQ ID NO: 357) GGVQI (SEQ
ID NO: 358) GKVQI (SEQ ID NO: 359) KVQI (SEQ ID NO: 360) VQI
QPGGGKVQII (SEQ ID NO: 362) PGGGGKVQII (SEQ ID NO: 363) GGGKVQII
(SEQ ID NO: 365) GGVQII (SEQ ID NO: 366) GKVQII (SEQ ID NO: 367)
KVQII (SEQ ID NO: 368) VQII (SEQ ID NO: 369) QII PGGGGKVQIIN (SEQ ID
NO: 371) GGGKVQIIN (SEQ ID NO: 372) GGVQIIN (SEQ ID NO: 373)

GKVQIIN (SEQ ID NO: 374) KVQIIN (SEQ ID NO: 375) VQIIN (SEQ ID NO: 376) QIIN (SEQ ID NO: 377) IIN GGGKVQIINK (SEQ ID NO: 379) GGGKVQIINK (SEQ ID NO: 380) GKVQIINK (SEQ ID NO: 381) KVQIINK (SEQ ID NO: 382) IINK (SEQ ID NO: 383) INK GGGKVQIINKK (SEQ ID NO: 385) GKVQIINKK (SEQ ID NO: 386) KVQIINKK (SEQ ID NO: 387) IINKK (SEQ ID NO: 388) INKK (SEQ ID NO: 389) NKK GKVQIINKKL (SEQ ID NO: 391) KVQIINKKL (SEQ ID NO: 392) IINKKL (SEQ ID NO: 393) INKKL (SEQ ID NO: 394) NKKL (SEQ ID NO: 395) KKL KVQIINKKLD (SEQ ID NO: 397) VQIINKKLD (SEQ ID NO: 398) QIINKKLD (SEQ ID NO: 399) IINKKLD (SEQ ID NO: 400) INKKLD (SEQ ID NO: 401) NKKLD (SEQ ID NO: 402) KKLD (SEQ ID NO: 403) KLD VQIINKKLDL (SEQ ID NO: 405) QIINKKLDL (SEQ ID NO: 406) IINKKLDL (SEQ ID NO: 407) INKKLDL (SEQ ID NO: 408) NKKLDL (SEQ ID NO: 409) KKLDL (SEQ ID NO: 410) KLDL (SEQ ID NO: 411) LDL QIINKKLDLS (SEQ ID NO: 413) IINKKLDLS (SEQ ID NO: 414) INKKLDLS (SEQ ID NO: 415) NKKLDLS (SEQ ID NO: 416) KKLDLS (SEQ ID NO: 417) KLDLS (SEQ ID NO: 418) LDLS (SEQ ID NO: 419) IINKKLDLSN (SEQ ID NO: 420) INKKLDLSN (SEQ ID NO: 421) NKKLDLSN (SEQ ID NO: 422) KKLDLSN (SEQ ID NO: 423) KLDLSN (SEQ ID NO: 424) LDLSN (SEQ ID NO: 425) DLSN (SEQ ID NO: 426) LSN INKKLDLSNV (SEQ ID NO: 428) NKKLDLSNV (SEQ ID NO: 429) KKLDLSNV (SEQ ID NO: 430) KLDLSNV (SEQ ID NO: 431) LDLSNV (SEQ ID NO: 432) DLSNV (SEQ ID NO: 433) LSNV (SEQ ID NO: 434) SNV NKKLDLSNVQ (SEQ ID NO: 436) KKLDLSNVQ (SEQ ID NO: 437) KLDLSNVQ (SEQ ID NO: 438) LDLSNVQ (SEQ ID NO: 439) DLSNVQ (SEQ ID NO: 440) LSNVQ (SEQ ID NO: 441) SNVQ (SEQ ID NO: 442) NVQ KKLDLSNVQS (SEQ ID NO: 444) KLDLSNVQS (SEQ ID NO: 445) LDLSNVQS (SEQ ID NO: 446) DLSNVQS (SEQ ID NO: 447) LSNVQS (SEQ ID NO: 448) SNVQS (SEQ ID NO: 449) NVQS (SEQ ID NO: 450) VOS SKCGSKDNIK (SEQ ID NO: 452) KCGSKDNIK (SEQ ID NO: 453) CGSKDNIK (SEQ ID NO: 454) GSKDNIK (SEQ ID NO: 455) SKDNIK (SEQ ID NO: 456) KDNIK (SEQ ID NO: 457) DNIK (SEQ ID NO: 458) NIK KCGSKDNIKH (SEQ ID NO: 460) CGSKDNIKH (SEQ ID NO: 461) GSKDNIKH (SEQ ID NO: 462) SKDNIKH (SEQ ID NO: 463) KDNIKH (SEQ ID NO: 464) DNIKH (SEQ ID NO: 465) NIKH (SEQ ID NO: 466) IKH CGSKDNIKHV (SEQ ID NO: 468) GSKDNIKHV (SEQ ID NO: 469) SKDNIKHV (SEQ ID NO: 470) KDNIKHV (SEQ ID NO: 471) DNIKHV (SEQ ID NO: 472) NIKHV (SEQ ID NO: 473) IKHV (SEQ ID NO: 474) KHV GSKDNIKHVP (SEQ ID NO: 476) SKDNIKHVP (SEQ ID NO: 477) KDNIKHVP (SEQ ID NO: 478) DNIKHVP (SEQ ID NO: 479) IKHVP (SEQ ID NO: 480) KHVP (SEQ ID NO: 481) HVP SKDNIKHVPG (SEQ ID NO: 483) KDNIKHVPG (SEQ ID NO: 484) DNIKHVPG (SEQ ID NO: 485) NIKHVPG (SEQ ID NO: 486) IKHVPG (SEQ ID NO: 487) KHVP (SEQ ID NO: 488) HVP (SEQ ID NO: 489) VPG KDNIKHVPGG (SEQ ID NO: 491) DNIKHVPGG (SEQ ID NO: 492) NIKHVPGG (SEQ ID NO: 493) IKHVPGG (SEQ ID NO: 494) KHVP (SEQ ID NO: 495) VPGG (SEQ ID NO: 496) PGG DNIKHVPGGG (SEQ ID NO: 498) NIKHVPGGG (SEQ ID NO: 499) IKHVPGGG (SEQ ID NO: 500) VPGGG (SEQ ID NO: 501) PGGG (SEQ ID NO: 502) GGG NIKHVPGGGGS (SEQ ID NO: 504) IKHVPGGGGS (SEQ ID NO: 505) KHVP (SEQ ID NO: 506) HVP (SEQ ID NO: 507) VPGGGGS (SEQ ID NO: 508) PGGGS (SEQ ID NO: 509) GGGGS (SEQ ID NO: 510) GGS IKHVPGGGGSV (SEQ ID NO: 512) KHVP (SEQ ID NO: 513) HVP (SEQ ID NO: 514) VPGGGGSV (SEQ ID NO: 515) PGGGSV (SEQ ID NO: 516) GGGSV (SEQ ID NO: 517)

GSV (SEQ ID NO: 518) GSVHPGGGSVQ (SEQ ID NO: 520) HVPGGGSVQ (SEQ ID NO: 521) VPGGGSVQ (SEQ ID NO: 522) PGGGSVQ (SEQ ID NO: 523) GGGSVQ (SEQ ID NO: 524) GGSVQ (SEQ ID NO: 525) GSVQ (SEQ ID NO: 526) SVQ HVPGGGSVQI (SEQ ID NO: 528) VPGGGSVQI (SEQ ID NO: 529) PGGGSVQI (SEQ ID NO: 530) GGGSVQI (SEQ ID NO: 531) GGSVQI (SEQ ID NO: 532) GSVQI (SEQ ID NO: 533) SVQI (SEQ ID NO: 534) GGSVQIVYKS (SEQ ID NO: 535) GSVQIVYKS (SEQ ID NO: 536) SVQIVYKS (SEQ ID NO: 537) VQIVYKS (SEQ ID NO: 538) QIVYKS (SEQ ID NO: 539) IVYKS (SEQ ID NO: 540) VYKS (SEQ ID NO: 541) YKS GSVQIVYKSV (SEQ ID NO: 543) SVQIVYKSV (SEQ ID NO: 544) VQIVYKSV (SEQ ID NO: 545) QIVYKSV (SEQ ID NO: 546) IVYKSV (SEQ ID NO: 547) VYKSV (SEQ ID NO: 548) YKSV (SEQ ID NO: 549) SVQIVYKSVD (SEQ ID NO: 550) VQIVYKSVD (SEQ ID NO: 551) QIVYKSVD (SEQ ID NO: 552) IVYKSVD (SEQ ID NO: 553) VYKSVD (SEQ ID NO: 554) YKSVD (SEQ ID NO: 555) KSVD (SEQ ID NO: 556) SVD VQIVYKSVDL (SEQ ID NO: 558) QIVYKSVDL (SEQ ID NO: 559) IVYKSVDL (SEQ ID NO: 560) VYKSVDL (SEQ ID NO: 561) YKSVDL (SEQ ID NO: 562) KSVDL (SEQ ID NO: 563) SVDL (SEQ ID NO: 564) QIVYKSVDLS (SEQ ID NO: 565) IVYKSVDLS (SEQ ID NO: 566) VYKSVDLS (SEQ ID NO: 567) YKSVDLS (SEQ ID NO: 568) KSVDLS (SEQ ID NO: 569) SVDLS (SEQ ID NO: 570) IVYKSVDLSK (SEQ ID NO: 571) VYKSVDLSK (SEQ ID NO: 572) YKSVDLSK (SEQ ID NO: 573) KSVDLSK (SEQ ID NO: 574) SVDLSK (SEQ ID NO: 364) VYKSVDLSKV (SEQ ID NO: 575) YKSVDLSKV (SEQ ID NO: 576) KSVDLSKV (SEQ ID NO: 577) SVDLSKV (SEQ ID NO: 578) YKSVDLSKVT (SEQ ID NO: 579) KSVDLSKVT (SEQ ID NO: 580) SVDLSKVT (SEQ ID NO: 581) DLSKVT (SEQ ID NO: 582) LSKVT (SEQ ID NO: 583) SKVT (SEQ ID NO: 584) KVT HGAEIVYKSV (SEQ ID NO: 586) GAEIVYKSV (SEQ ID NO: 587) AEIVYKSV (SEQ ID NO: 588) GAEIVYKSVV (SEQ ID NO: 589) AEIVYKSVV (SEQ ID NO: 590) EIVYKSVV (SEQ ID NO: 591) IVYKSVV (SEQ ID NO: 592) VYKSVV (SEQ ID NO: 593) YKSVV (SEQ ID NO: 594) KSVV (SEQ ID NO: 595) SVV AEIVYKSVVS (SEQ ID NO: 597) EIVYKSVVS (SEQ ID NO: 598) IVYKSVVS (SEQ ID NO: 599) VYKSVVS (SEQ ID NO: 600) YKSVVS (SEQ ID NO: 601) KSVVS (SEQ ID NO: 602) SVVS (SEQ ID NO: 603) EIVYKSVVSG (SEQ ID NO: 604) IVYKSVVSG (SEQ ID NO: 605) VYKSVVSG (SEQ ID NO: 606) YKSVVSG (SEQ ID NO: 607) KSVVSG (SEQ ID NO: 608) SVVSG (SEQ ID NO: 609) VVSG (SEQ ID NO: 610) VSG IVYKSVVSGD (SEQ ID NO: 612) VYKSVVSGD (SEQ ID NO: 613) YKSVVSGD (SEQ ID NO: 614) KSVVSGD (SEQ ID NO: 615) SVVSGD (SEQ ID NO: 616) VVSGD (SEQ ID NO: 617) VYKSVVSGDT (SEQ ID NO: 618) YKSVVSGDT (SEQ ID NO: 619) KSVVSGDT (SEQ ID NO: 620) SVVSGDT (SEQ ID NO: 621) YKSVVSGDTS (SEQ ID NO: 622) KSVVSGDTS (SEQ ID NO: 623) SVVSGDTS (SEQ ID NO: 624) YKSVVSGDTS (SEQ ID NO: 625) KSVVSGDTS (SEQ ID NO: 626) SVVSGDTS (SEQ ID NO: 627) VVSGDTS (SEQ ID NO: 628) KSVVSGDTSP (SEQ ID NO: 629) SVVSGDTSP (SEQ ID NO: 630) SVVSGDTSPR (SEQ ID NO: 631) DHGAEIVYKP (SEQ ID NO: 632) HGAEIVYKP (SEQ ID NO: 633) GAEIVYKP (SEQ ID NO: 634) AEIVYKP (SEQ ID NO: 635) HGAEIVYKPV (SEQ ID NO: 636) GAEIVYKPV (SEQ ID NO: 637) AEIVYKPV (SEQ ID NO: 638) GAEIVYKPVV (SEQ ID NO: 639) AEIVYKPVV (SEQ ID NO: 640) EIVYKPVV (SEQ ID NO: 641) IVYKPVV (SEQ ID NO: 642) VYKPVV (SEQ ID NO: 643) YKPVV (SEQ ID NO: 644) KPVV (SEQ ID NO: 645) AEIVYKPVVS (SEQ ID NO: 646) EIVYKPVVS (SEQ ID NO: 647) IVYKPVVS (SEQ ID NO: 648)

VYKPVVS (SEQ ID NO: 649) YKPVVS (SEQ ID NO: 650) KPVVS (SEQ ID NO: 651) EIVYKPVVSG (SEQ ID NO: 652) IVYKPVVSG (SEQ ID NO: 653) VYKPVVSG (SEQ ID NO: 654) YKPVVSG (SEQ ID NO: 655) KPVVSG (SEQ ID NO: 656) IVYKPVVSGD (SEQ ID NO: 657) VYKPVVSGD (SEQ ID NO: 658) YKPVVSGD (SEQ ID NO: 659) KPVVSGD (SEQ ID NO: 660) VYKPVVSGDT (SEQ ID NO: 661) YKPVVSGDT (SEQ ID NO: 662) KPVVSGDT (SEQ ID NO: 663) YKPVVSGDTS (SEQ ID NO: 664) KPVVSGDTS (SEQ ID NO: 665) PVVSGDTS (SEQ ID NO: 666) VVSGDTS (SEQ ID NO: 667) KPVVSGDTSP (SEQ ID NO: 668) CNIK (SEQ ID NO: 669) CNIKH (SEQ ID NO: 670) CNIKHV (SEQ ID NO: 671) CNIKHVPGG (SEQ ID NO: 672) CNIKHVPGGG (SEQ ID NO: 673) CNIKHVPGGGG (SEQ ID NO: 674) ENLKHQPGGG (SEQ ID NO: 675) NLKHQPGGG (SEQ ID NO: 676) LKHQPGGG (SEQ ID NO: 677) KHQPGGG (SEQ ID NO: 678) HQPGGG (SEQ ID NO: 679) QPGGG (SEQ ID NO: 680) TENLKHQPGG (SEQ ID NO: 681) ENLKHQPGG (SEQ ID NO: 682) NLKHQPGG (SEQ ID NO: 683) LKHQPGG (SEQ ID NO: 684) KHQPGG (SEQ ID NO: 685) HQPGG (SEQ ID NO: 686) QPGG (SEQ ID NO: 687) TENLKHQPG (SEQ ID NO: 688) ENLKHQPG (SEQ ID NO: 689) NLKHQPG (SEQ ID NO: 690) LKHQPG (SEQ ID NO: 691) KHQPG (SEQ ID NO: 692) HQPG (SEQ ID NO: 693) QPG TENLKHQP (SEQ ID NO: 695) ENLKHQP (SEQ ID NO: 696) NLKHQP (SEQ ID NO: 697) LKHQP (SEQ ID NO: 698) KHQP (SEQ ID NO: 699) HQP TENLKHQ (SEQ ID NO: 701) ENLKHQ (SEQ ID NO: 702) NLKHQ (SEQ ID NO: 703) LKHQ (SEQ ID NO: 704) KHQ TENLKH (SEQ ID NO: 706) ENLKH (SEQ ID NO: 707) NLKH (SEQ ID NO: 708) LKH TENLK (SEQ ID NO: 710) ENLK (SEQ ID NO: 711) NLK TENI (SEQ ID NO: 713) ENL TEN KDNIKHVPGGG (SEQ ID NO: 716) KDNI (SEQ ID NO: 717) KDN IKHVGGG (SEQ ID NO: 719) IKHVGG (SEQ ID NO: 720) IKHVG (SEQ ID NO: 721) KHVGGG (SEQ ID NO: 722) KHVGG (SEQ ID NO: 723) KHVG (SEQ ID NO: 724) LGNIHHKPGGG (SEQ ID NO: 725) GNIHHKPGGG (SEQ ID NO: 726) NIHHKPGGG (SEQ ID NO: 727) IHHKPGGG (SEQ ID NO: 728) HHKPGGG (SEQ ID NO: 729) KPGGG (SEQ ID NO: 730) LGNIHHKPGG (SEQ ID NO: 731) GNIHHKPGG (SEQ ID NO: 732) NIHHKPGG (SEQ ID NO: 733) IHHKPGG (SEQ ID NO: 734) HHKPGG (SEQ ID NO: 735) KPGG (SEQ ID NO: 736) LGNIHHKPG (SEQ ID NO: 737) GNIHHKPG (SEQ ID NO: 738) NIHHKPG (SEQ ID NO: 739) IHHKPG (SEQ ID NO: 740) HHKPG (SEQ ID NO: 741) HKPG (SEQ ID NO: 742) KPG LGNIHHKP (SEQ ID NO: 744) GNIHHKP (SEQ ID NO: 745) NIHHKP (SEQ ID NO: 746) IHHKP (SEQ ID NO: 747) HHKP (SEQ ID NO: 748) HKP LGNIHHK (SEQ ID NO: 750) GNIHHK (SEQ ID NO: 751) NIHHK (SEQ ID NO: 752) IHHK (SEQ ID NO: 753) HHK LGNIHH (SEQ ID NO: 755) GNIHH (SEQ ID NO: 756) NIHH (SEQ ID NO: 757) IHH LGNIH (SEQ ID NO: 759) GNIH (SEQ ID NO: 760) NIH LGNI (SEQ ID NO: 762) GNI LGN LDNITHVPGGG (SEQ ID NO: 765) DNITHVPGGG (SEQ ID NO: 766) NITHVPGGG (SEQ ID NO: 767) ITHVPGGG (SEQ ID NO: 768) THVPGGG (SEQ ID NO: 769) LDNITHVPGG (SEQ ID NO: 770) DNITHVPGG (SEQ ID NO: 771) NITHVPGG (SEQ ID NO: 772) ITHVPGG (SEQ ID NO: 773) THVPGG (SEQ ID NO: 774) LDNITHVPG (SEQ ID NO: 775) DNITHVPG (SEQ ID NO: 776) NITHVPG (SEQ ID NO: 777) ITHVPG (SEQ ID NO: 778) THVPG (SEQ ID NO: 779) LDNITHVP (SEQ ID NO: 780) DNITHVP (SEQ ID NO: 781) NITHVP (SEQ ID NO: 782) ITHVP (SEQ ID NO: 783) THVP (SEQ ID NO: 784) LDNITHV (SEQ ID NO: 785) DNITHV (SEQ ID NO: 786) NITHV (SEQ ID NO: 787) ITHV (SEQ ID NO: 788) THV LDNITH (SEQ ID NO: 790) DNITH

(SEQ ID NO: 791) NITH (SEQ ID NO: 792) ITH LDNIT (SEQ ID NO: 794)
DNIT (SEQ ID NO: 795) NIT LDNI (SEQ ID NO: 797) LDN KNVKSKIGST (SEQ
ID NO: 799) NVKSKIGST (SEQ ID NO: 800) VKSKIGST (SEQ ID NO: 801)
KSKIGST (SEQ ID NO: 802) SKIGST (SEQ ID NO: 803) KIGST (SEQ ID NO:
804) IGST (SEQ ID NO: 805) GST NVKSKIGSTE (SEQ ID NO: 807) VKSKIGSTE
(SEQ ID NO: 808) KSKIGSTE (SEQ ID NO: 809) SKIGSTE (SEQ ID NO: 810)
KIGSTE (SEQ ID NO: 811) IGSTE (SEQ ID NO: 812) GSTE (SEQ ID NO: 813)
STE VKSKIGSTEN (SEQ ID NO: 815) KSKIGSTEN (SEQ ID NO: 816) SKIGSTEN
(SEQ ID NO: 817) KIGSTEN (SEQ ID NO: 818) IGSTEN (SEQ ID NO: 819)
GSTEN (SEQ ID NO: 820) STEN (SEQ ID NO: 821) KSKIGSTENL (SEQ ID
NO: 822) SKIGSTENL (SEQ ID NO: 823) KIGSTENL (SEQ ID NO: 824) IGSTENL
(SEQ ID NO: 825) GSTENL (SEQ ID NO: 826) STENL (SEQ ID NO: 827)
SKIGSTENLK (SEQ ID NO: 828) KIGSTENLK (SEQ ID NO: 829) IGSTENLK
(SEQ ID NO: 830) GSTENLK (SEQ ID NO: 831) STENLK (SEQ ID NO: 832)
KIGSTENLKH (SEQ ID NO: 833) IGSTENLKH (SEQ ID NO: 834) GSTENLKH
(SEQ ID NO: 835) STENLKH (SEQ ID NO: 836) IGSTENLKHQ (SEQ ID NO:
837) GSTENLKHQ (SEQ ID NO: 838) STENLKHQ (SEQ ID NO: 839)
GSTENLKHQP (SEQ ID NO: 840) STENLKHQP (SEQ ID NO: 841) STENLKHQPG
(SEQ ID NO: 842) SNVQSKCGSK (SEQ ID NO: 843) NVQSKCGSK (SEQ ID
NO: 844) VQSKCGSK (SEQ ID NO: 845) QSKCGSK (SEQ ID NO: 846) SKCGSK
(SEQ ID NO: 847) KCGSK (SEQ ID NO: 848) CGSK (SEQ ID NO: 849) GSK
NVQSKCGSKD (SEQ ID NO: 851) VQSKCGSKD (SEQ ID NO: 852) QSKCGSKD
(SEQ ID NO: 853) SKCGSKD (SEQ ID NO: 854) KCGSKD (SEQ ID NO: 855)
CGSKD (SEQ ID NO: 856) GSKD (SEQ ID NO: 857) SKD VQSKCGSKDN (SEQ
ID NO: 859) QSKCGSKDN (SEQ ID NO: 860) SKCGSKDN (SEQ ID NO: 861)
KCGSKDN (SEQ ID NO: 862) CGSKDN (SEQ ID NO: 863) GSKDN (SEQ ID
NO: 864) SKDN (SEQ ID NO: 865) QSKCGSKDNI (SEQ ID NO: 866)
SKCGSKDNI (SEQ ID NO: 867) KCGSKDNI (SEQ ID NO: 868) CGSKDNI (SEQ
ID NO: 869) GSKDNI (SEQ ID NO: 870) SKDNI (SEQ ID NO: 871)
SKVTSKCGSL (SEQ ID NO: 872) KVTSKCGSL (SEQ ID NO: 873) VTSKCGSL
(SEQ ID NO: 874) TSKCGSL (SEQ ID NO: 875) SKCGSL (SEQ ID NO: 876)
KCGSL (SEQ ID NO: 877) CGSL (SEQ ID NO: 878) GSL KVTSKCGSLG (SEQ
ID NO: 880) VTSKCGSLG (SEQ ID NO: 881) TSKCGSLG (SEQ ID NO: 882)
SKCGSLG (SEQ ID NO: 883) KCGSLG (SEQ ID NO: 884) CGSLG (SEQ ID
NO: 885) GSLG (SEQ ID NO: 886) SLG VTSKCGSLGN (SEQ ID NO: 888)
TSKCGSLGN (SEQ ID NO: 889) SKCGSLGN (SEQ ID NO: 890) KCGSLGN
(SEQ ID NO: 891) CGSLGN (SEQ ID NO: 892) GSLGN (SEQ ID NO: 893)
SLGN (SEQ ID NO: 894) TSKCGSLGNI (SEQ ID NO: 895) SKCGSLGNI (SEQ
ID NO: 896) KCGSLGNI (SEQ ID NO: 897) CGSLGNI (SEQ ID NO: 898)
GSLGNI (SEQ ID NO: 899) SLGNI (SEQ ID NO: 900) SKCGSLGNIH (SEQ ID
NO: 901) KCGSLGNIH (SEQ ID NO: 902) CGSLGNIH (SEQ ID NO: 903)
GSLGNIH (SEQ ID NO: 904) SLGNIH (SEQ ID NO: 905) KCGSLGNIHH (SEQ
ID NO: 906) CGSLGNIHH (SEQ ID NO: 907) GSLGNIHH (SEQ ID NO: 908)
SLGNIHH (SEQ ID NO: 909) CGSLGNIHHK (SEQ ID NO: 910) GSLGNIHHK
(SEQ ID NO: 911) SLGNIHHK (SEQ ID NO: 912) GSLGNIHHKP (SEQ ID
NO: 913) SLGNIHHKP (SEQ ID NO: 914) SLGNIHHKPG (SEQ ID NO: 915)
DRVQSKIGSL (SEQ ID NO: 916) RVQSKIGSL (SEQ ID NO: 917) VQSKIGSL
(SEQ ID NO: 918) QSKIGSL (SEQ ID NO: 919) SKIGSL (SEQ ID NO: 920)
KIGSL (SEQ ID NO: 921) IGSL (SEQ ID NO: 922) RVQSKIGSLD (SEQ ID
NO: 923) VQSKIGSLD (SEQ ID NO: 924) QSKIGSLD (SEQ ID NO: 925)

SKIGSLD (SEQ ID NO: 926) KIGSLD (SEQ ID NO: 927) IGSLD (SEQ ID NO: 928) GSLD (SEQ ID NO: 929) SLD VQSKIGSLDN (SEQ ID NO: 931) QSKIGSLDN (SEQ ID NO: 932) SKIGSLDN (SEQ ID NO: 933) KIGSLDN (SEQ ID NO: 934) IGSLDN (SEQ ID NO: 935) GSLDN (SEQ ID NO: 936) SLDN (SEQ ID NO: 937) QSKIGSLDNI (SEQ ID NO: 938) SKIGSLDNI (SEQ ID NO: 939) KIGSLDNI (SEQ ID NO: 940) IGSLDNI (SEQ ID NO: 941) GSLDNI (SEQ ID NO: 942) SKIGSLDNIT (SEQ ID NO: 943) KIGSLDNIT (SEQ ID NO: 944) IGSLDNIT (SEQ ID NO: 945) GSLDNIT (SEQ ID NO: 946) SLDNIT (SEQ ID NO: 947) KIGSLDNITH (SEQ ID NO: 948) IGSLDNITH (SEQ ID NO: 949) GSLDNITH (SEQ ID NO: 950) SLDNITH (SEQ ID NO: 951) IGSLDNITHV (SEQ ID NO: 952) GSLDNITHV (SEQ ID NO: 953) SLDNITHV (SEQ ID NO: 954) GSLDNITHVP (SEQ ID NO: 955) SLDNITHVP (SEQ ID NO: 956) SLDNITHVPG (SEQ ID NO: 957) PDLKNVKS (SEQ ID NO: 958) DLKNVKS (SEQ ID NO: 959) LKNVKS (SEQ ID NO: 960) KNVKSKIG (SEQ ID NO: 961) NVKSKIGS (SEQ ID NO: 962) DLSNVQSK (SEQ ID NO: 963) LSNVQSKC (SEQ ID NO: 964) SNVQSKCG (SEQ ID NO: 965) NVQSKCGS (SEQ ID NO: 966) VDLSKVTS (SEQ ID NO: 967) DLSKVTSK (SEQ ID NO: 968) LSKVTSKC (SEQ ID NO: 969) SKVTSKCG (SEQ ID NO: 970) KVTSKCGS (SEQ ID NO: 971) LDFKDRVQ (SEQ ID NO: 972) DFKDRVQS (SEQ ID NO: 973) FKDRVQSK (SEQ ID NO: 974) KDRVQSKI (SEQ ID NO: 975) DRVQSKIG (SEQ ID NO: 976) RVQSKIGS (SEQ ID NO: 977) SKIGSTENLKH (SEQ ID NO: 978) SKIGSTENIKH (SEQ ID NO: 979) SKIGSKDNLKH (SEQ ID NO: 980) SKIGSKENIKH (SEQ ID NO: 981) SKIGSLENLKH (SEQ ID NO: 982) SKIGSLENIKH (SEQ ID NO: 983) SKIGSTDNLKH (SEQ ID NO: 984) SKIGSTDNIKH (SEQ ID NO: 985) SKIGSKDNIKH (SEQ ID NO: 986) SKIGSLDNLKH (SEQ ID NO: 987) SKIGSLDNIKH (SEQ ID NO: 988) SKIGSTGNLKH (SEQ ID NO: 989) SKIGSTGNIKH (SEQ ID NO: 990) SKIGSKGNLKH (SEQ ID NO: 991) SKIGSKGNIKH (SEQ ID NO: 992) SKIGSLGNLKH (SEQ ID NO: 993) SKIGSLGNIKH (SEQ ID NO: 994)

Lys Xaa.sub.1 Xaa.sub.2 Ser Xaa.sub.3 Xaa.sub.4 Asn Xaa.sub.5 Xaa.sub.6 His (SEQ ID NO: 995), wherein [0209] Xaa.sub.1 is I or C; [0210] Xaa.sub.2 is G; [0211] Xaa.sub.3 is T, K or L; [0212] Xaa.sub.4 is E, D or G; [0213] Xaa.sub.5 is L or I; [0214] Xaa.sub.6 is K, H or T.

TABLE-US-00034 (SEQ ID NO: 996) (Q/E)IVYK(S/P) (SEQ ID NO: 57) DAEFRHDDRQIVYKPV (SEQ ID NO: 58) DAEFRHDRREIVYKSV (SEQ ID NO: 59) DAEFRHDDRQIVYKPVGGC (SEQ ID NO: 60) DAEFRHDDRQIVYKPV (SEQ ID NO: 61) DAEFRHDDRQIVYKPVAA (SEQ ID NO: 62) DAEFRHDDRQIVYKPVKKC (SEQ ID NO: 63) DAEFRHDRREIVYKSPGGC (SEQ ID NO: 64) DAEFRHDRREIVYKSPAAC (SEQ ID NO: 65) DAEFRHDRREIVYKSPC (SEQ ID NO: 66) DAEFRHDRREIVYKSPKKC (SEQ ID NO: 67) DAEFRHDSGYEVHHQKLFFAEDVGSNKG (SEQ ID NO: 68) GGGSVQIVYKPV (DLS (RVRR; SEQ ID NO: 69) Arg-Val-Arg-Arg (SEQ ID NO: 80) Gly-Ala-Gly-Ala (AGAG; SEQ ID NO: 81) Ala-Gly-Ala-Gly (KGKG; SEQ ID NO: 82) Lys -Gly-Lys-Gly, (SEQ ID NO: 70) DAEFRHDDRQIVYKPVXXC, If present, XX can be GG or AA or KK or SS (SEQ ID NO: 79) DAEFRHDRREIVYKSVXXC, If present, XX can be GG or AA or KK or SS (SEQ ID NO: 71) DAEFRHDC (SEQ ID NO: 72) QIVYKPVGGC (SEQ ID NO: 73) GGSQIVYKPV (DLS (SEQ ID NO: 74) DAEFRHDRREIVYKSVGGC (SEQ ID NO: 75) DAEFRHDRREIVYKSVAAC (SEQ ID NO: 76) DAEFRHDRREIVYKSVC (SEQ ID NO: 77) DAEFRHDRREIVYKSVKKC (SEQ ID NO: 78) DAEFRHDDRNIKHVP (GGC (SEQ ID NO: 997) DAEFRHDDRNVKSKIGSTGGC (SEQ

ID NO: 998) DAEFRHDDRISKIGSTENGGC (SEQ ID NO: 999)
DAEFRHDDRRTENLKHQPGGC (SEQ ID NO: 1000) DAEFRHDDRRENLKHQPGGGC
(SEQ ID NO: 1001) DAEFRHDDRISKIGSKDNIKHGGC MTBR peptide 1 (SEQ ID
NO: 1058) QTAPVPMPDLKNVSKIGSTENLKHQPGGGK, MTBR peptide 2 (SEQ
ID NO: 1059) VQIINKKLDLSNVQSKCGSKDNIKHVPGGGS, MTBR peptide 3 (SEQ
ID NO: 1060) VQIVYKPVDLSKVTSKCGSLGNIHHKPGGGQ, MTBR peptide 4
(SEQ ID NO: 1061) VEVKSEKLDFKDRVOSKIGSLDNITHVPGGGN, (SEQ ID NO:
1062) GGGG, (SEQ ID NO: 1063) GGGG,

[0215] The disclosure provides peptide compositions and immunotherapy compositions comprising an amyloid-beta (A β) peptide and a tau peptide. The disclosure also provides methods of treating or effecting prophylaxis of Alzheimer's disease or other diseases with beta-amyloid deposition in a subject, including methods of clearing deposits, inhibiting or reducing aggregation of A β and/or tau, blocking the uptake by neurons, clearing amyloid, and inhibiting propagation of tau seeds in a subject having or at risk of developing Alzheimer's disease or other diseases containing tau and/or amyloid-beta accumulations. The methods include administering to such patients the compositions comprising an amyloid-beta (A β) peptide and a tau peptide.

Claims

1-89. (canceled)

90. A polypeptide comprising a first peptide comprising 3-10 contiguous amino acids from residues 1-10 of SEQ ID NO:01, wherein the 3-10 contiguous amino acids comprise an amino acid sequence selected from DAE, EFR, FRH, RHD, HDS, DSG, and SGY linked to a second peptide comprising 3-13 contiguous amino acids from residues 258-272 of SEQ ID NO: 02, wherein the 3-13 contiguous amino acids comprise an amino acid sequence selected from SKI, KIG, IGS, GST, STE, TEN, ENL, NLK, LKH, KHQ, HQP, QPG, PGG and GGG.

91. The polypeptide of claim 90, wherein one or both of the 3-10 contiguous amino acids of the first peptide and the 3-13 contiguous amino acids of the second peptide is at least four contiguous amino acids.

92. The polypeptide of claim 90, wherein one or both of the 3-10 contiguous amino acids of the first peptide and the 3-13 contiguous amino acids of the second peptide is at least five contiguous amino acids.

93. The polypeptide of claim 90, wherein one or both of the 3-10 contiguous amino acids of the first peptide and the 3-13 contiguous amino acids of the second peptide is at least six contiguous amino acids.

94. The polypeptide of claim 90, wherein one or both of the 3-10 contiguous amino acids of the first peptide and the 3-13 contiguous amino acids of the second peptide is at least seven contiguous amino acids.

95. The polypeptide of claim 90, wherein: (a) the 3-10 contiguous amino acids of the first peptide comprises an amino acid sequence selected from TABLE-US-00035 (SEQ ID NO: 03) DAEFRHDSGY, (SEQ ID NO: 04) DAEFRHDSG, (SEQ ID NO: 05) DAEFRHDS, (SEQ ID NO: 06) DAEFRHD, (SEQ ID NO: 07) DAEFRH, (SEQ ID NO: 08) DAEFR, (SEQ ID NO: 09) DAEF, (SEQ ID NO: 11) AEFRHDSGY, (SEQ ID NO: 12) AEFRHDSG, (SEQ ID NO: 13) AEFRHDS, (SEQ ID NO: 14) AEFRHD, (SEQ ID NO: 15) AEFRH, (SEQ ID NO: 16) AEFR, (SEQ ID NO: 18) EFRHDSGY, (SEQ ID NO: 19) EFRHDSG, (SEQ ID NO: 20) EFRHDS, (SEQ ID NO: 21) EFRHD, (SEQ ID NO: 22) EFRH, (SEQ ID NO: 24) FRHDSGY, (SEQ ID NO: 25) FRHDSG, (SEQ ID NO: 26) FRHDS, (SEQ ID NO: 27) FRHD, (SEQ ID NO: 29) RHDSGY, (SEQ ID NO: 30) RHDSG, (SEQ ID NO: 31) RHDS, (SEQ ID NO: 33) HDSGY, (SEQ ID NO: 34) HDSG, and (SEQ ID NO:

36) DSGY, and (b) the 3-13 contiguous amino acids of the second peptide comprises an amino acid sequence selected from TABLE-US-00036 (SEQ ID NO: 675) ENLKHQPGGG, (SEQ ID NO: 676) NLKHQPGGG, (SEQ ID NO: 677) LKHQPGGG, (SEQ ID NO: 678) KHQPGGG, (SEQ ID NO: 679,) HQPGGG, (SEQ ID NO: 680) QPGGG, (SEQ ID NO: 681) TENLKHQPGG, (SEQ ID NO: 682) ENLKHQPGG, (SEQ ID NO: 683) NLKHQPGG, (SEQ ID NO: 684) LKHQPGG, (SEQ ID NO: 685) KHQPGG, (SEQ ID NO: 686) HQPGG, (SEQ ID NO: 687) QPGG, (SEQ ID NO: 688) TENLKHQPG, (SEQ ID NO: 689) ENLKHQPG, (SEQ ID NO: 690) NLKHQPG, (SEQ ID NO: 691) LKHQPG, (SEQ ID NO: 692) KHQPG, (SEQ ID NO: 693) HQPG, (SEQ ID NO: 695) TENLKHQP, (SEQ ID NO: 696) ENLKHQP, (SEQ ID NO: 697) NLKHQP, (SEQ ID NO: 698) LKHQP, (SEQ ID NO: 699) KHQP, (SEQ ID NO: 701) TENLKHQ, (SEQ ID NO: 702) ENLKHQ, (SEQ ID NO: 703) NLKHQ, (SEQ ID NO: 704) LKHQ, (SEQ ID NO: 706,) TENLKH, (SEQ ID NO: 707) ENLKH, (SEQ ID NO: 708) NLKH, (SEQ ID NO: 710) TENLK, (SEQ ID NO: 711) ENLK, (SEQ ID NO: 713) TENL, (SEQ ID NO: 803) SKIGST, (SEQ ID NO: 804) KIGST, (SEQ ID NO: 805) IGST, (SEQ ID NO: 808) VKSKIGSTE, (SEQ ID NO: 809) KSKIGSTE, (SEQ ID NO: 810) SKIGSTE, (SEQ ID NO: 811) KIGSTE, (SEQ ID NO: 812) IGSTE, (SEQ ID NO: 813) GSTE, (SEQ ID NO: 817) SKIGSTEN, (SEQ ID NO: 818) KIGSTEN, (SEQ ID NO: 819) IGSTEN, (SEQ ID NO: 820) GSTEN, (SEQ ID NO: 821) STEN, (SEQ ID NO: 823) SKIGSTENL, (SEQ ID NO: 824) KIGSTENL, (SEQ ID NO: 825) IGSTENL, (SEQ ID NO: 826) GSTENL, (SEQ ID NO: 827) STENL, (SEQ ID NO: 828) SKIGSTENLK, (SEQ ID NO: 829) KIGSTENLK, (SEQ ID NO: 830) IGSTENLK, (SEQ ID NO: 831) GSTENLK, (SEQ ID NO: 832) STENLK, (SEQ ID NO: 833) KIGSTENLKH, (SEQ ID NO: 834) IGSTENLKH, (SEQ ID NO: 835) GSTENLKH, (SEQ ID NO: 836) STENLKH, (SEQ ID NO: 837) IGSTENLKHQ, (SEQ ID NO: 838) GSTENLKHQ, (SEQ ID NO: 839) STENLKHQ, (SEQ ID NO: 840) GSTENLKHQP, (SEQ ID NO: 841) STENLKHQP, (SEQ ID NO: 842) STENLKHQPG, and (SEQ ID NO: 978) SKIGSTENLKH.

96. The polypeptide of claim 95, wherein the 3-10 contiguous amino acids of the first peptide comprises an amino acid sequence selected from DAEFRHD (SEQ ID NO:06), DAEFR (SEQ ID NO: 08), EFRHD (SEQ ID NO:21).

97. The polypeptide of claim 95, wherein the 3-13 contiguous amino acids of the second peptide comprises an amino acid sequence selected from SKIGSTEN (SEQ ID NO:817), TENLKHQP (SEQ ID NO:695), ENLKHQP (SEQ ID NO:696), and ENLKHQPG (SEQ ID NO:689).

98. The polypeptide of claim 95, wherein the 3-10 contiguous amino acids of the first peptide comprises an amino acid sequence selected from DAEFRHD (SEQ ID NO:06), DAEFR (SEQ ID NO: 08), and EFRHD (SEQ ID NO:21), and the wherein the 3-13 contiguous amino acids of the second peptide comprises an amino acid sequence selected from SKIGSTEN (SEQ ID NO: 817), TENLKHQP (SEQ ID NO:695), ENLKHQP (SEQ ID NO:696), and ENLKHQPG (SEQ ID NO:689).

99. The polypeptide of claim 95, wherein the first peptide and second peptide are linked by a cleavable linker.

100. The polypeptide of claim 99, wherein the cleavable linker comprises an amino acid sequence comprising arginine-arginine (Arg-Arg), arginine-valine-arginine-arginine (Arg-Val-Arg-Arg (SEQ ID NO:69)), valine-citrulline (Val-Cit), valine-arginine (Val-Arg), valine-lysine (Val-Lys), valine-alanine (Val-Ala), phenylalanine-lysine (Phe-Lys), glycine-alanine-glycine-alanine (Gly-Ala-Gly-Ala; SEQ ID NO:80), alanine-glycine-alanine-glycine (Ala-Gly-Ala-Gly; SEQ ID NO:81), or lysine-glycine-lysine-glycine (Lys-Gly-Lys-Gly; SEQ ID NO:82).

101. An immunotherapy composition, comprising the polypeptide of claim 99, a carrier, and a

linker to the carrier located at either a C-terminus or a N-terminus of the polypeptide.

102. The immunotherapy composition of claim 101, wherein the linker to the carrier comprises an amino acid sequence selected from the group consisting of GG, GGG, AA, AAA, KK, KKK, SS, SSS, GAGA (SEQ ID NO:80), AGAG (SEQ ID NO:81), and KGKG (SEQ ID NO: 82).

103. The immunotherapy composition of claim 102, wherein the polypeptide, or linker to the carrier, if present, further comprises a C-terminal cysteine (C).

104. The immunotherapy composition of claim 103, wherein the carrier comprises serum albumins, immunoglobulin molecules, thyroglobulin, ovalbumin, tetanus toxoid (TT), diphtheria toxoid (DT), a genetically modified cross-reacting material (CRM) of diphtheria toxin, CRM197, meningococcal outer membrane protein complex (OMPC) and *H. influenzae* protein D (HiD), rEPA (*Pseudomonas aeruginosa* exotoxin A), KLH (keyhole limpet hemocyanin), and flagellin.

105. A pharmaceutical formulation comprising the polypeptide of claim 1 and at least one adjuvant.

106. The pharmaceutical formulation of claim 105, wherein the adjuvant is selected from the group consisting of aluminum hydroxide, aluminum phosphate, aluminum sulfate, 3 De-O-acylated monophosphoryl lipid A (MPL), QS-21, TQL1055, QS-18, QS-17, QS-7, Complete Freund's Adjuvant (CFA), Incomplete Freund's Adjuvant (IFA), oil in water emulsions (such as squalene or peanut oil), CpG, polyglutamic acid, polylysine, AddaVax™, MF59®, combinations thereof, or liposomal formulation or a multiple antigen presenting system (MAP).

107. A pharmaceutical composition comprising the immunotherapy composition of claim 101 and at least one adjuvant.

108. A nucleic acid comprising a nucleic acid sequence encoding the polypeptide of claim 90.

109. A nucleic acid immunotherapy composition comprising the nucleic acid of claim 108 and at least one adjuvant.

110. An immunization kit comprising the immunotherapy composition of claim 101 and an adjuvant, wherein the immunotherapy composition is in a first container and the adjuvant is in a second container.

111. A immunization kit comprising the nucleic acid immunotherapy composition of claim 109 and an adjuvant, wherein the nucleic acid is in a first container and the adjuvant is in a second container.
